

Owner:	Ingredion (Thailand)	Bid Package:	MEI Work
Project:	Sprint 18K TPA Spray Dryer in Thailand	Doc. No.:	RFA-IM-IGD-D25/001-000-014 REV.B
Consultant:	AEC Industrial Engineering	Date:	28/08/2026
Contractor:	Deline Engineering Co., Ltd.	Total Page(s):	66 (including this page)

Function:	☐ Architecture	☐ Civil / Structure	☐ HVAC	☐ Fire Protection
	☐ Sanitary	☐ Electrical	☐ Process	☐ Utilities
	☐ Mechanical / Piping	☒ Control / Instrument	☐ General	

Subject: ☒ Material for Approval ☐ _____

☐ Shop Drawings for Approval

Description: PLC Control Panel Wiring Diagram

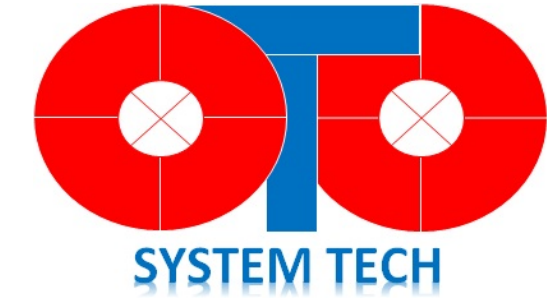
1 REQUESTED BY CONTRACTOR :		Authorized Signature
Attachment:	IM-3-IGD-D25001-000-007 Rev.B	 Name: <u>Mr. Theeraporn K.</u> Title: _____ Date: <u>28/08/2026</u>
	<u>PLC Control Panel Wiring Diagram</u>	

2 REVIEWED BY CONSULTANT : AEC INDUSTRIAL ENGINEERING		Authorized Signature
☐ Approved ☐ Approved as noted ☐ Rejected		Name: _____ Title: _____ Date: _____
☐ Make corrections as noted and proceed		
☐ Make corrections as noted and resubmit		
Notes: _____		

3 ACKNOWLEDGED BY OWNER: INGREDION PROJECT TEAM		Authorized Signature
☐ Acknowledged ☐ Acknowledged with comments		Name: _____ Title: _____ Date: _____
Notes: _____		

RECORD		
Original	☐ Return to Contractor	
Soft-copy (PDF)	☐ Ingredion (SharePoint)	☐ AECIE

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EPLAN GmbH & Co. KG

An der alten Ziegelei 2
40789 Monheim am Rhein
Phone +49 (0)2173 - 39 64 - 0

Company / customer

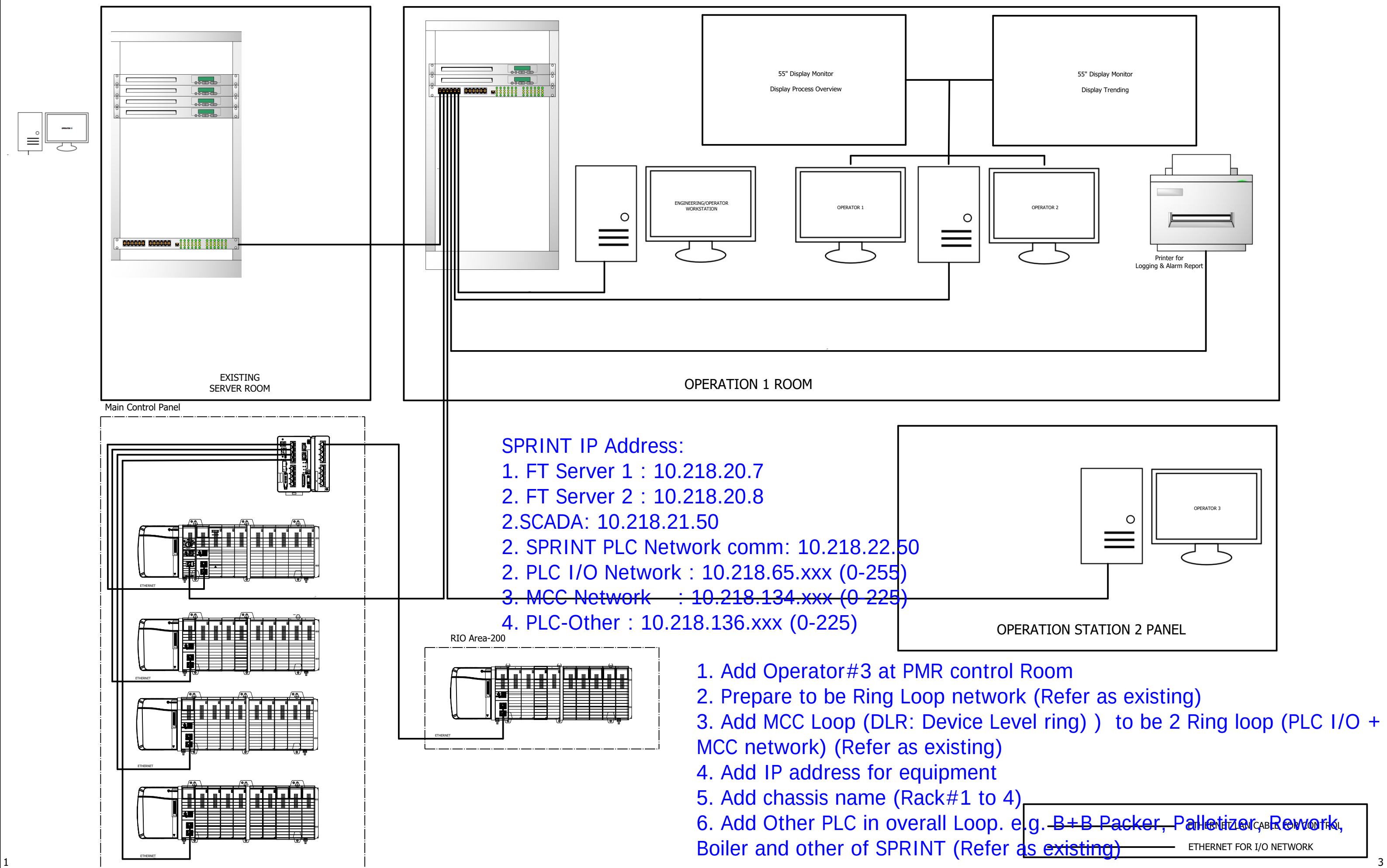
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Job number NFPA_mm_bas001
Commission EPLAN

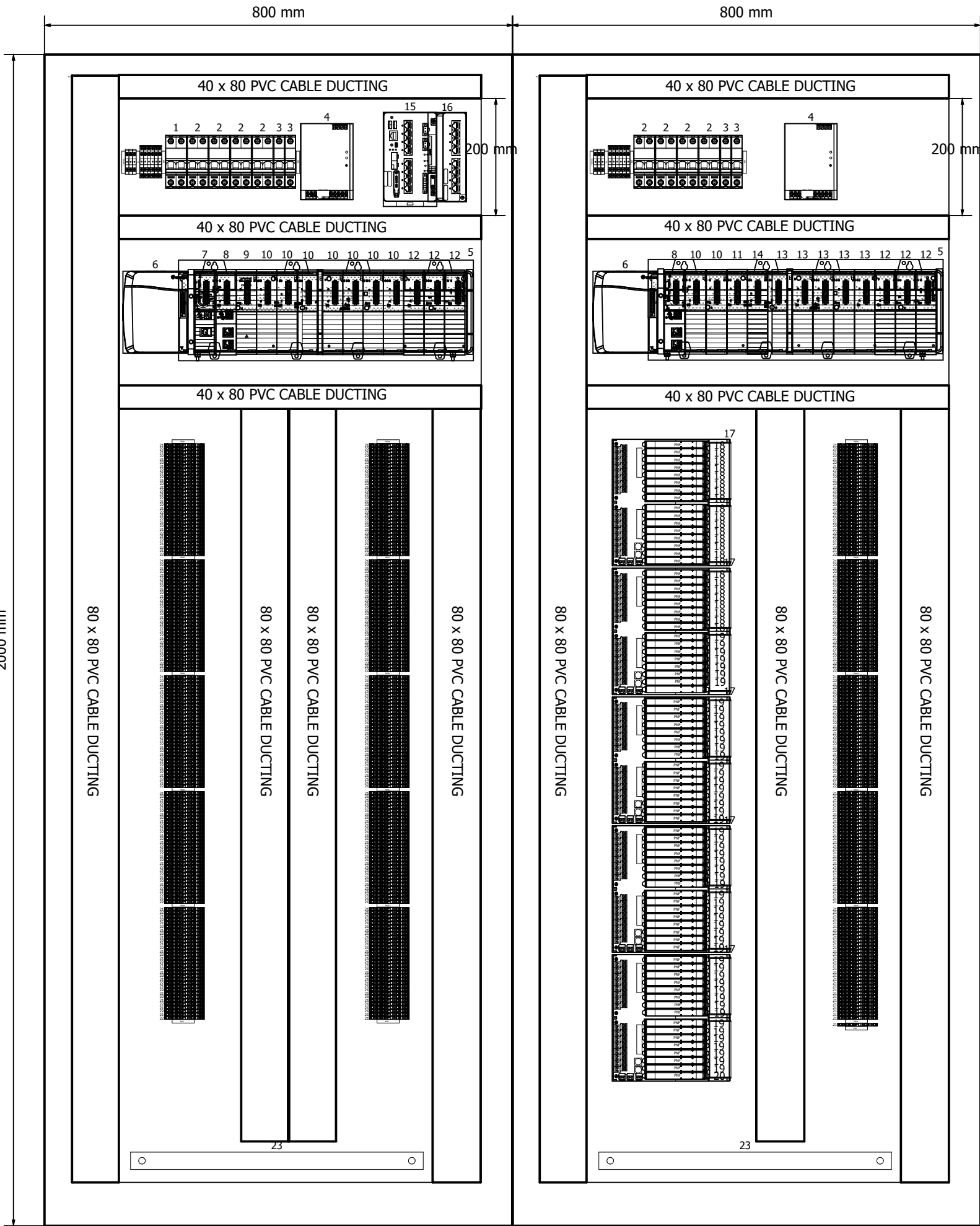
Manufacturer (company) EPLAN GmbH & Co. KG

Path EPLAN sample project
Project name SPRINT 18K SPRAY DRYER
Make
Type
Place of installation
Responsible for project
Part feature

Created on 07/14/26
Edit date 08/28/26 by (short name) THUMWH Number of pages 25

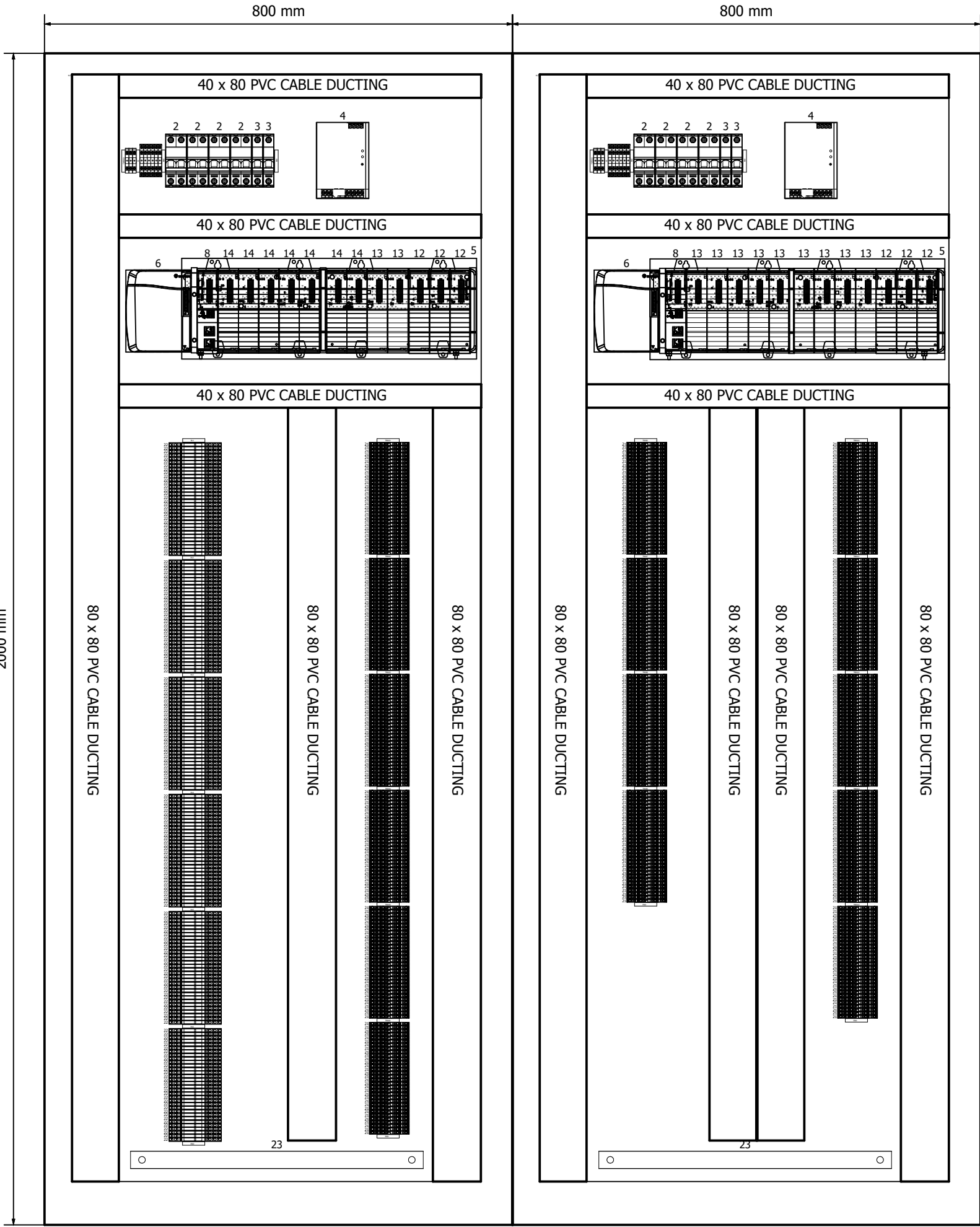
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			Ed.	THUMWH					+	
			Appr.		Basic project according to NFPA standard using a metric grid					
Modification	Date	Name	Original		Replacement of	Replaced by			NFPA_mm_bas001	Page 1 / 25





=CA1+MCP-M1 (FRONT)

=CA1+MCP-M2 (FRONT)



=CA1+MCP-M2 (REAR)

=CA1+MCP-M1 (REAR)

Enclosure legend

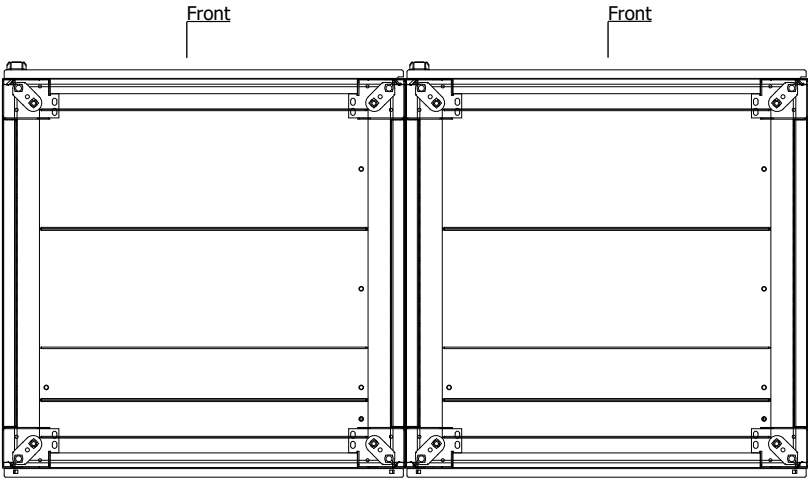
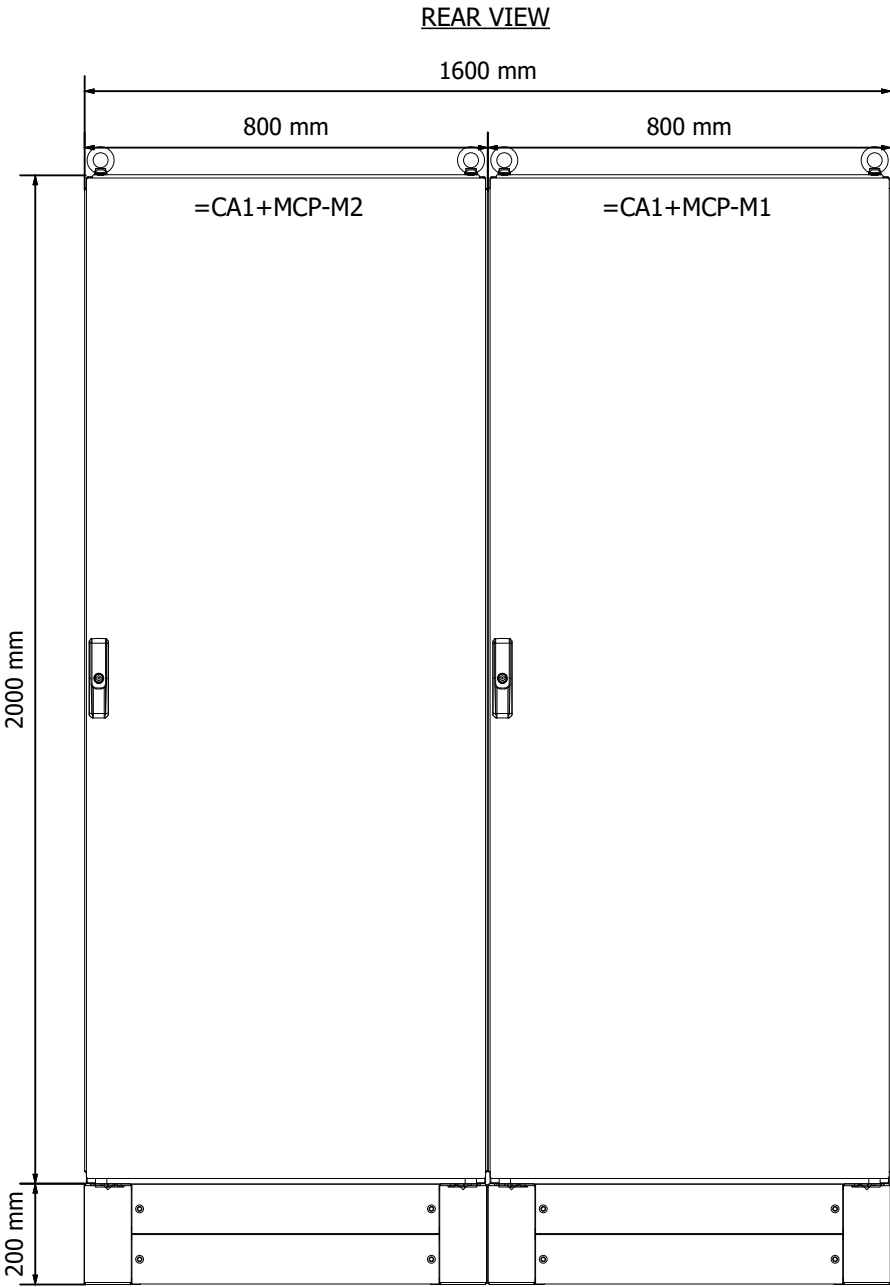
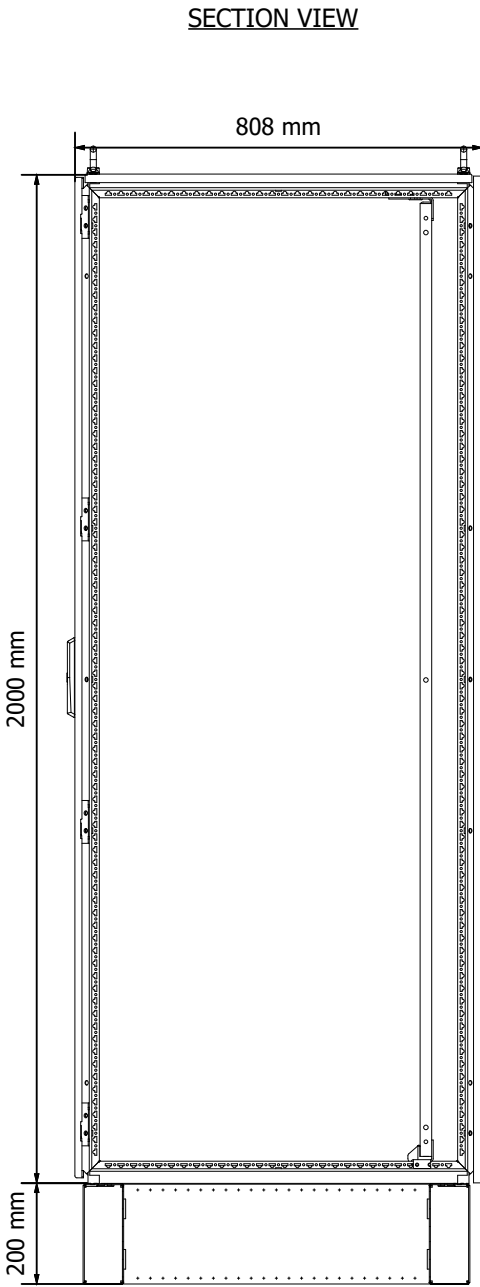
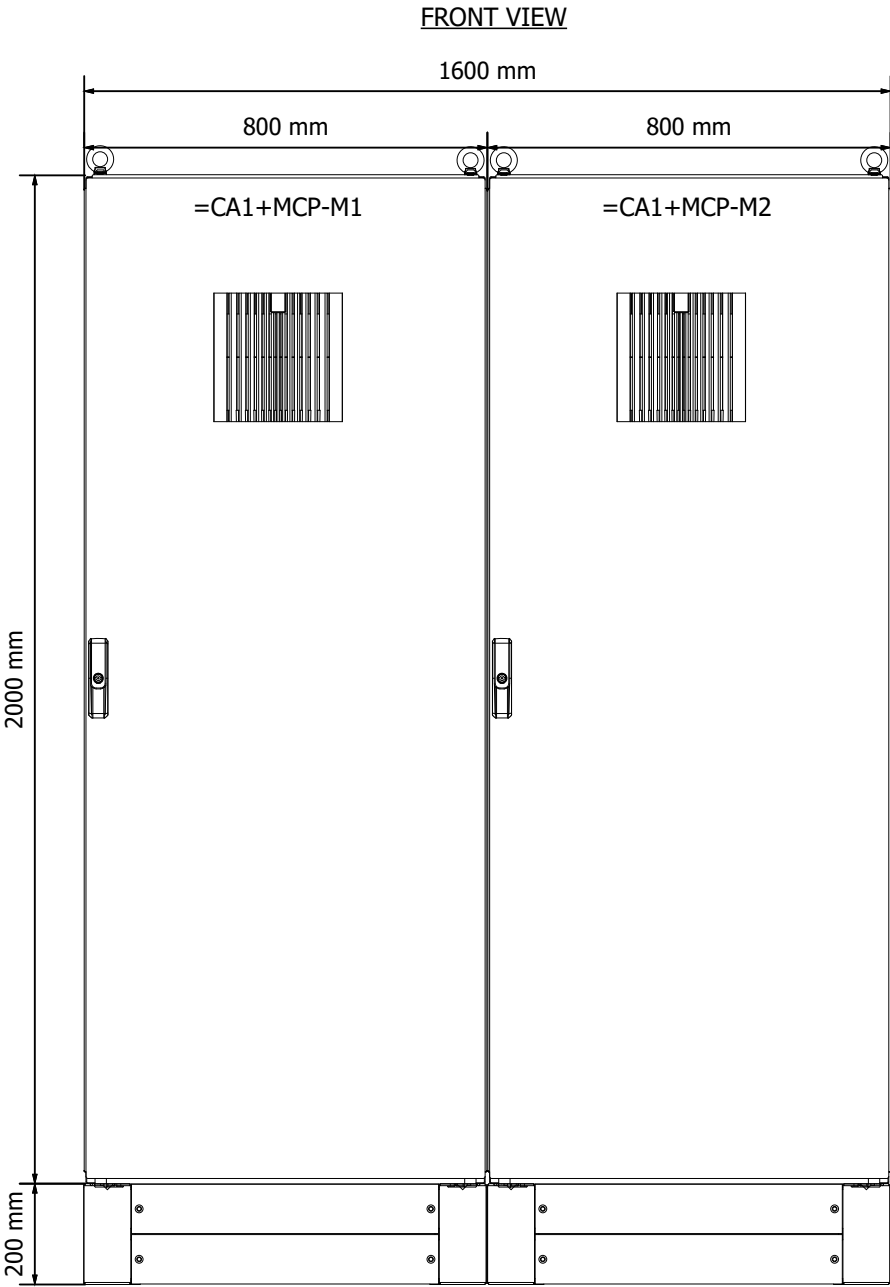
F18_005

Item number	Manufacturer	Part number	Description	Qty
2	Schneider Electric	A9F03210	Miniature circuit breaker - 2P- 10A - B curve Double terminal	4
3	Schneider Electric	A9F03106	Miniature circuit breaker - 1P - 6A - B curve Double terminal	2
4	Phoenix Contact	2866776	Power supply unit Review change to be "Allen Bradley"	1
5	Allen-Bradley	1756-A13	Chassis 13-slot	1
6	Allen-Bradley	1756-PA75	ControlLogix 120V/240VAC, 13A Power Supply	1
8	Allen-Bradley	1756-EN4TR	EtherNet/IP communication module, dual port, 10M/100M/1G twisted pair	1
12	Allen-Bradley	1756-N2	CLX Slot filler module for empty slots in standard CLX chassis	3
13	Allen-Bradley	1756-IB32	CLX DIG. INPUT MODULE	9
21	Phoenix Contact	3210567	Double-level terminal block Chang to be Fuse terminal block	288
23	Rittal	9686536	PE/PEN busbar E-Cu	1

Enclosure legend

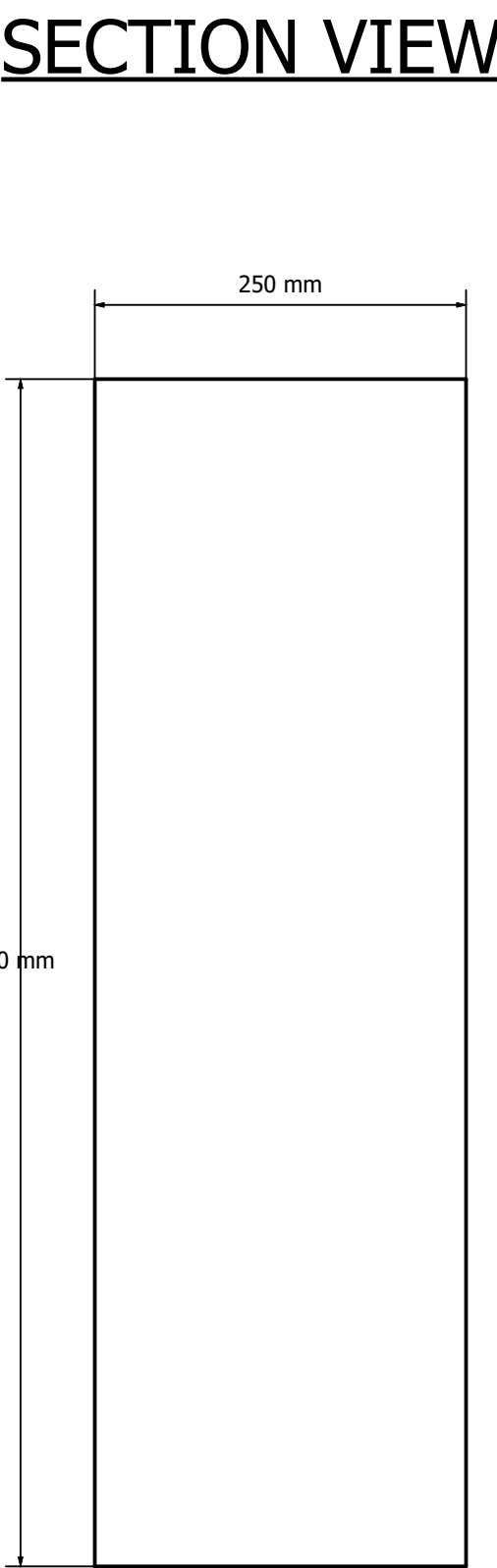
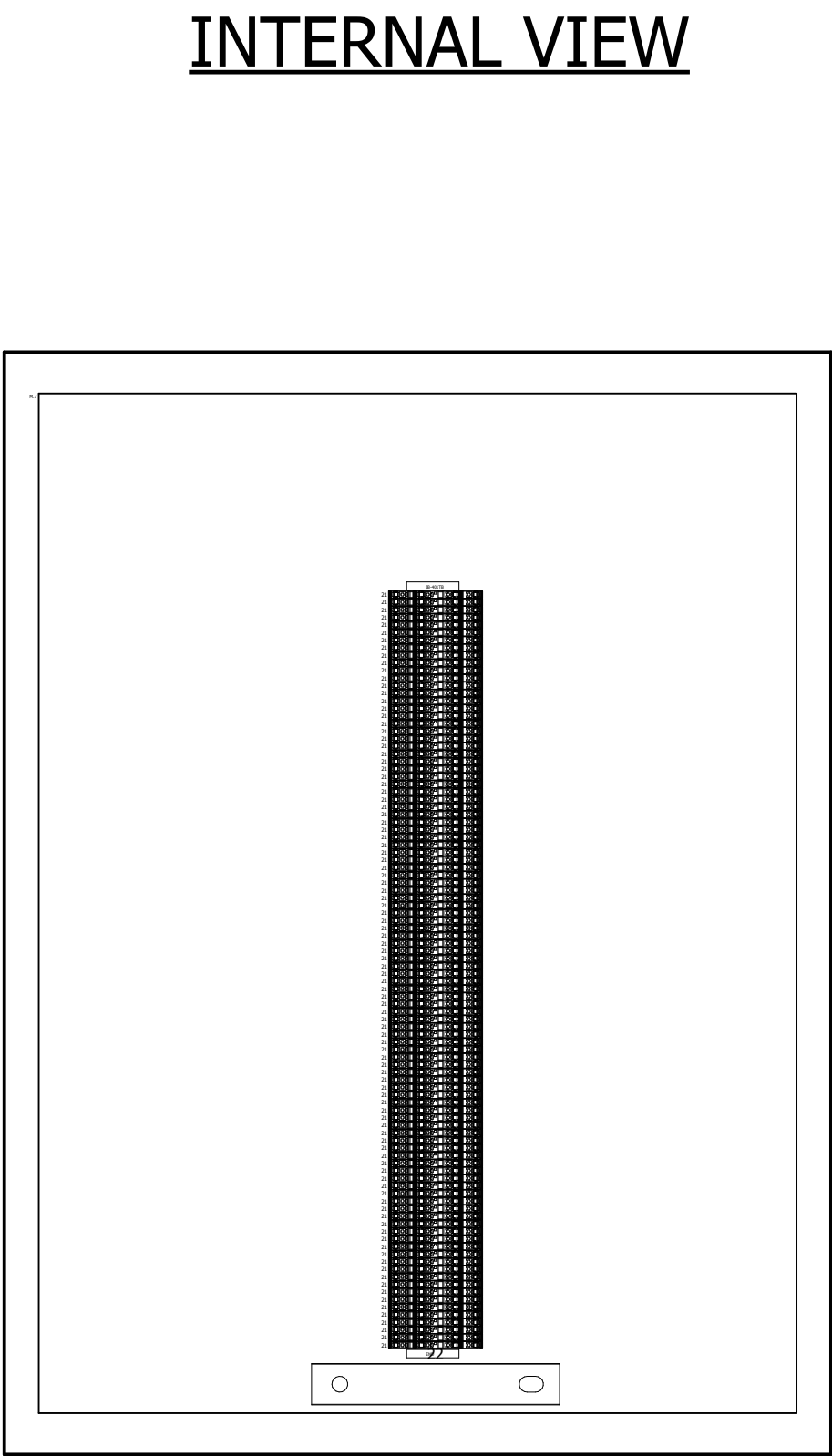
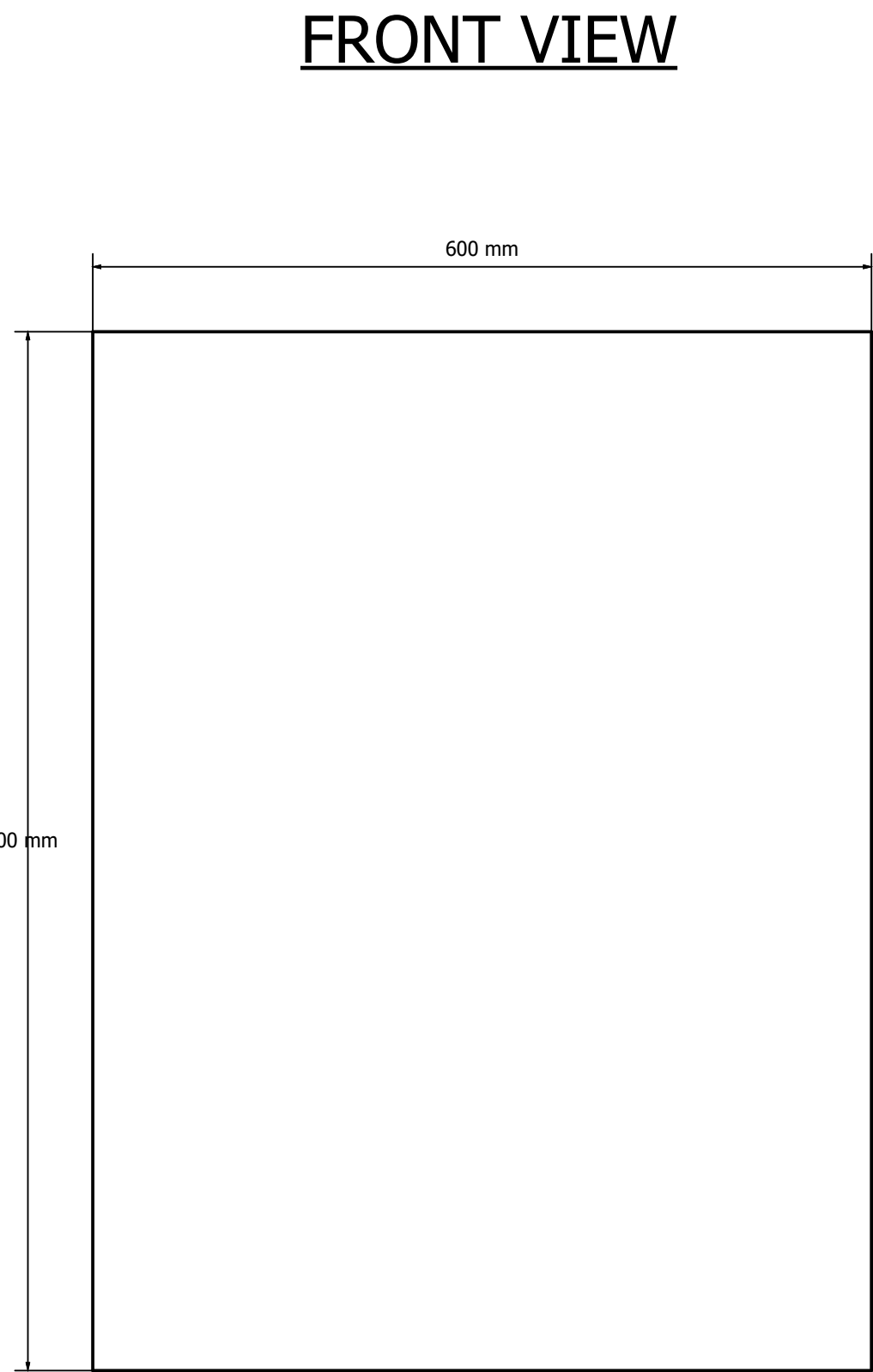
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Item number	Manufacturer	Part number	Description	Qty
2	Schneider Electric	A9F03210	Miniature circuit breaker - 2P- 10A - B curve Double terminal	4
3	Schneider Electric	A9F03106	Miniature circuit breaker - 1P - 6A - B curve Double terminal	2
4	Phoenix Contact	2866776	Power supply unit Review change to be fuse terminal block	1
5	Allen-Bradley	1756-A13	Chassis 13-slot	1
6	Allen-Bradley	1756-PA75	ControlLogix 120V/240VAC, 13A Power Supply	1
8	Allen-Bradley	1756-EN4TR	EtherNet/IP communication module, dual port, 10M/100M/1G twisted pair	1
12	Allen-Bradley	1756-N2	CLX Slot filler module for empty slots in standard CLX chassis	3
13	Allen-Bradley	1756-IB32	CLX DIG. INPUT MODULE	2
14	Allen-Bradley	1756-OB32	CLX DIG. OUTPUT MODULE	7
21	Phoenix Contact	3210567	Double-level terminal block Change to be fuse terminal block	192
22	OMRON	G2RV-SR500 DC24	Electromechanical relays, Industrial plug-in relays, G2RV	192
23	Rittal	9686536	PE/PEN busbar E-Cu	1



BOTTOM VIEW

0	1	2	3	4	5	6	7	8	9
Enclosure legend									
F18_005									
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21	Phoenix Contact	3210567	Double-level terminal block	100					
22	Rittal	9686539	PE/PEN baying bracket E-Cu	1					

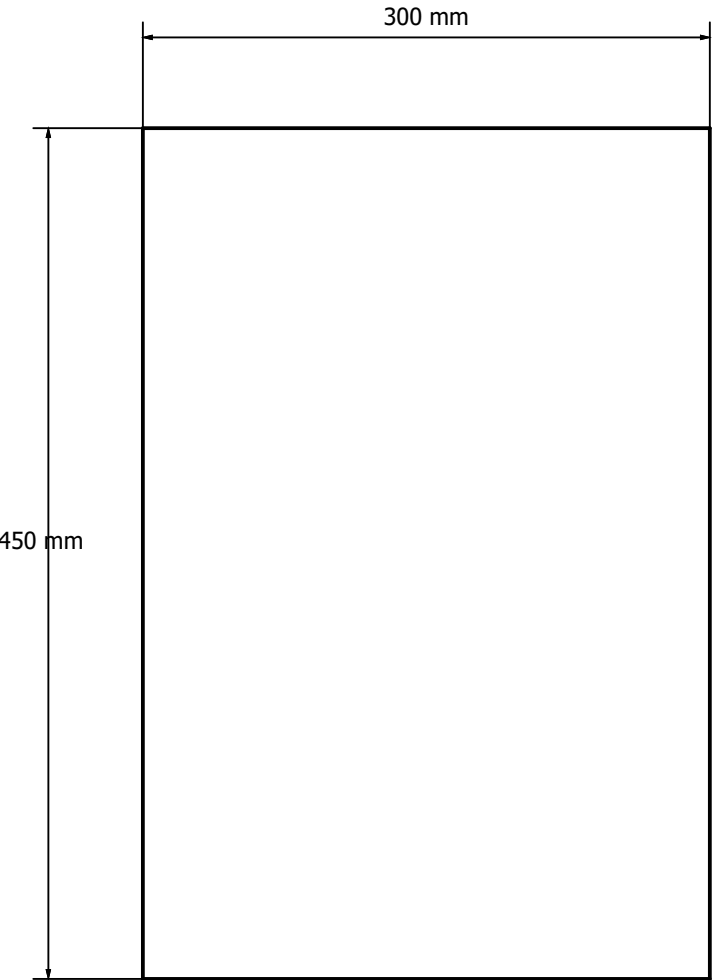


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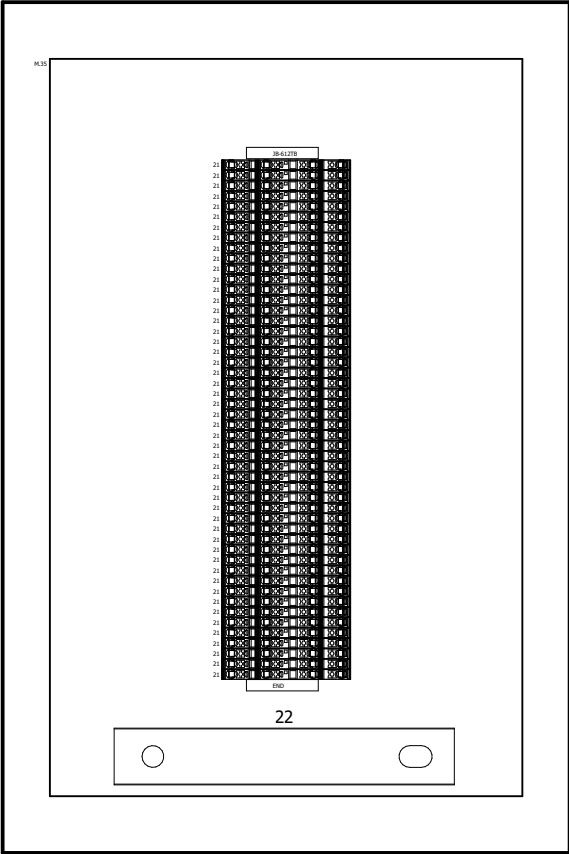
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22	Rittal	9686539	PE/PEN baying bracket E-Cu	1

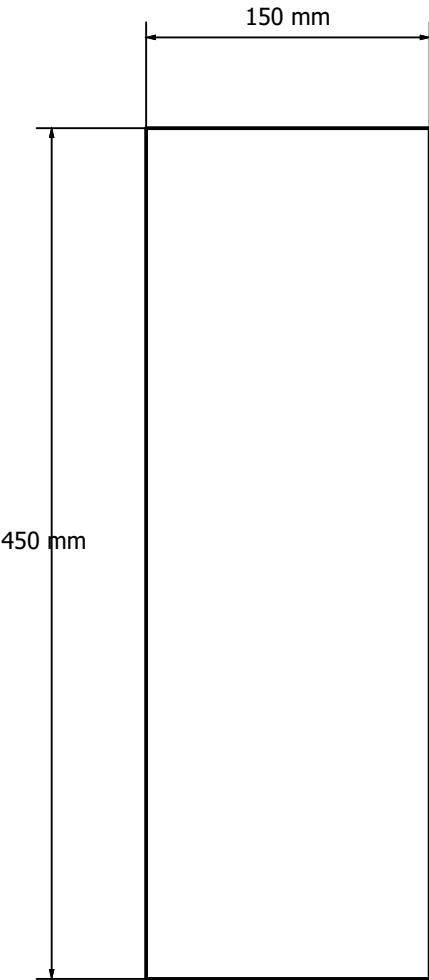
FRONT VIEW



INTERNAL VIEW



SECTION VIEW

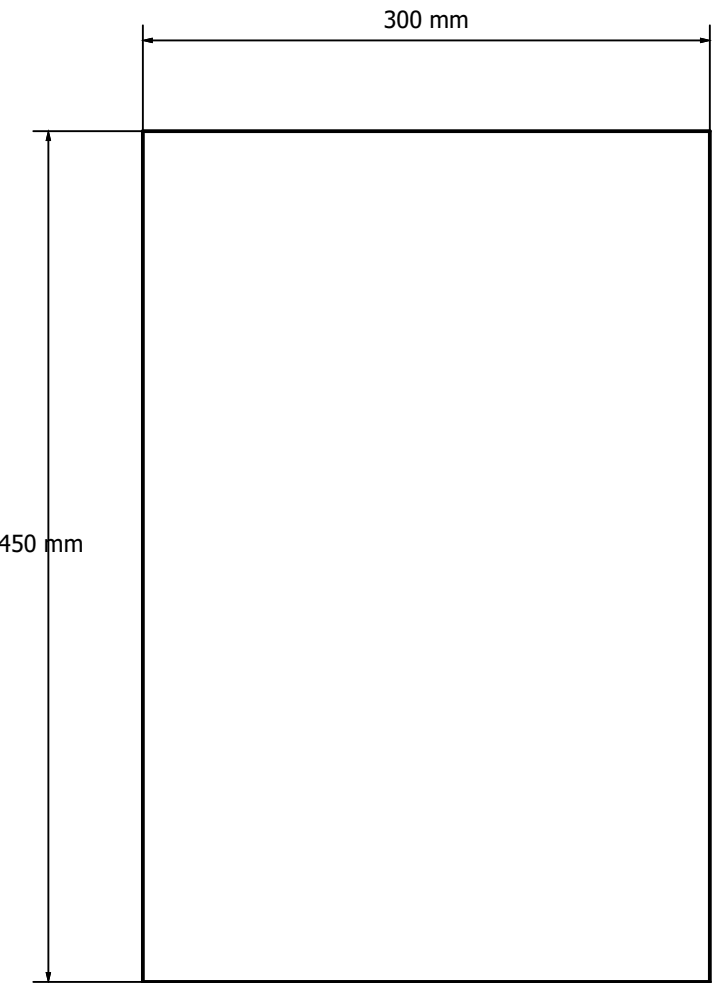


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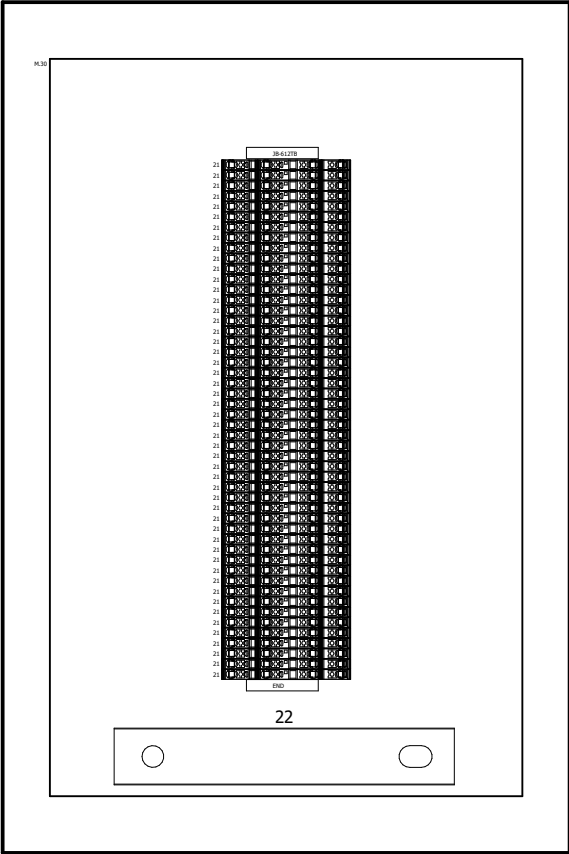
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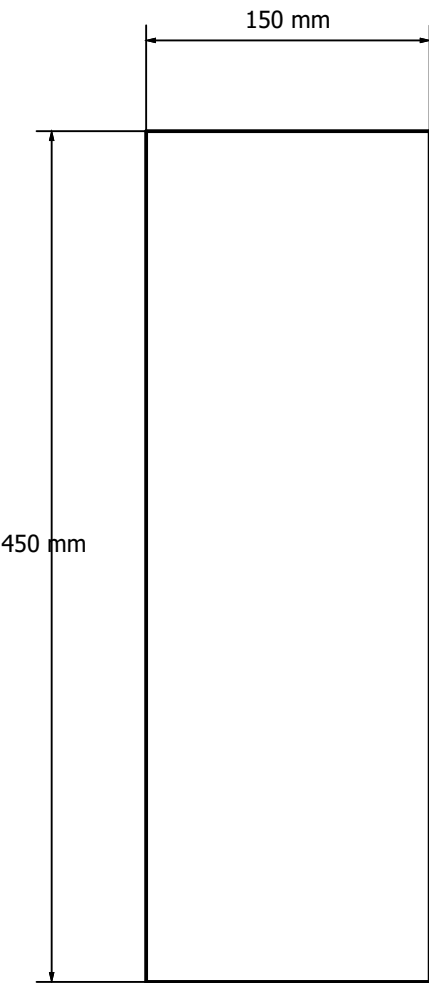
FRONT VIEW



INTERNAL VIEW



SECTION VIEW

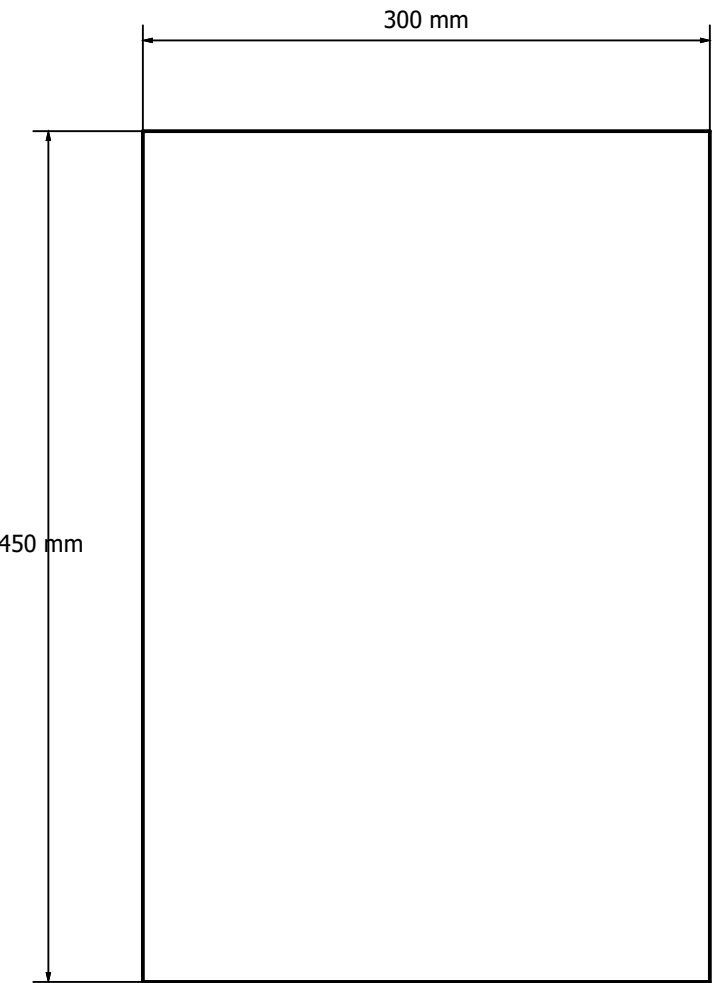


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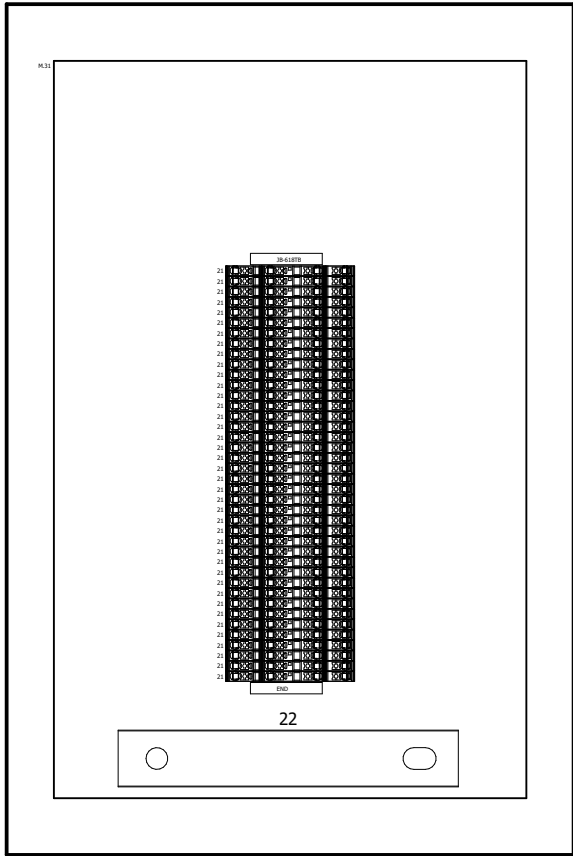
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21	Phoenix Contact	3210567	Double-level terminal block	40
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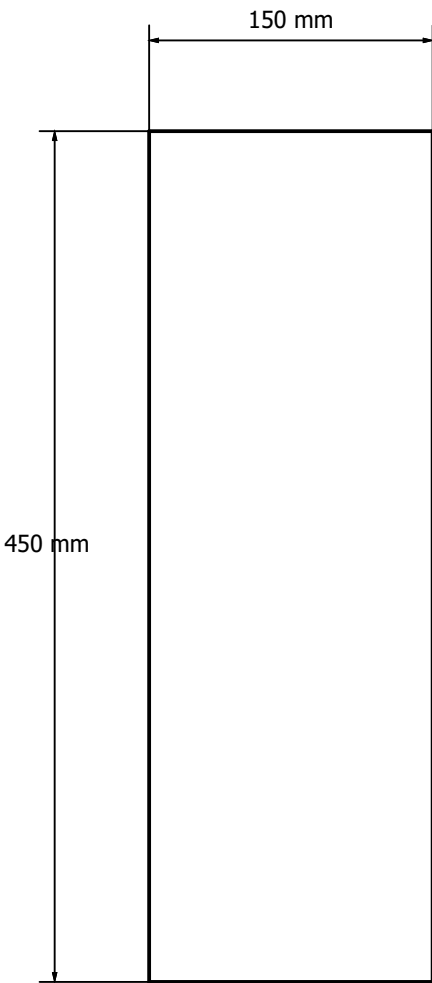
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INTERNAL VIEW



SECTION VIEW

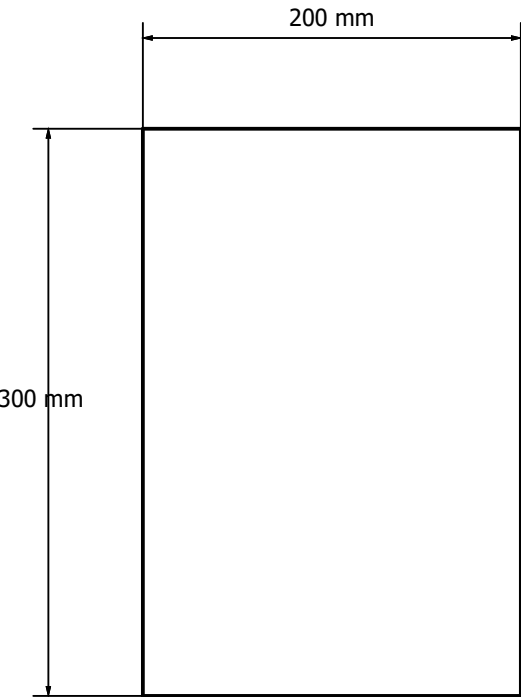


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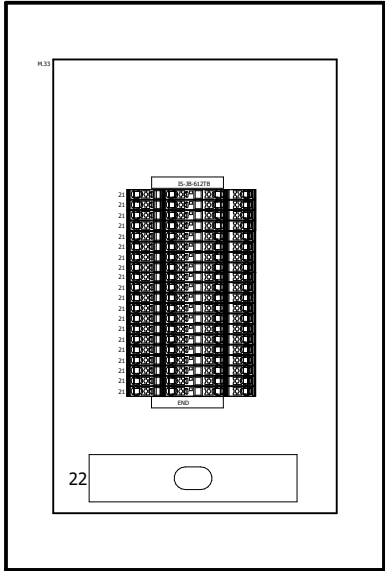
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21	Phoenix Contact	3210567	Double-level terminal block	20
22	Rittal	9686820	Filler piece	1

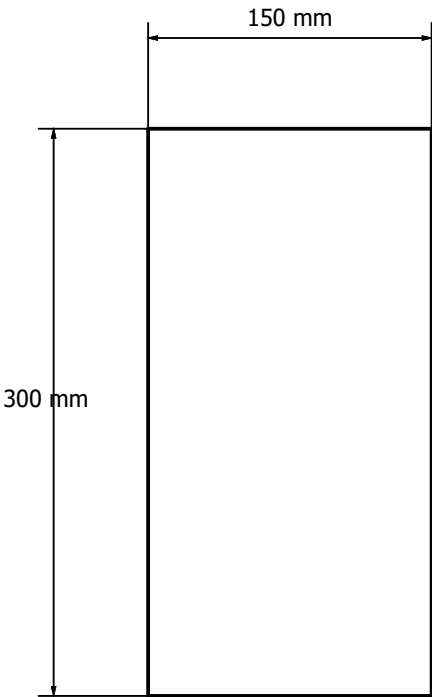
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INTERNAL VIEW



SECTION VIEW

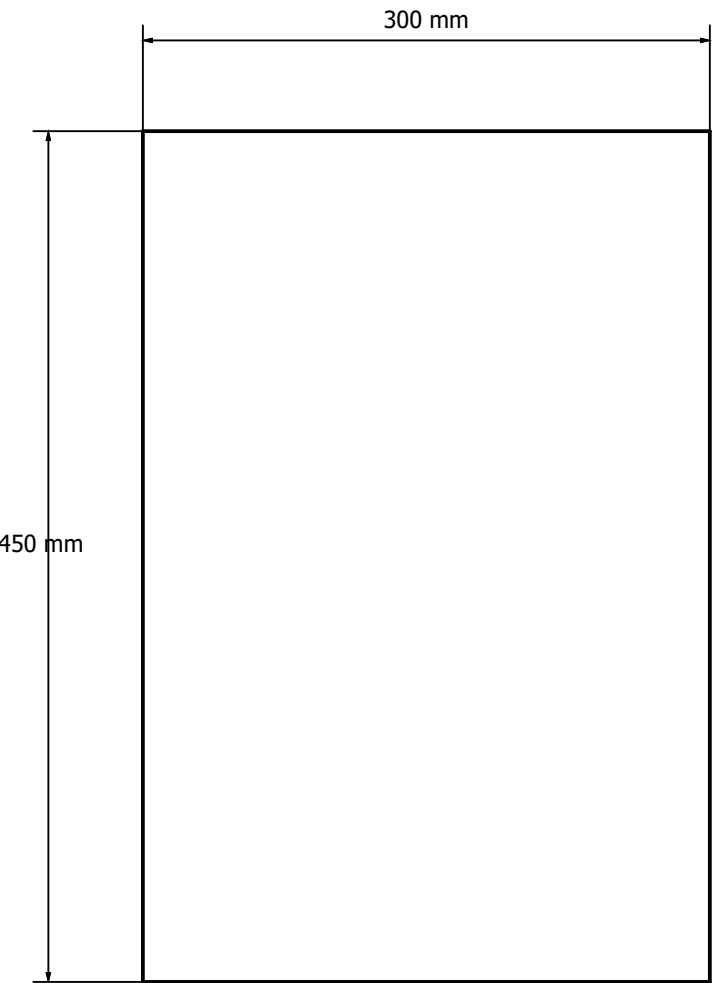


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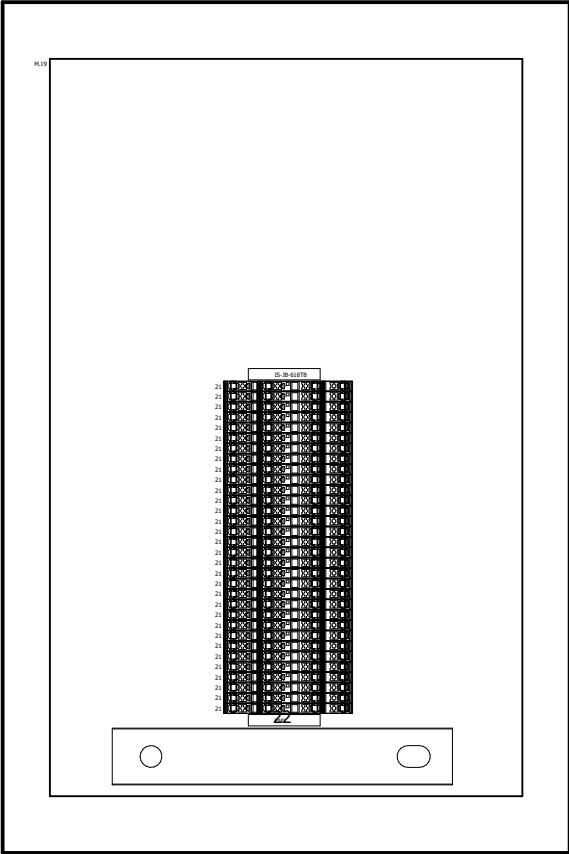
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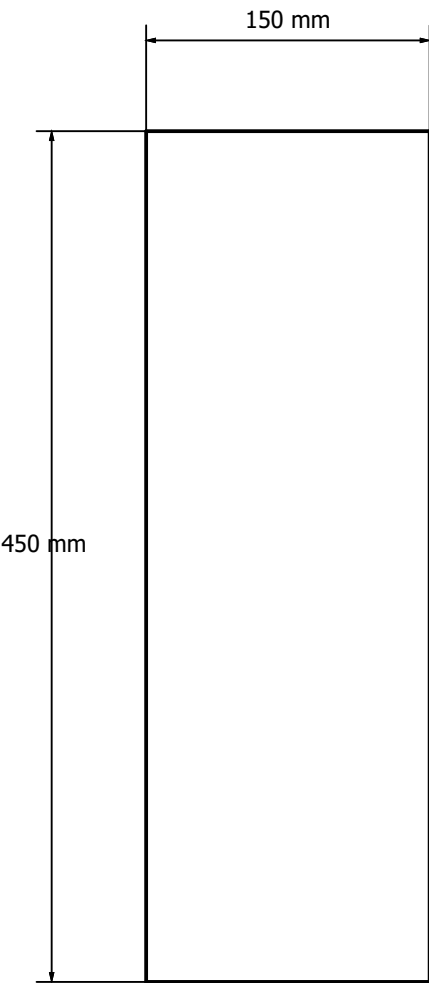
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INTERNAL VIEW



SECTION VIEW

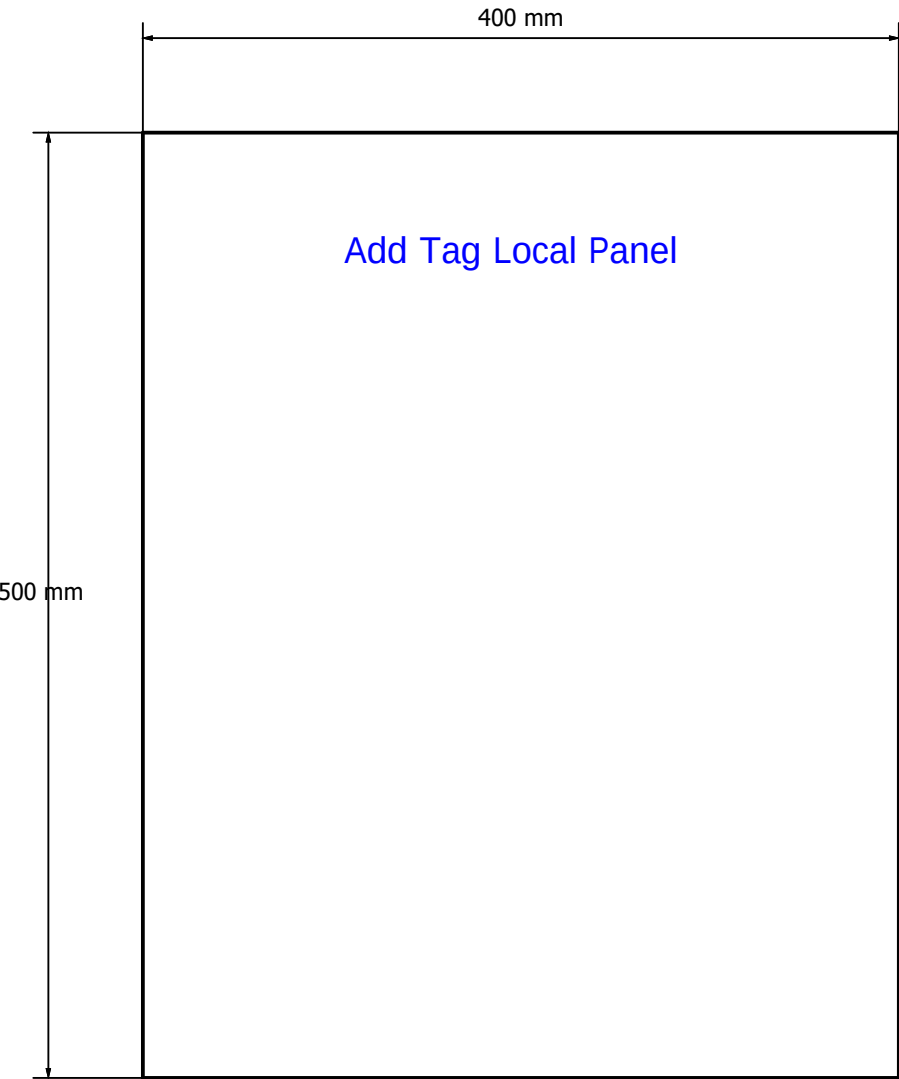


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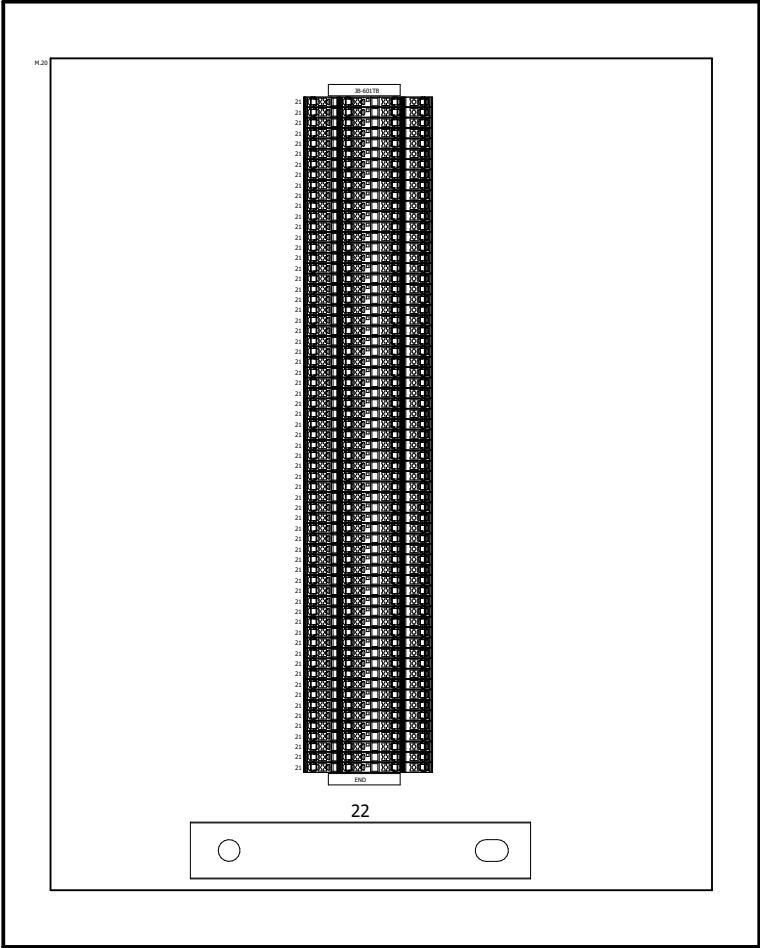
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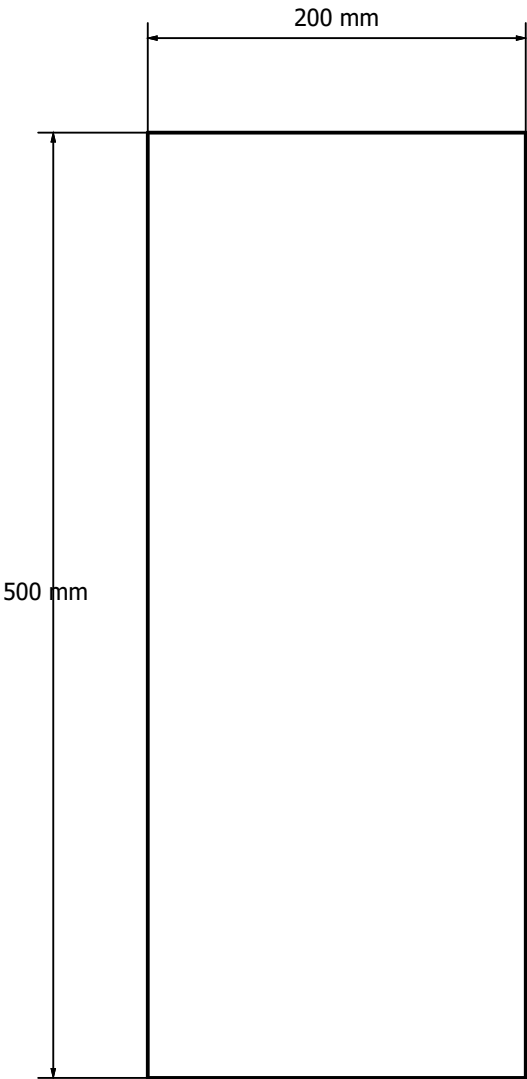
FRONT VIEW



INTERNAL VIEW



SECTION VIEW

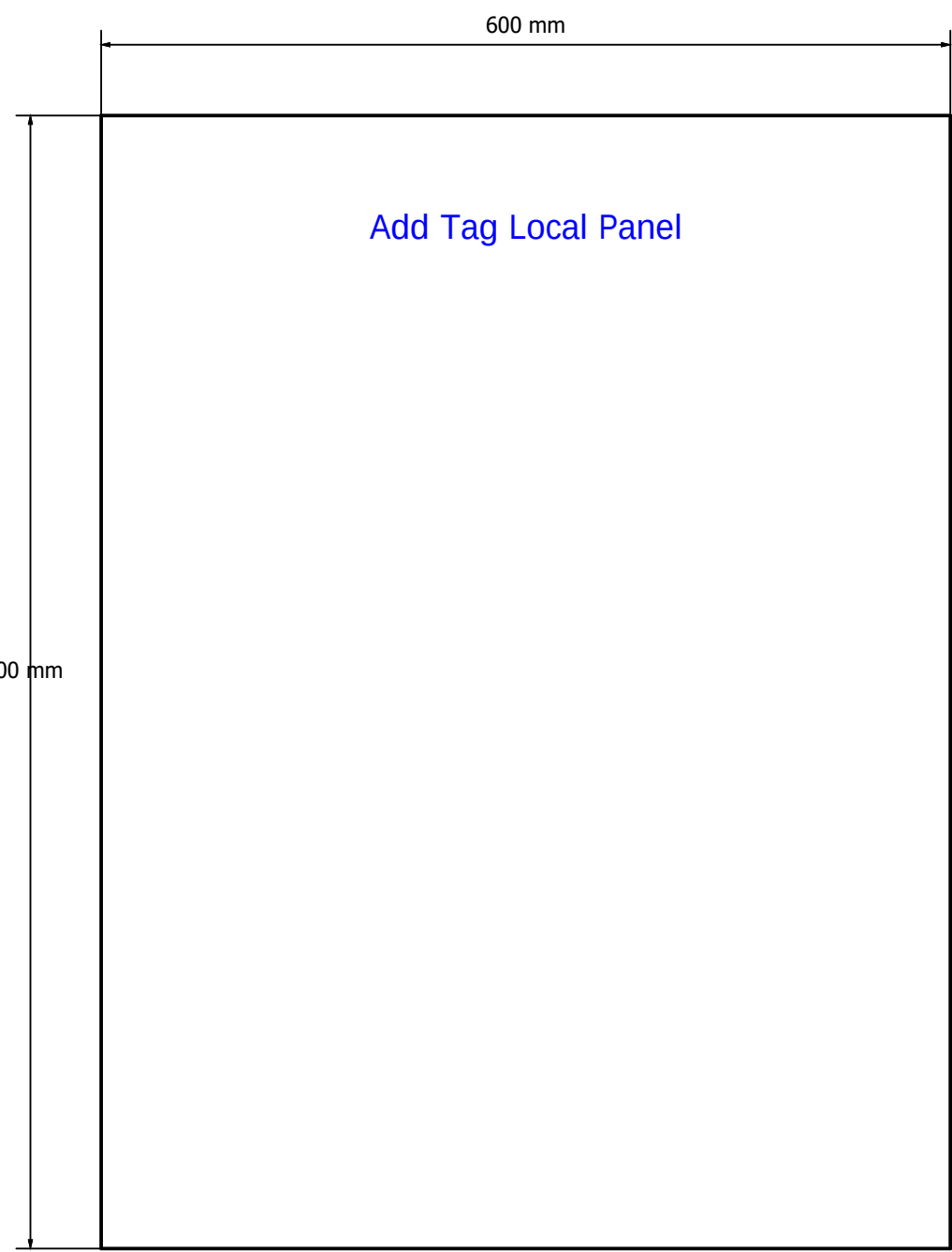


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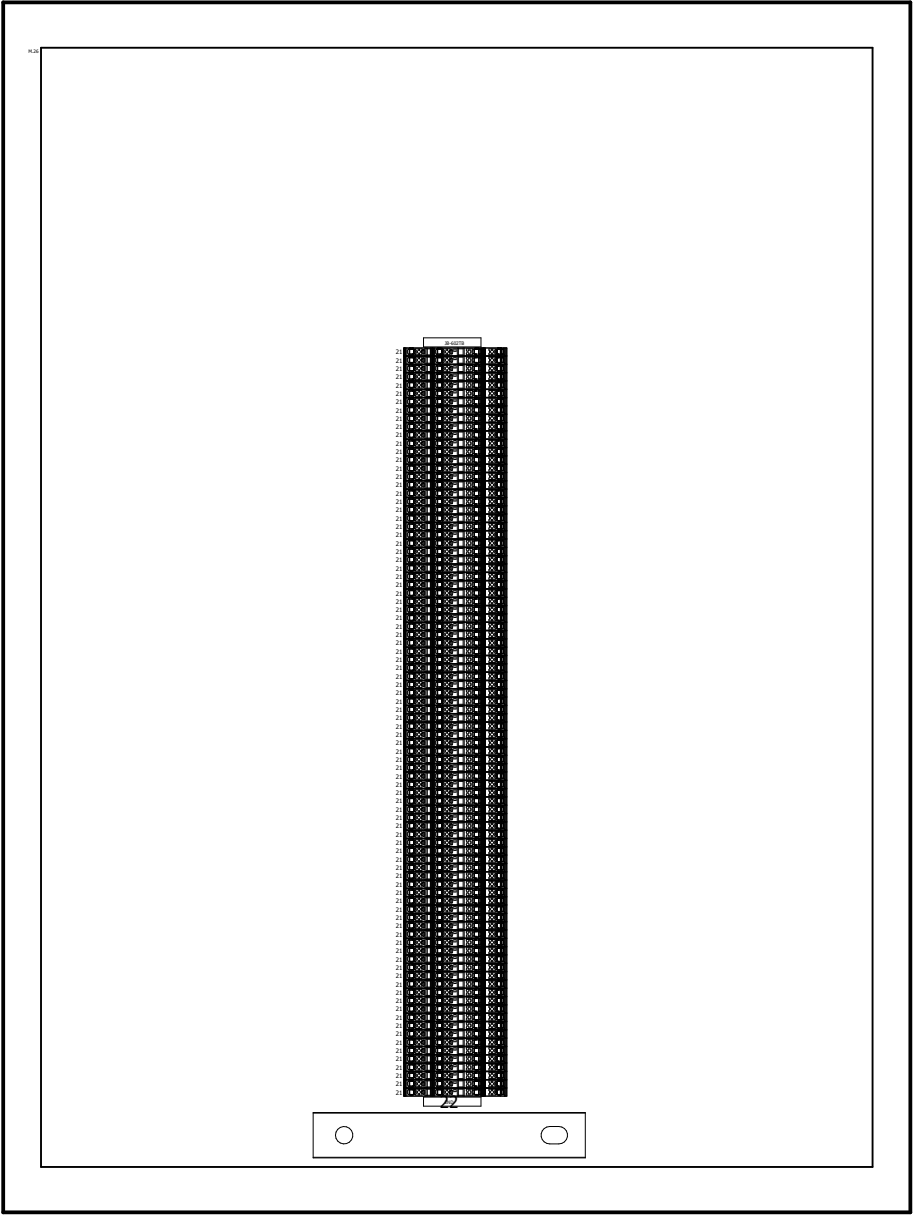
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22	Rittal	9686539	PE/PEN baying bracket E-Cu	1

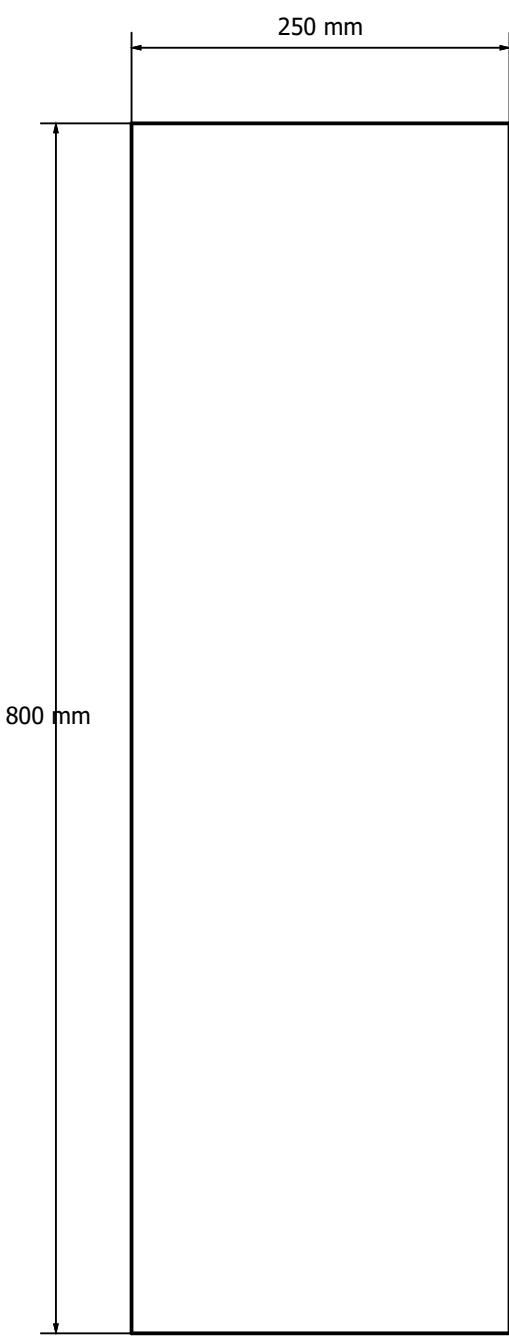
FRONT VIEW



INTERNAL VIEW



SECTION VIEW



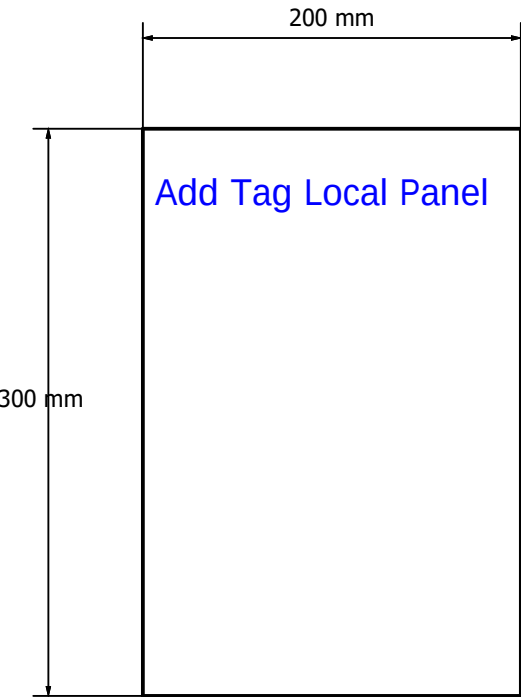
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Modification	Date	Name	Original		Replacement of	Replaced by			NFPA_mm_bas001	Page 19 / 25

Enclosure legend

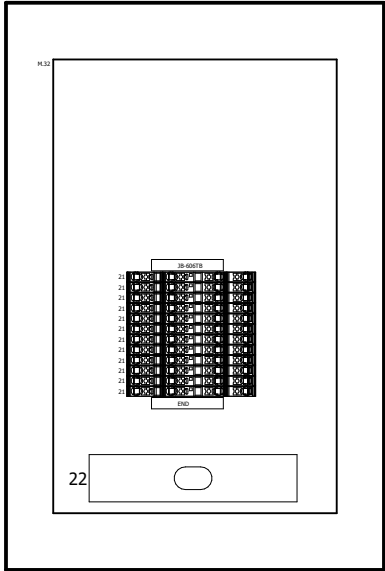
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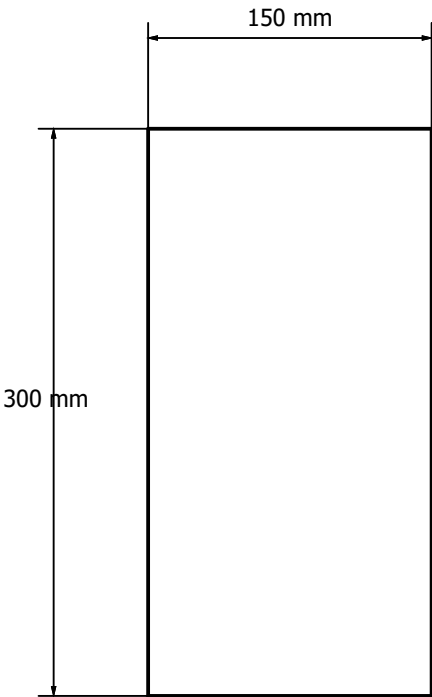
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INTERNAL VIEW



SECTION VIEW

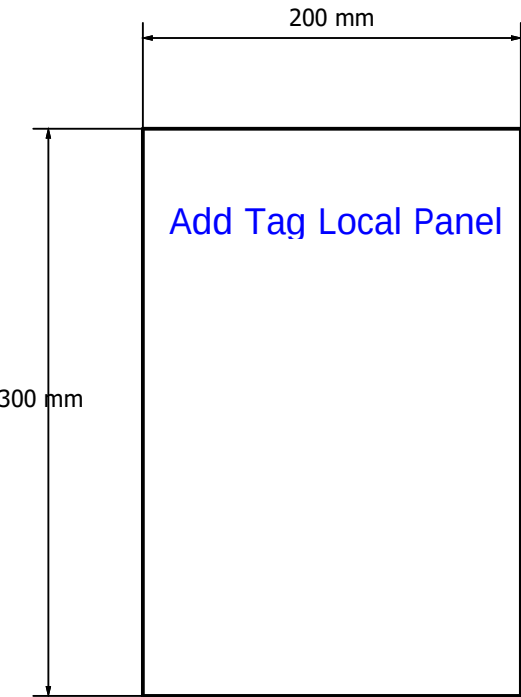


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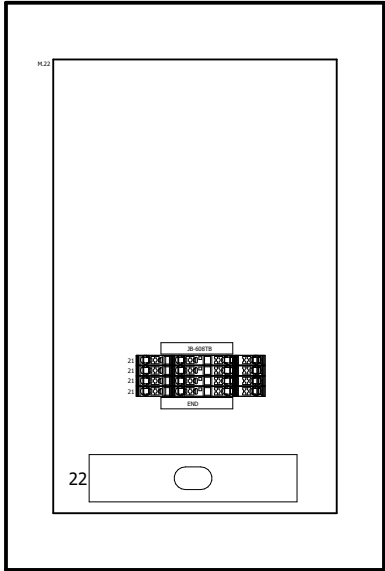
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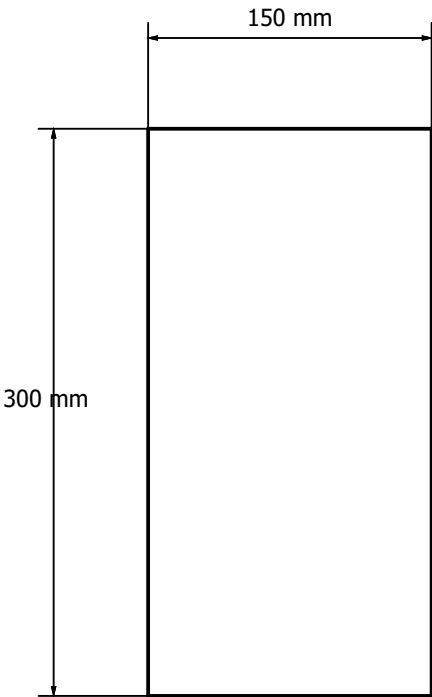
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INTERNAL VIEW



SECTION VIEW

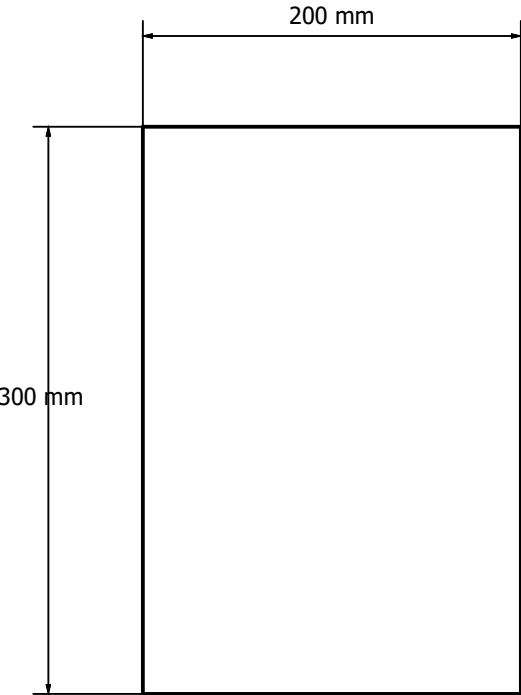


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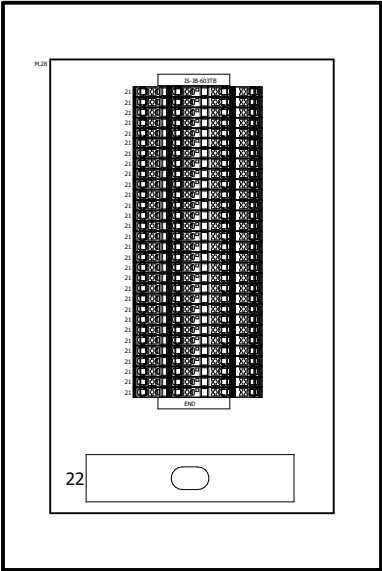
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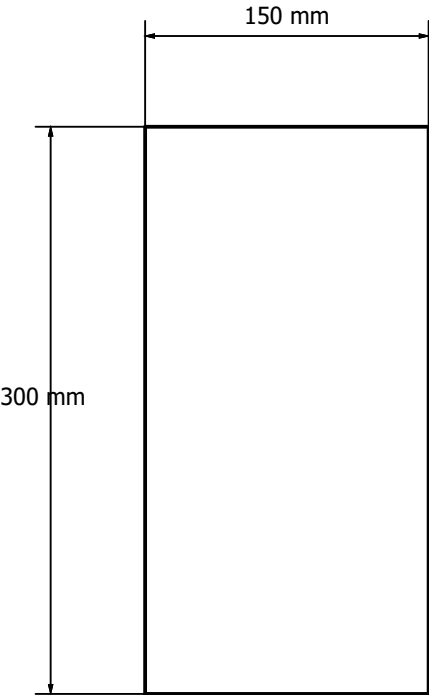
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INTERNAL VIEW



SECTION VIEW



Enclosure legend

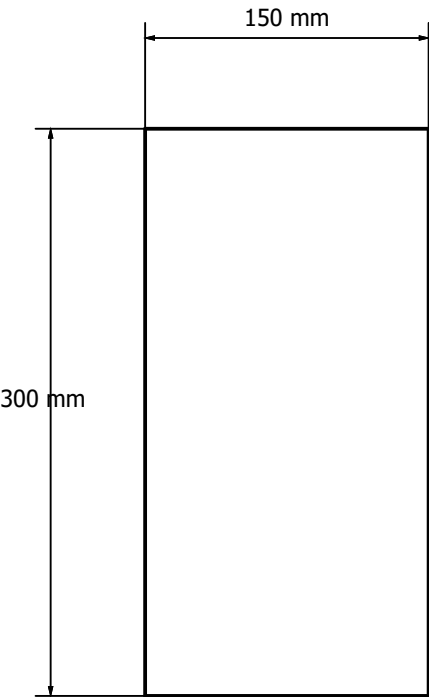
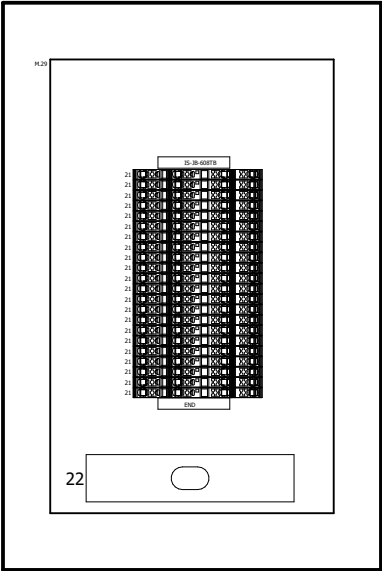
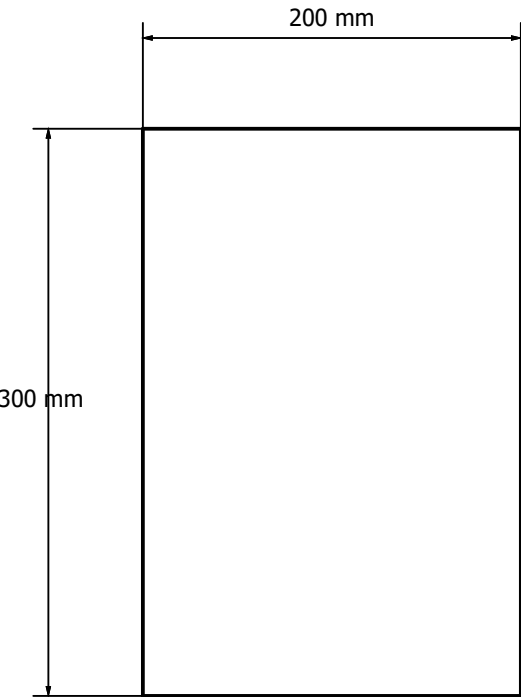
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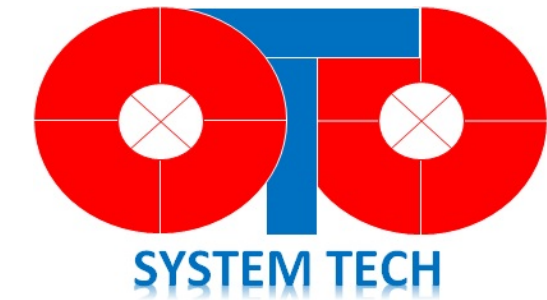
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21	Phoenix Contact	3210567	Double-level terminal block	22
22	Rittal	9686820	Filler piece	1

FRONT VIEW

INTERNAL VIEW

SECTION VIEW





EPLAN GmbH & Co. KG

An der alten Ziegelei 2
40789 Monheim am Rhein
Phone +49 (0)2173 - 39 64 - 0

Company / customer

Project description Basic project according to NFPA standard using a metric grid
Job number NFPA_mm_bas001
Commission EPLAN

Manufacturer (company) EPLAN GmbH & Co. KG

Path EPLAN sample project
Project name SPRINT 18K SPRAY DRYER ELECTRICAL
Make
Type
Place of installation
Responsible for project
Part feature

Created on 07/16/26
Edit date 08/27/26 by (short name) THUMWH
Number of pages 39

			Date	03/30/21	EPLAN	EPLAN GmbH & Co. KG	Title page		=	
			Ed.	EPL					+	
			Appr.		Basic project according to NFPA standard using a metric grid					
Modification	Date	Name	Original		Replacement of	Replaced by				
								NFPA_mm_bas001		Page 1 / 39

Pos.	Code name	Designation	Quantity	Remark
	A-B.1756-A13	Chassis 13-slot	4	
	PNF.HiC2025	HiC2025 - Transmitter Power Supplies	29	
	PNF.HiC2821	Switch Amplifier HiC2821	56	
	PNF.HiC2871	Solenoid Driver	2	
	RIT.9686536	PE/PEN busbar E-Cu	4	PE/PEN Rail System 30x10 mm, Suitable for width, 600 mm, E-Cu
	RIT.9686539	PE/PEN baying bracket E-Cu	9	Baying bracket E-Cu for PE/PEN combinations, To fit busbars 30x10 mm, E-Cu
	RIT.9686820	Filler piece	5	Filler pieces, Length: 110 mm, Cross-section E-Cu busbars: 10 x 25 mm
	A-B.1756-PA75	ControlLogix 120V/240VAC, 13A Power Supply	5	(512 TCP connections, 256 axes), CIP
	A-B.1756-EN4TR	EtherNet/IP communication module, dual port, 10M/100M/1G twisted pair	4	motion module, 3 Ethernet Port with USB 1.1 port, (supports 128 TCP/IP connections,
	A-B.1756-A10	ControlLogix Chassis 10-Slot	1	For Power Supply Variant 10 ControlLogix Cards up to 255 axis)
	A-B.1756-EN2TR	ControlLogix EtherNet dual port 10-100M Interface Module	1	(supports 128 TCP/IP connections, up to 8 axis);
	PMV12.00	Cabinet Filter Fan	1	
	OMR.G2RV-SR500_DC24	Electromechanical relays, Industrial plug-in relays, G2RV	214	
	A-B.1783-MMS10	Industrial Ethernet Switch	1	
	A-B.1783-MMX8T	Industrial Ethernet Switch	1	
	SE.A9F03232	Miniature circuit breaker - 2P- 32A - B curve Double terminal	1	
	SE.A9F03210	Miniature circuit breaker - 2P- 10A - B curve Double terminal	17	
	SE.A9F03106	Miniature circuit breaker - 1P - 6A - B curve Double terminal	9	
	PXC.2866776	Power supply unit	5	
	PXC.3210567	Double-level terminal block	1604	Change to be "Allen Bradley"
	PXC.3210596	Double-level terminal block - PE	172	
	A-B.1756-IF16	CLX ANALOG INPUT MODULE	10	
	A-B.1756-OF8	CLX ANALOG OUTPUT MODULE	1	
	A-B.1756-IB32	CLX DIG. INPUT MODULE	18	
	PROSOFT.MV156E-MNET	ControlLogix® Platform Modbus TCP/IP Interface Module	1	
	A-B.1756-OB32	CLX DIG. OUTPUT MODULE	9	

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Table of contents

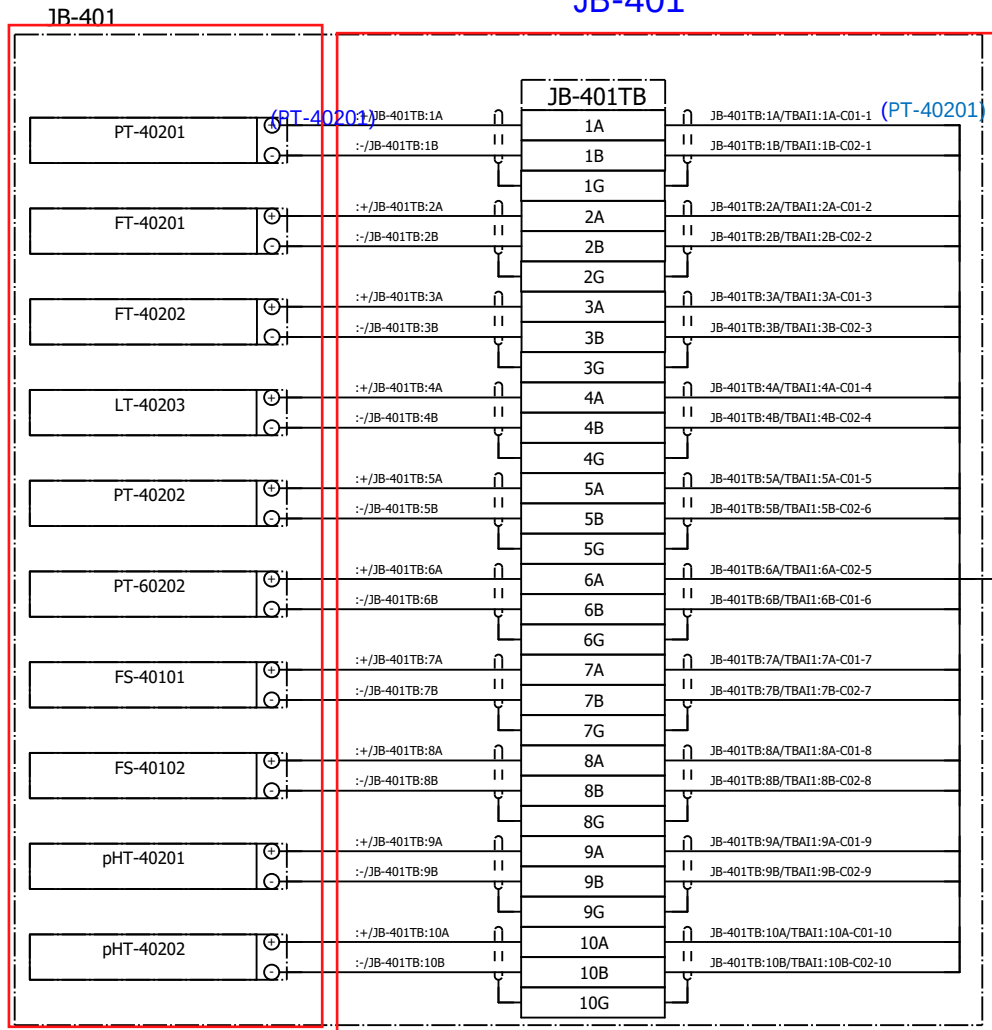
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2	Table of contents : 1 - 32		08/10/26	THUMWH	
2.1	Table of contents : 33 - 38		08/10/26	THUMWH	
3	Power distribution		08/07/26	THUMWH	
4	Chassis 1 Slot 3 (Analog Input)		08/09/26	THUMWH	
5	Chassis 1 Slot 4 (Analog Input)		08/09/26	THUMWH	
6	Chassis 1 Slot 5 (Analog Input)		08/07/26	THUMWH	
7	Chassis 1 Slot 6 (Analog input)		08/07/26	THUMWH	
8	Chassis 1 Slot 7 (Analog input)		08/07/26	THUMWH	
9	Chassis 1 Slot 8 (Analog input)		08/09/26	THUMWH	
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18	Chassis 2 Slot 8 (Digital input)		08/09/26	THUMWH	
19	Chassis 2 Slot 9 (Digital input)		08/09/26	THUMWH	
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22	Chassis 3 Slot 3 (Digital Input)		08/07/26	THUMWH	
23	Chassis 3 Slot 4 (Digital Input)		08/07/26	THUMWH	
24	Chassis 3 Slot 5 (Digital Input)		08/07/26	THUMWH	
25	Chassis 3 Slot 6 (Digital Input)		08/07/26	THUMWH	
26	Chassis 3 Slot 7 (Digital Input)		08/07/26	THUMWH	
27	Chassis 3 Slot 8 (Digital Input)		08/07/26	THUMWH	
28	Chassis 3 Slot 9 (Digital Input)		08/06/26	THUMWH	
29	Chassis 4 Slot 1 (Digital Output)		08/07/26	THUMWH	
30	Chassis 4 Slot 2 (Digital Output)		08/07/26	THUMWH	
31	Chassis 4 Slot 3 (Digital Output)		08/07/26	THUMWH	
32	Chassis 4 Slot 4 (Digital Output)		08/07/26	THUMWH	

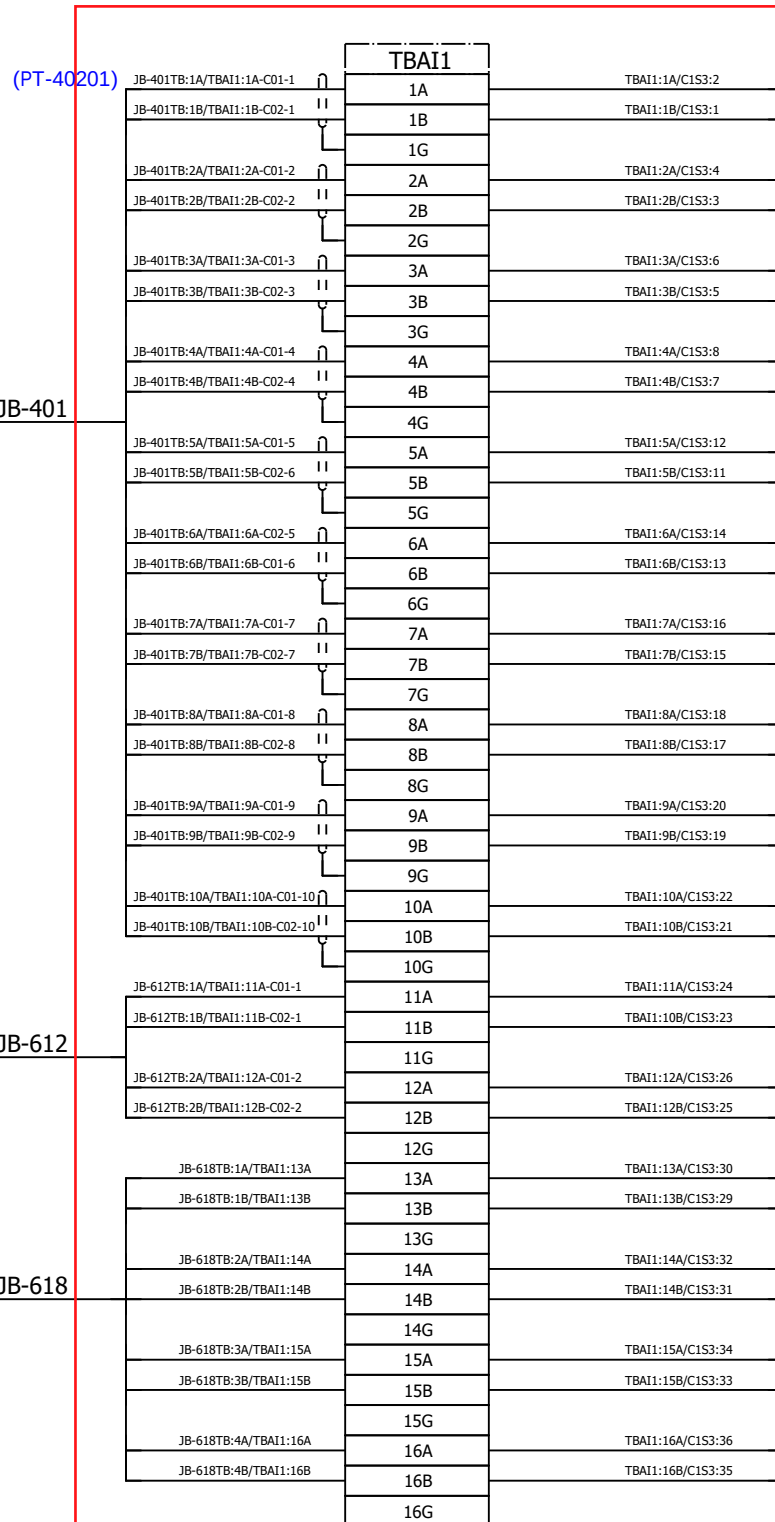
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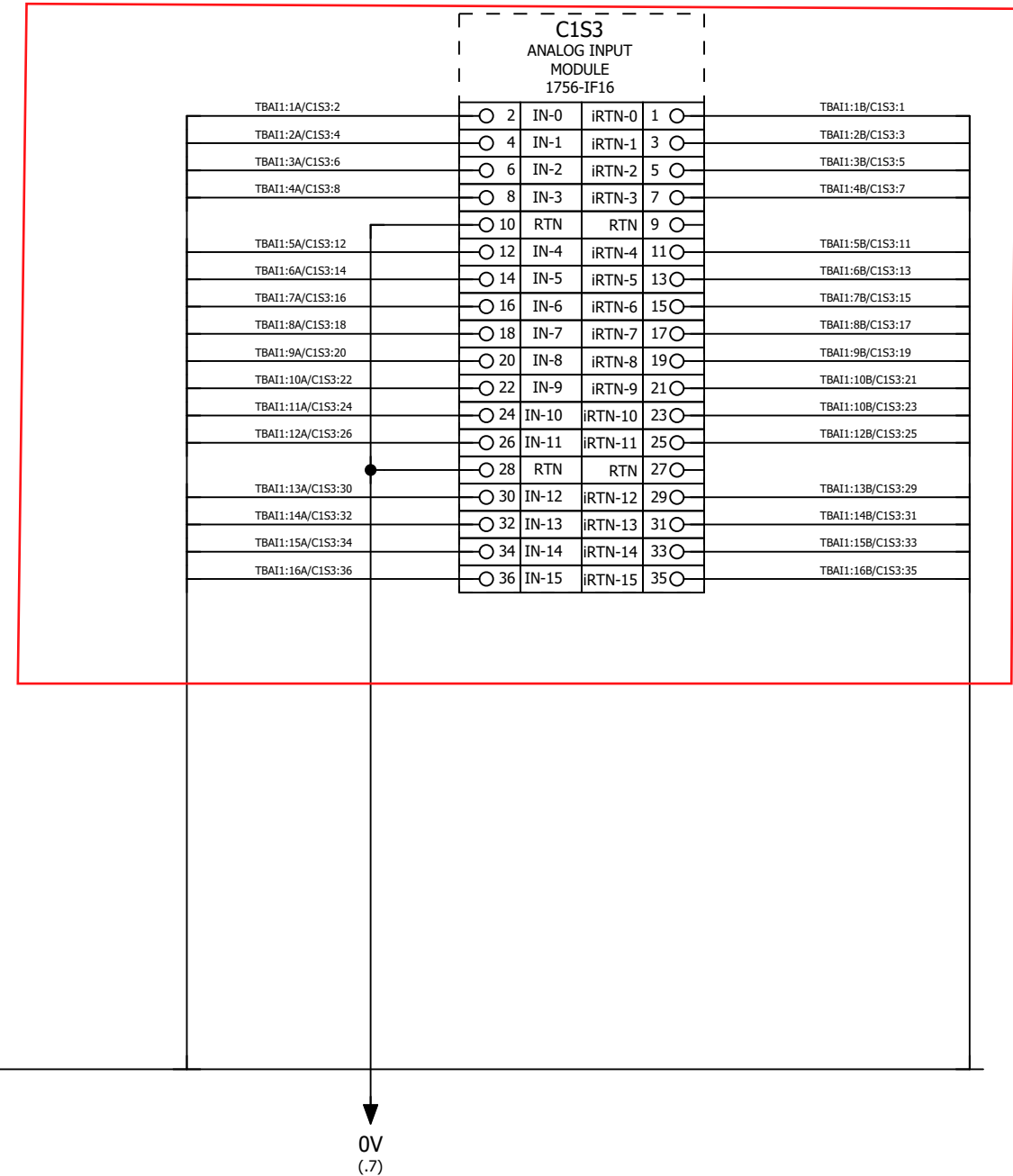
JB-401



PLC Panel

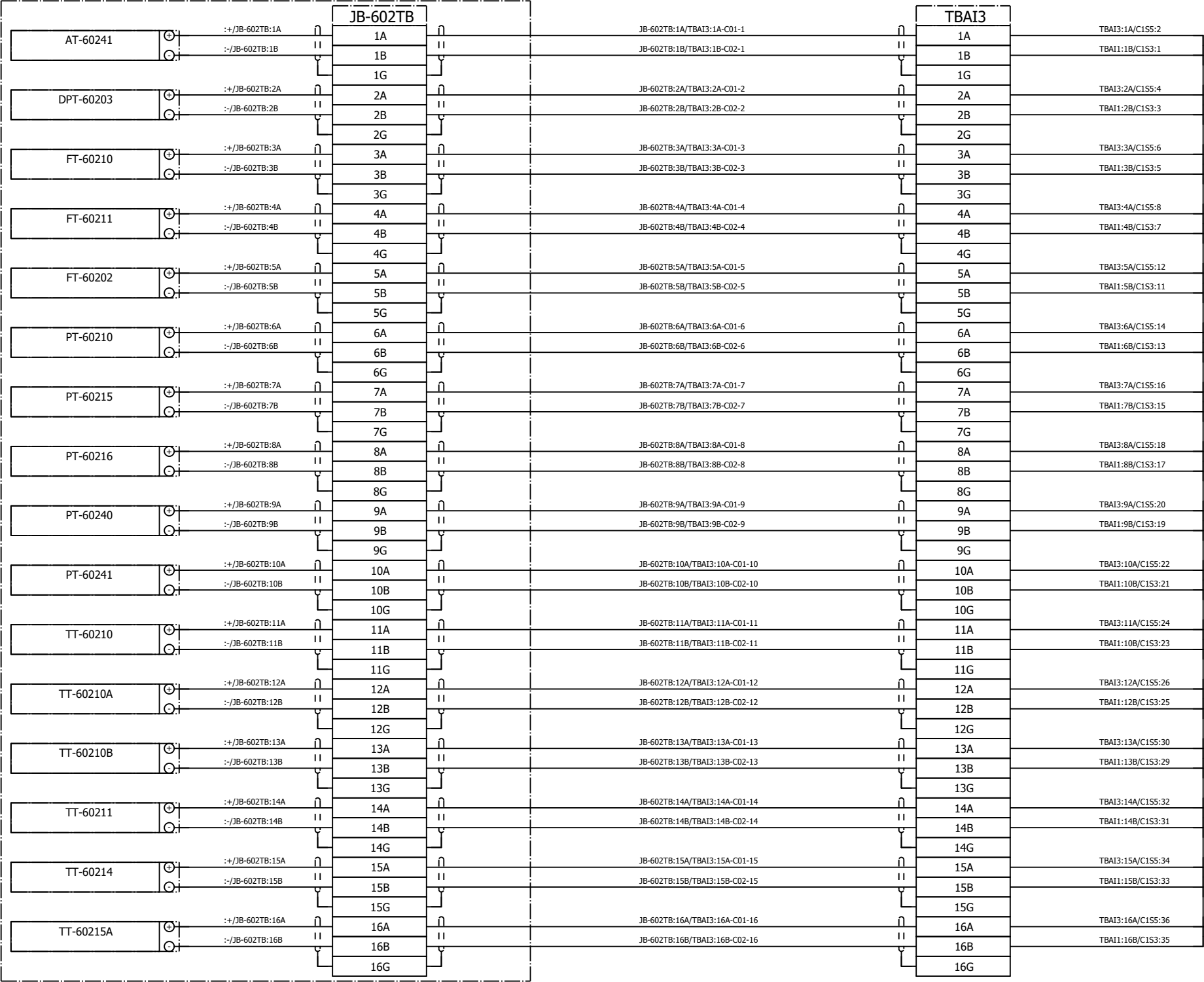


CHASSIS 1 SLOT 3

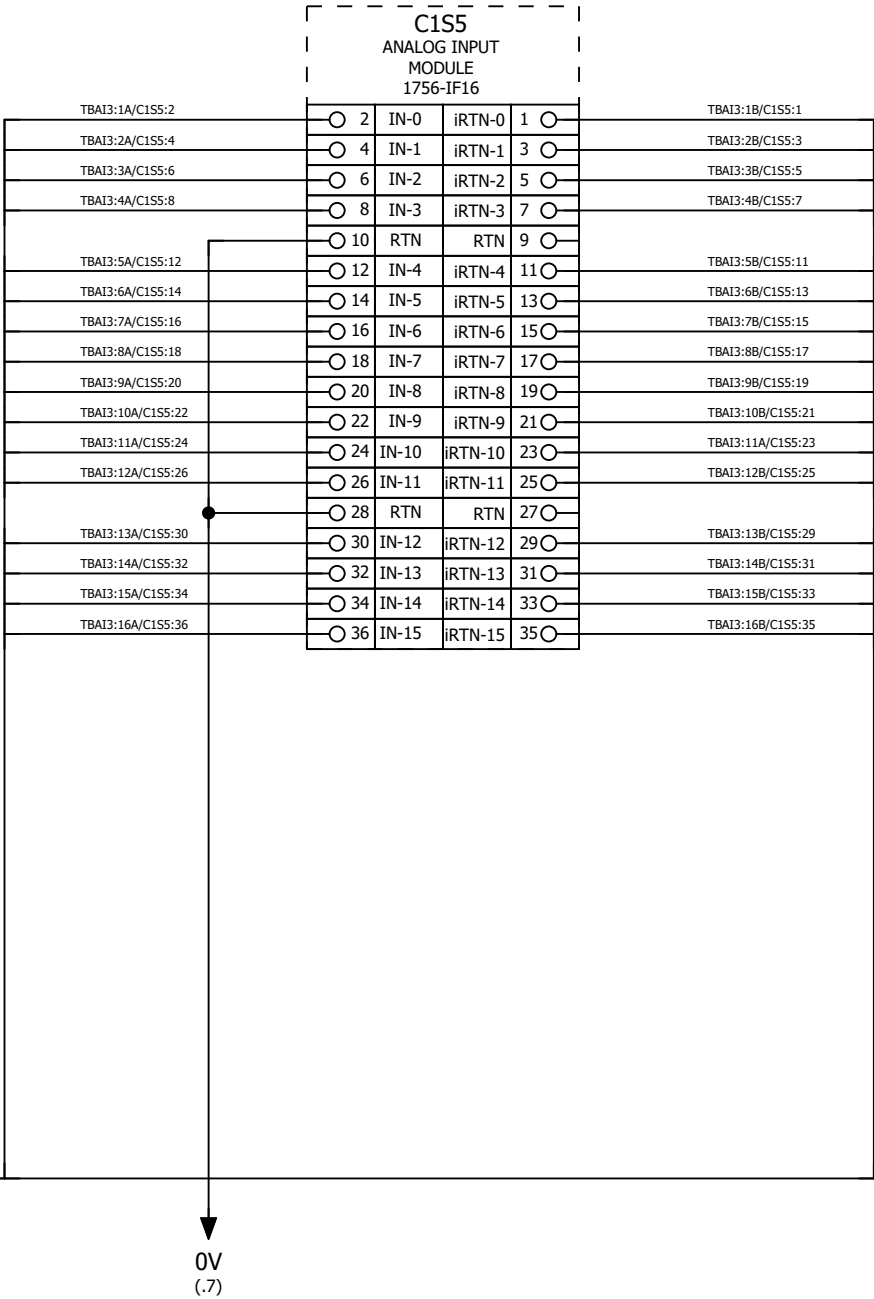


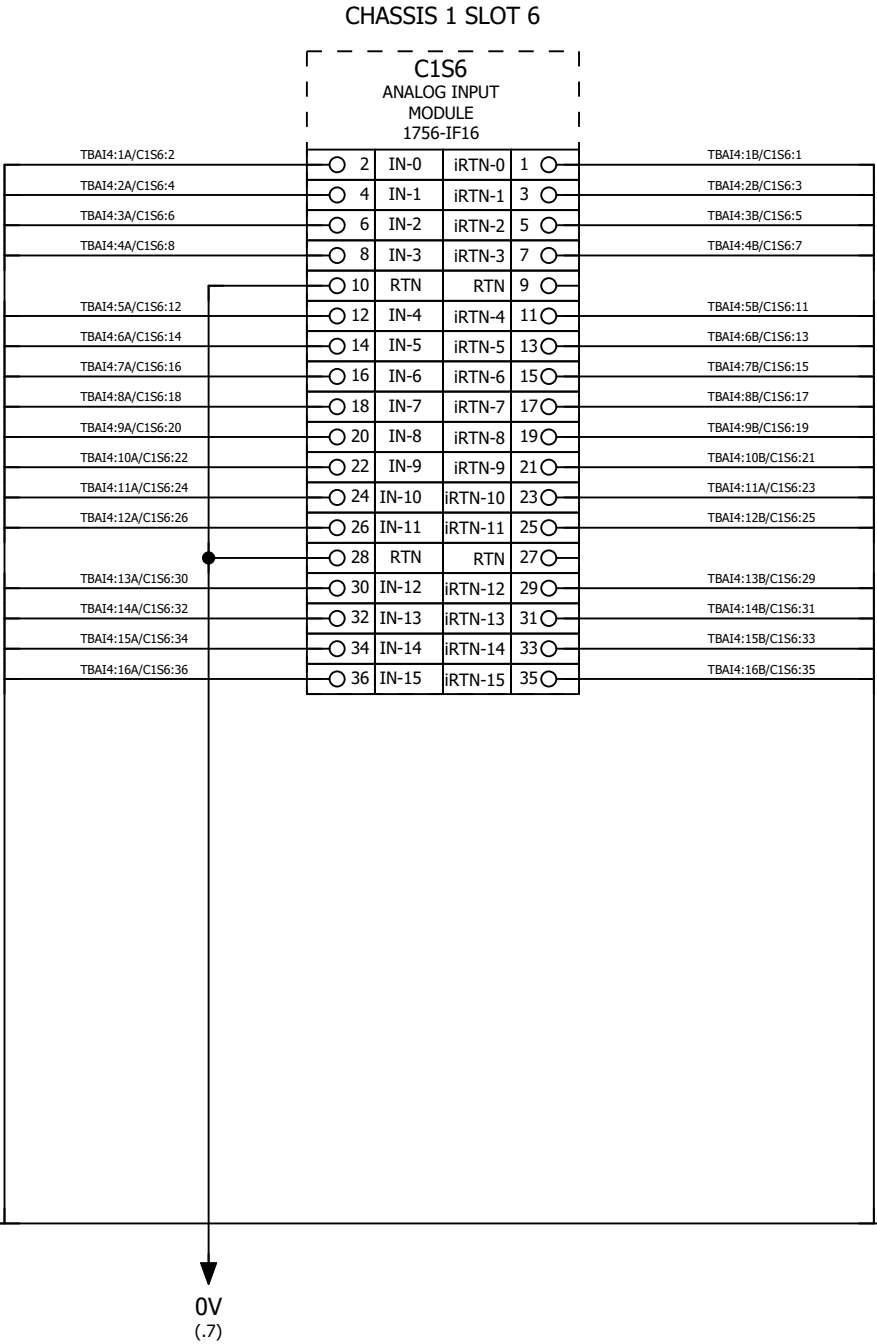
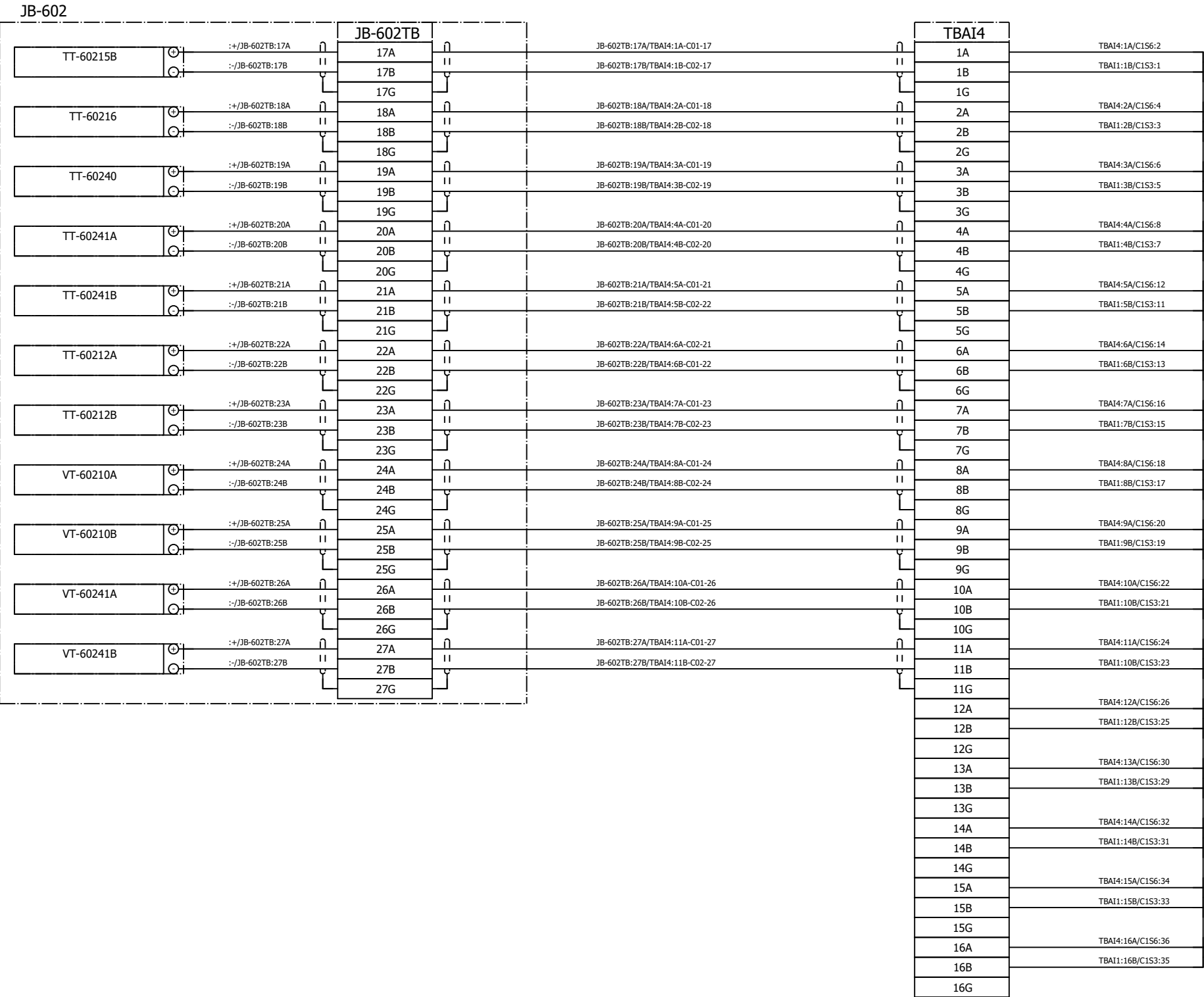
Change to be fuse terminal block

JB-602

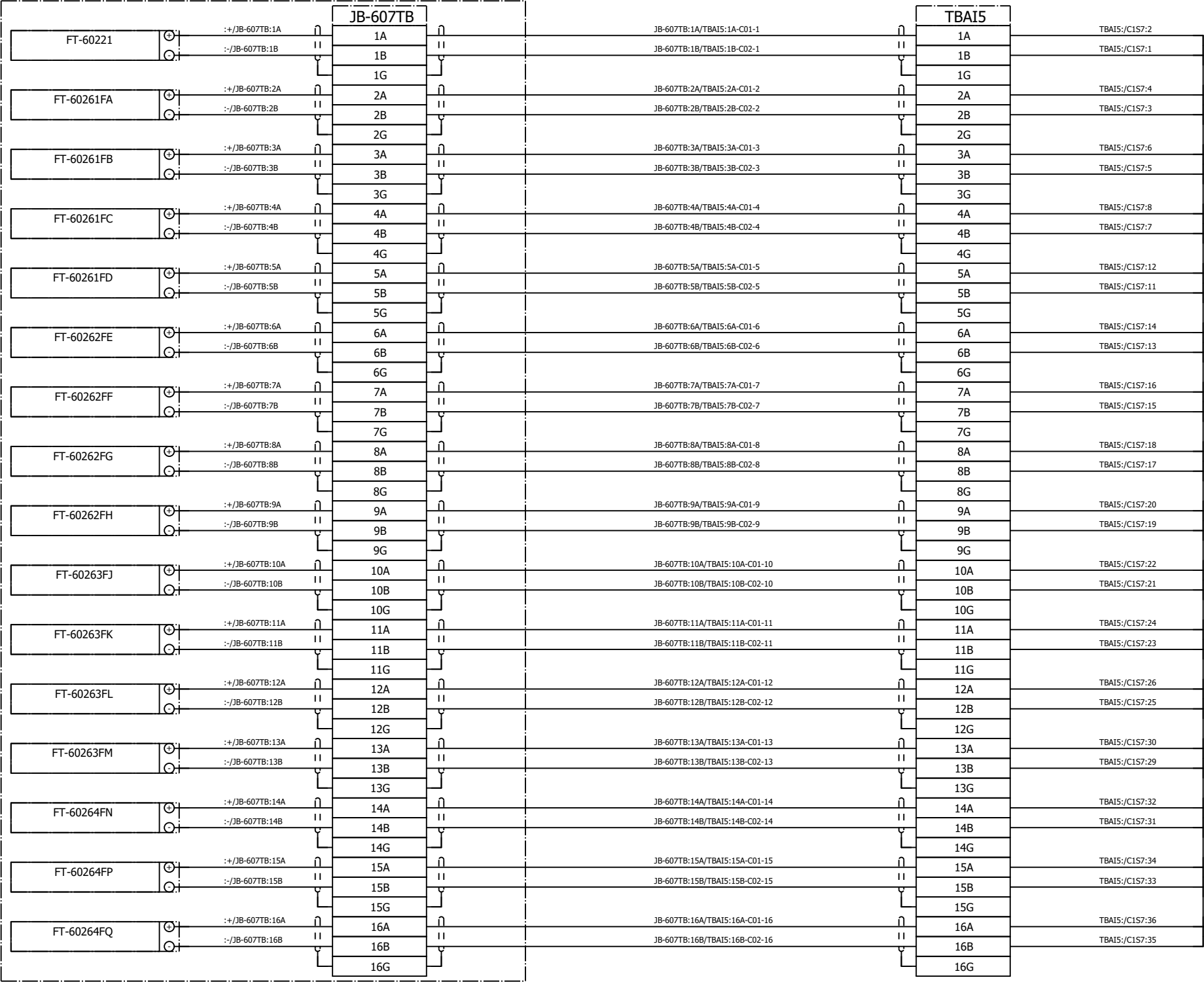


CHASSIS 1 SLOT 5

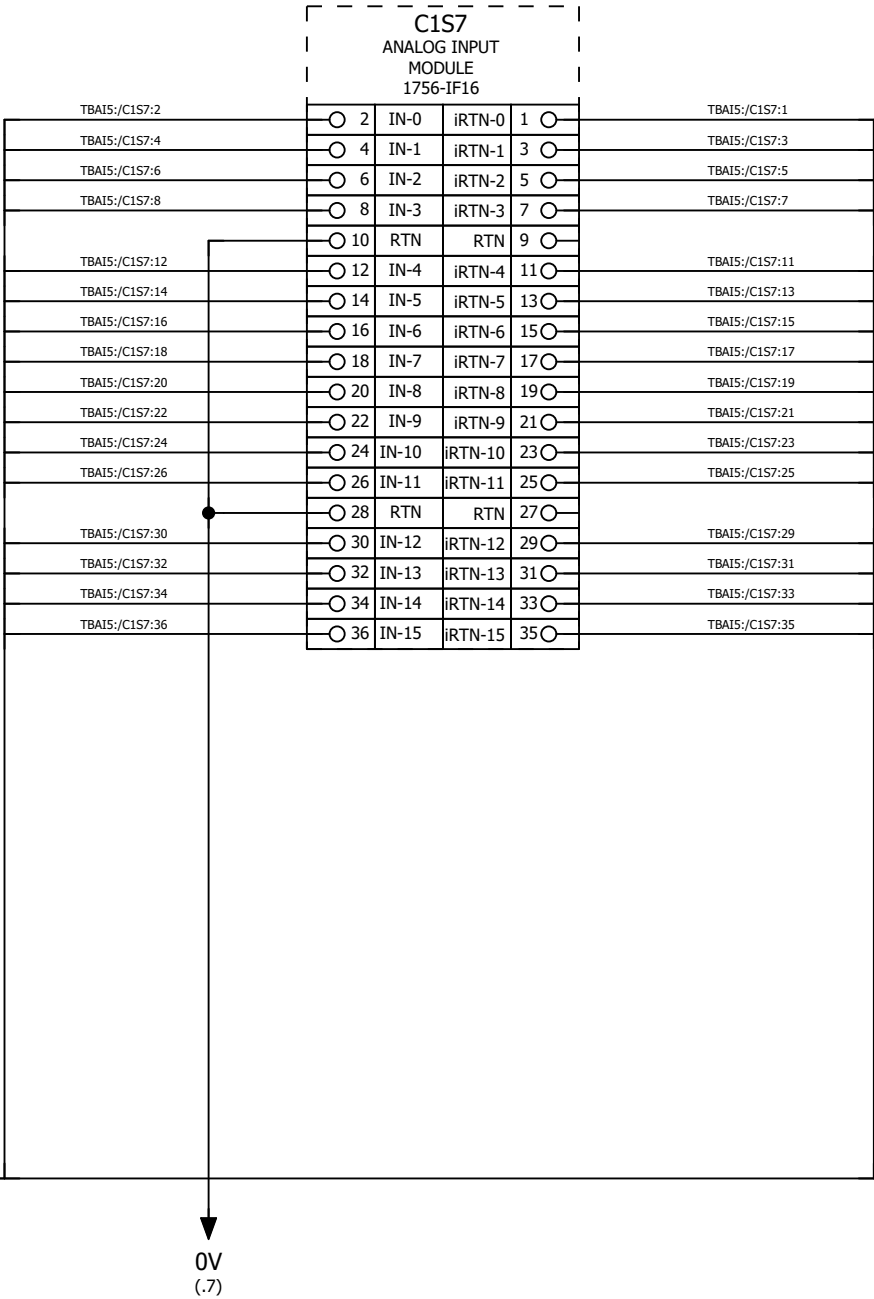




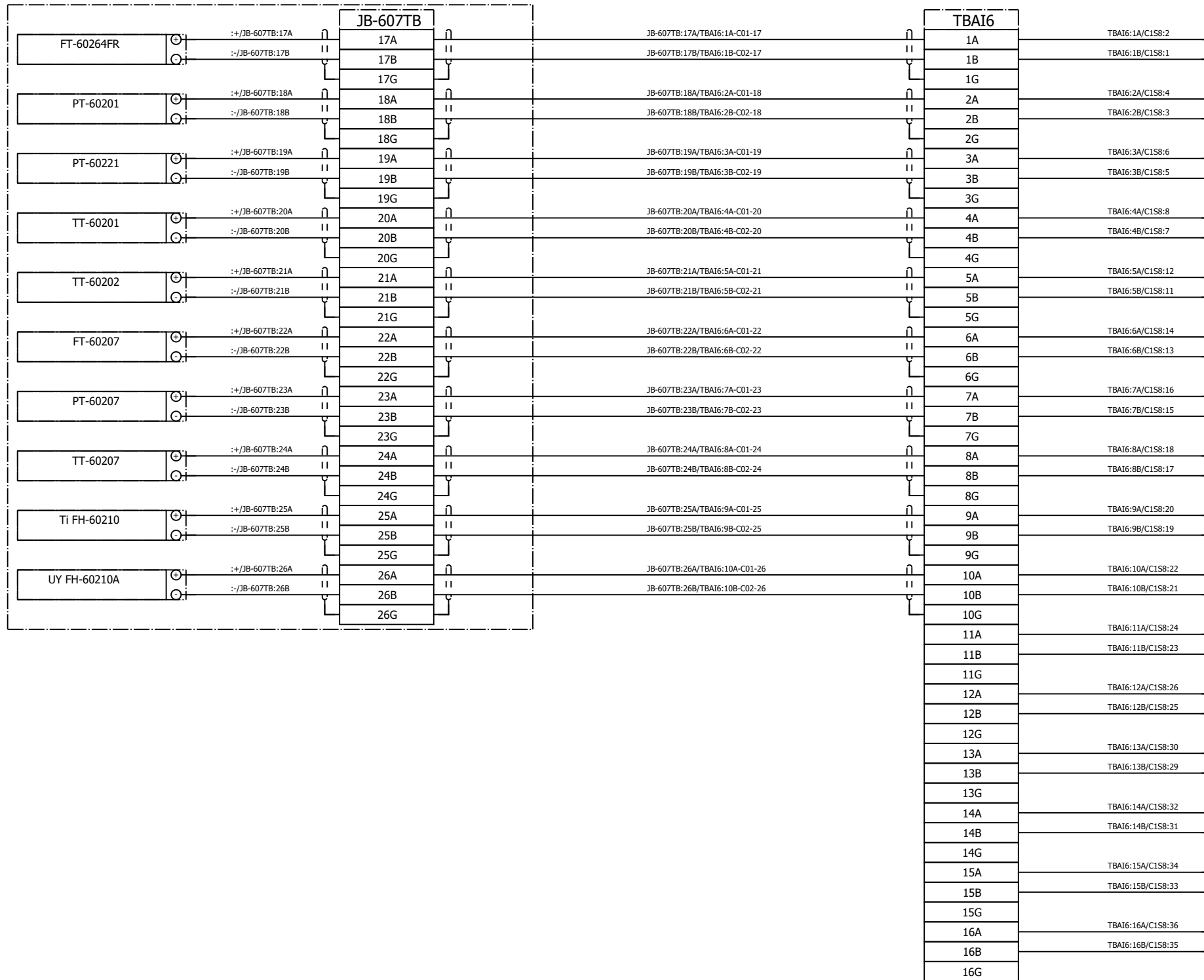
JB-607



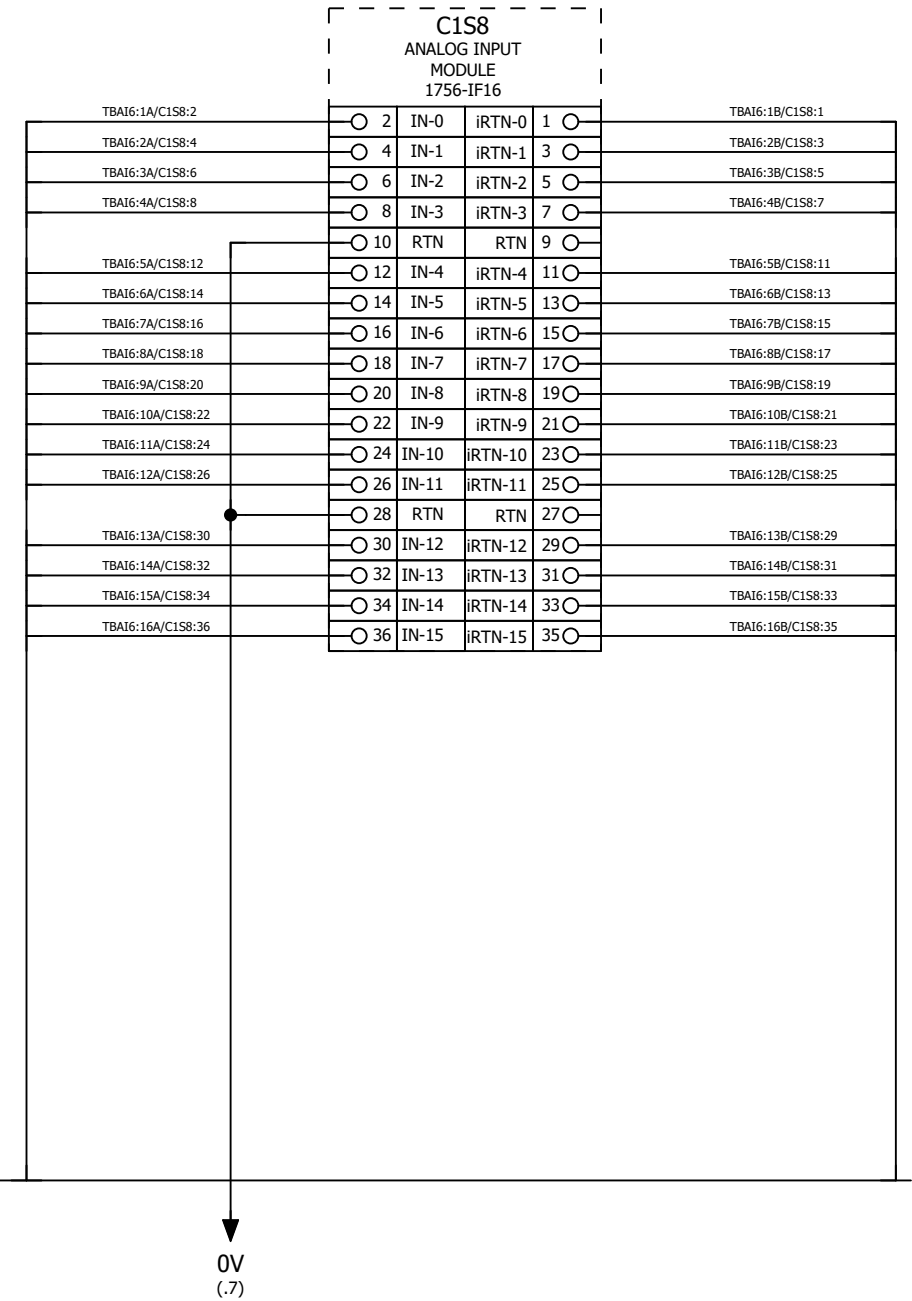
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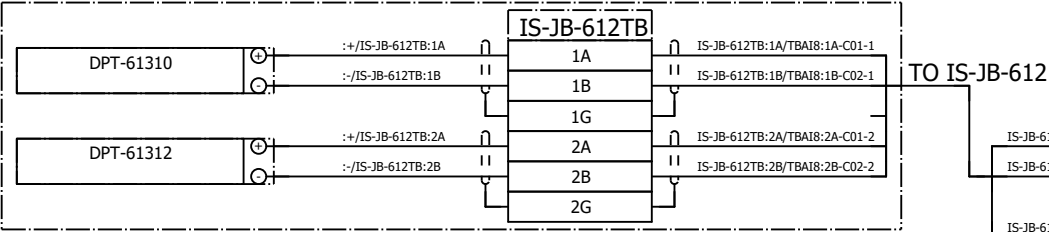
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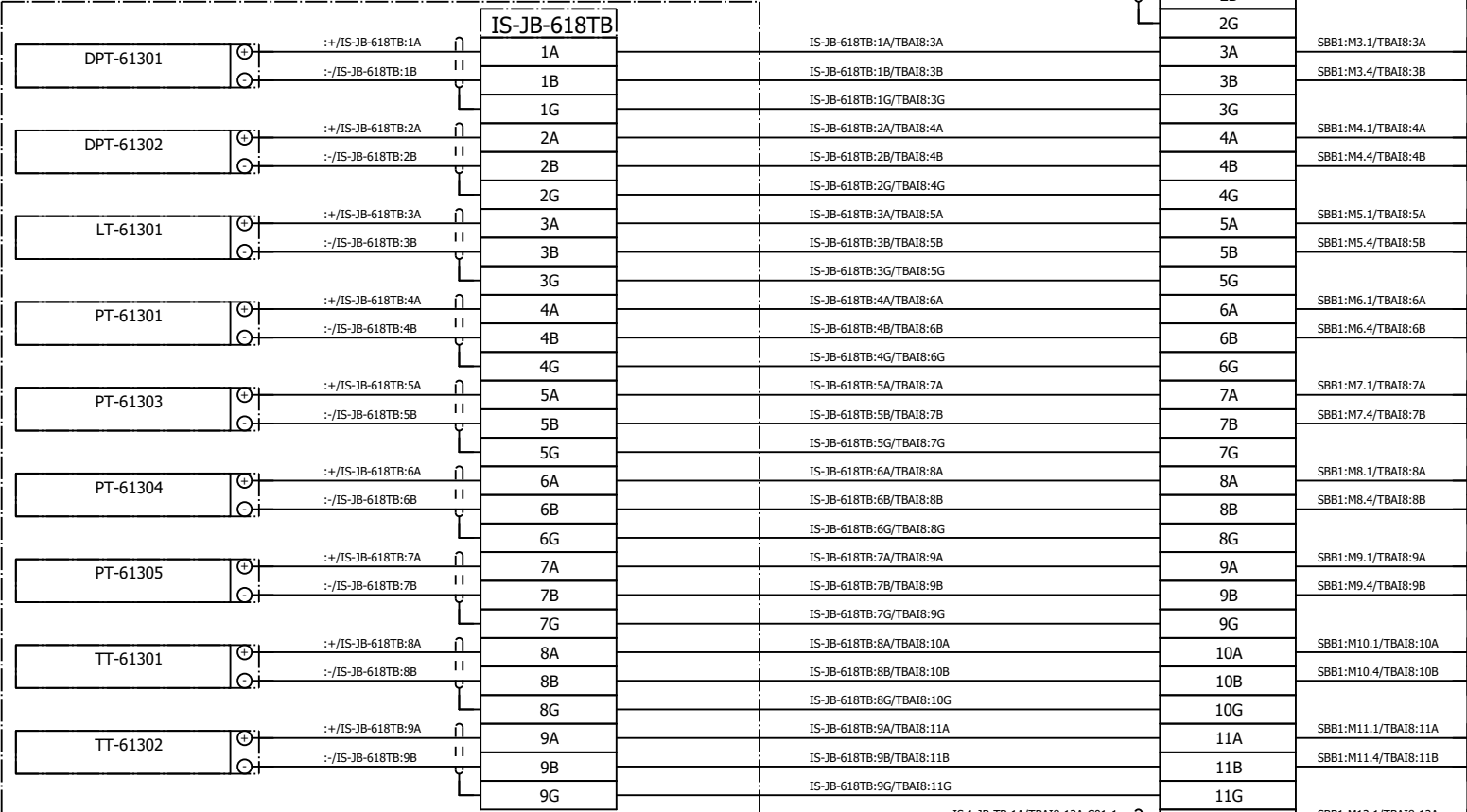
CHASSIS 1 SLOT 8



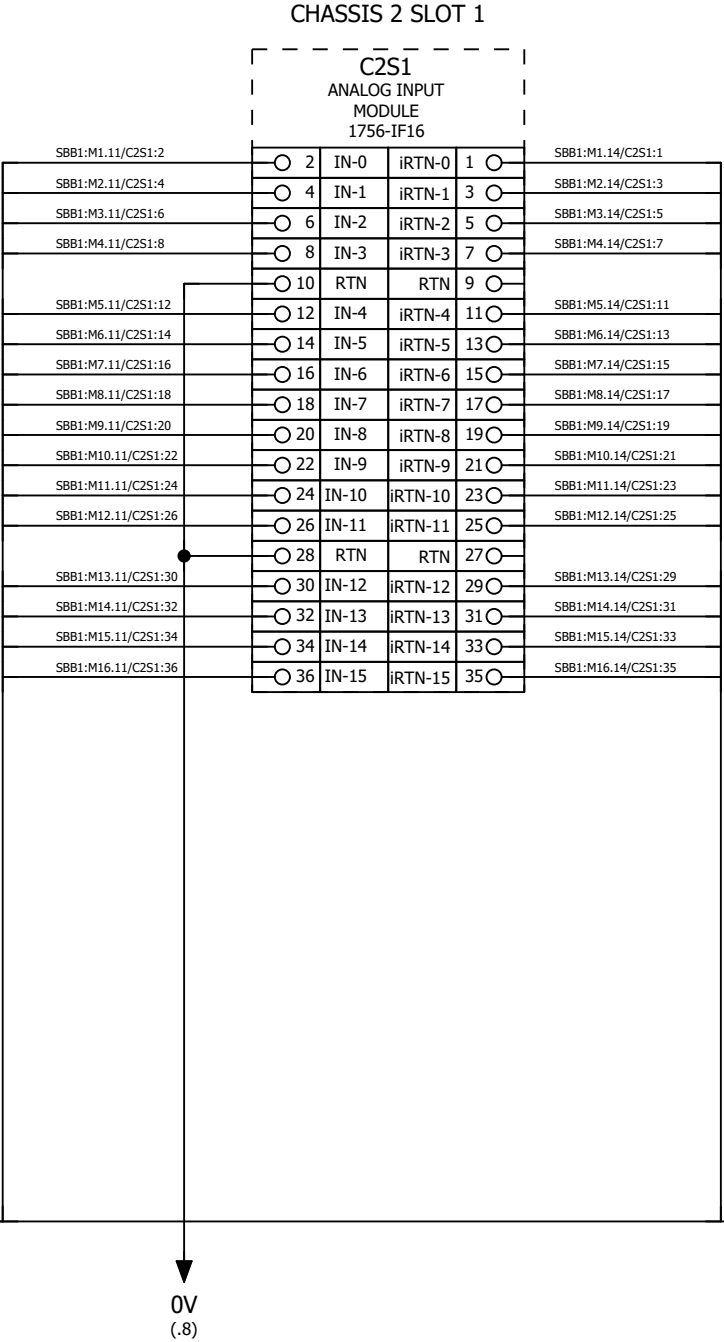
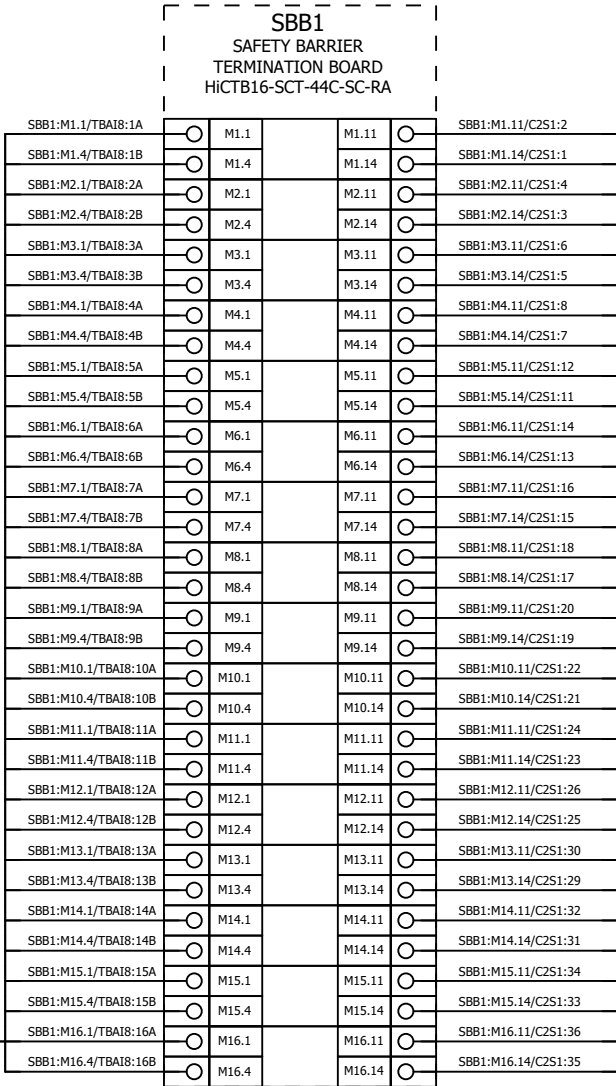
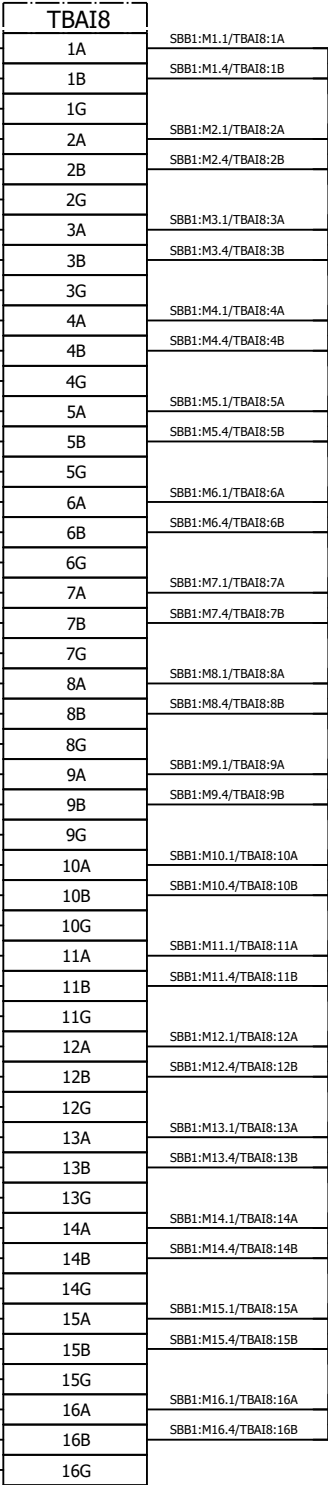
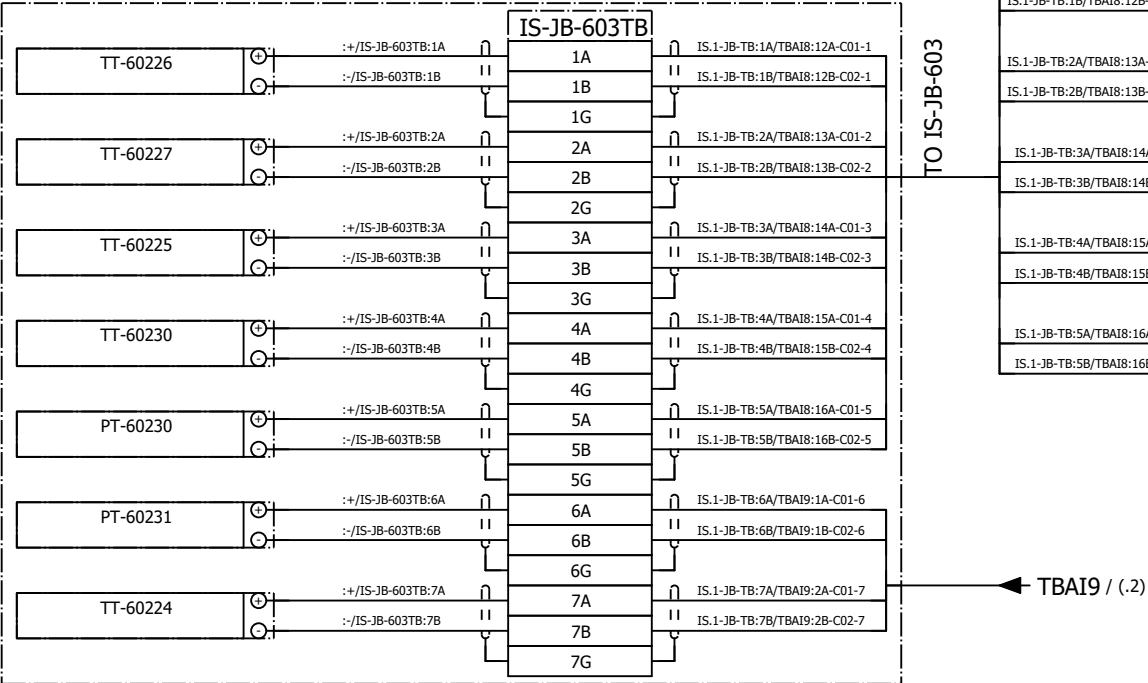
IS-JB-612



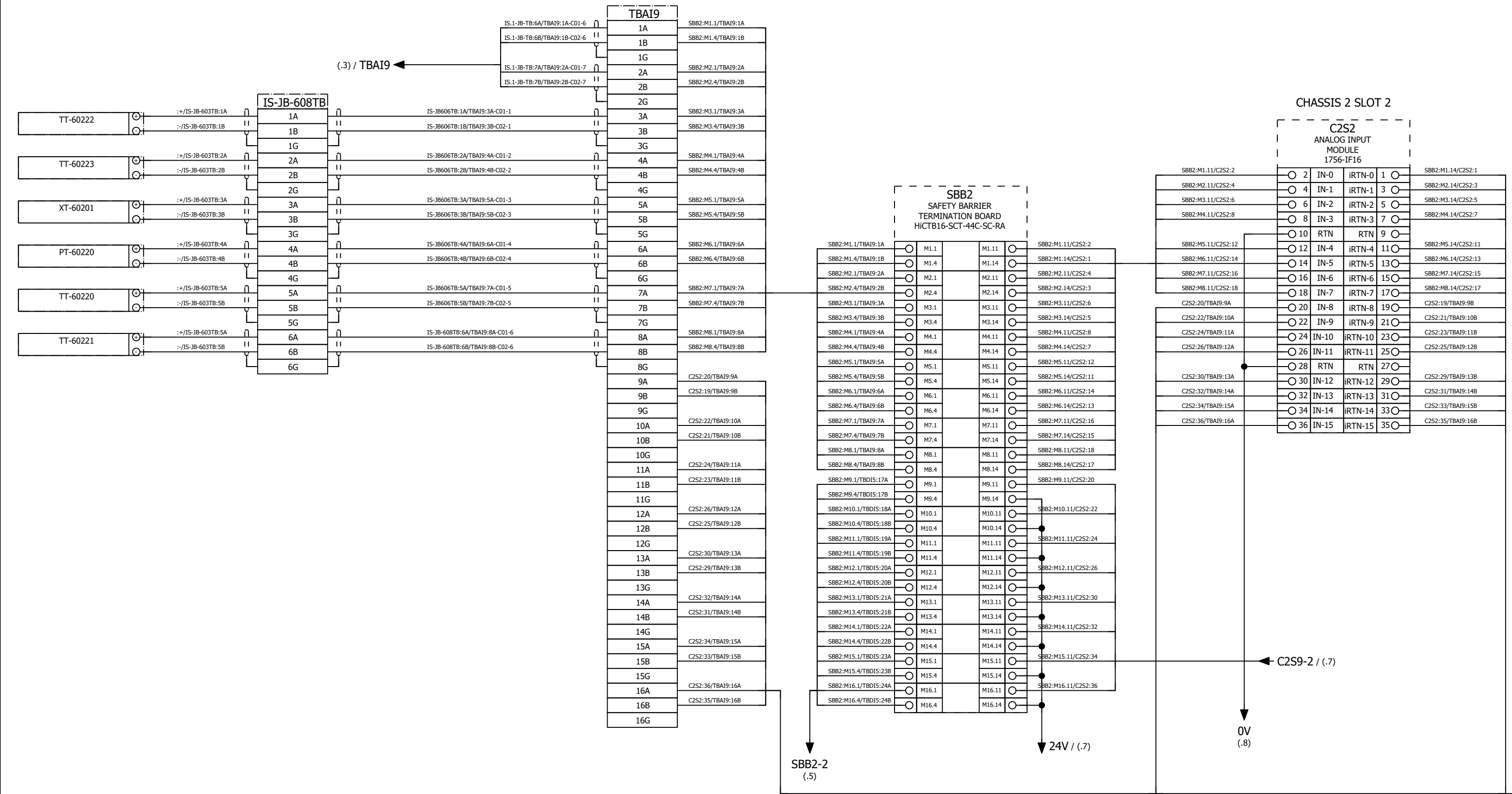
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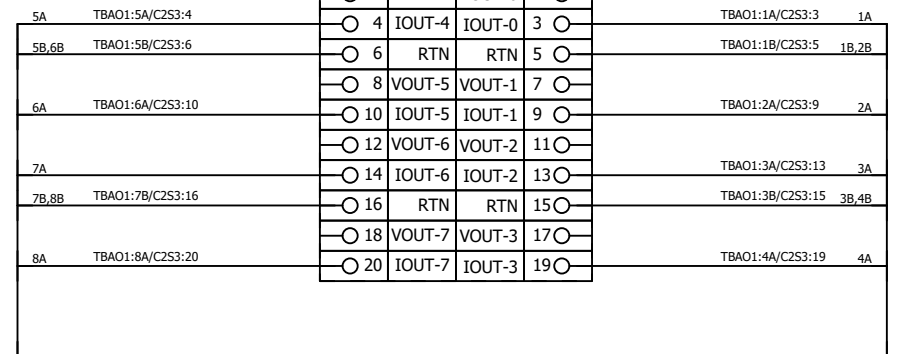
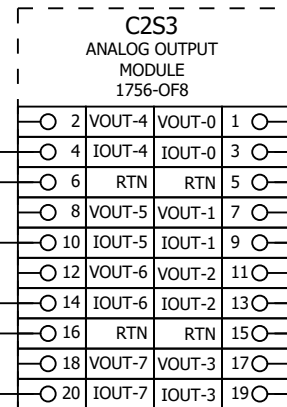
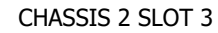
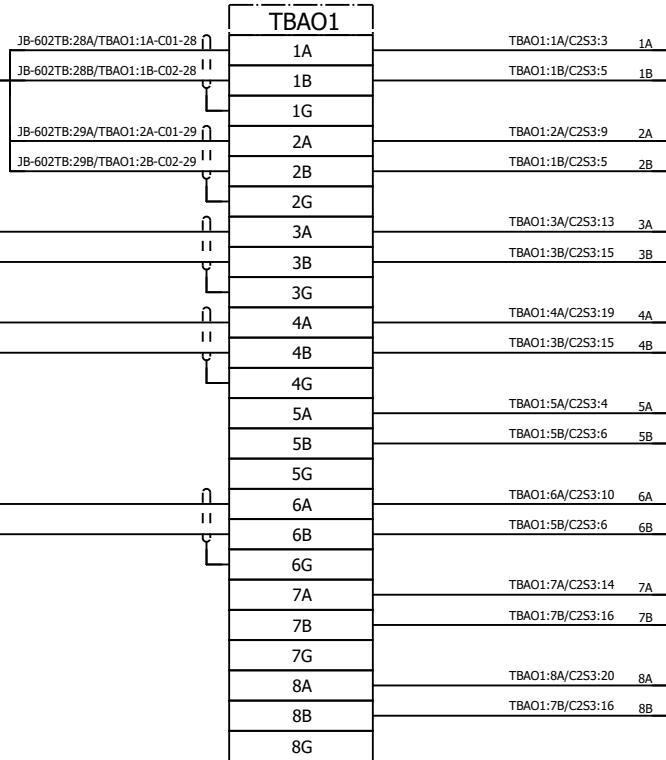
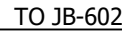
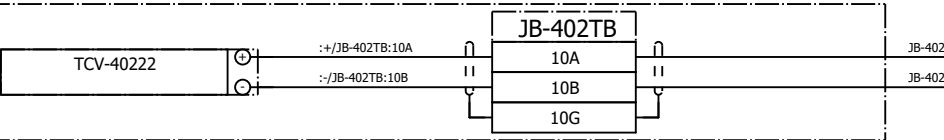
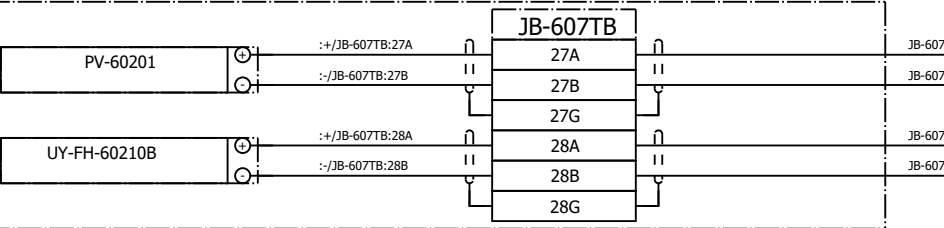
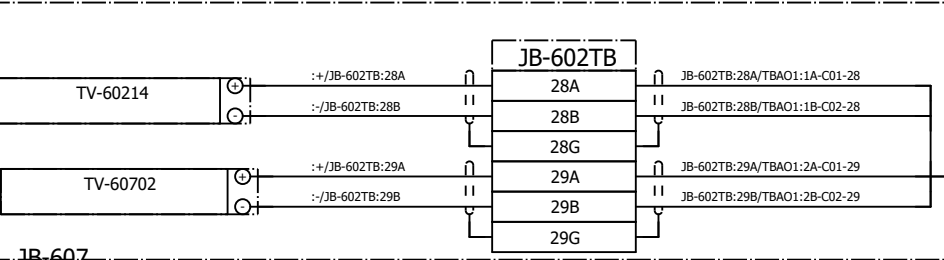


IS-JB-603

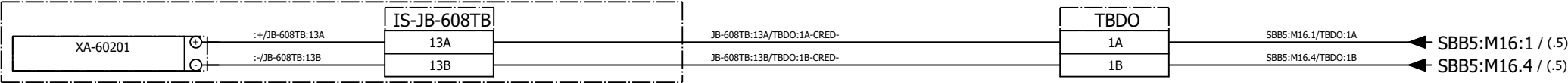


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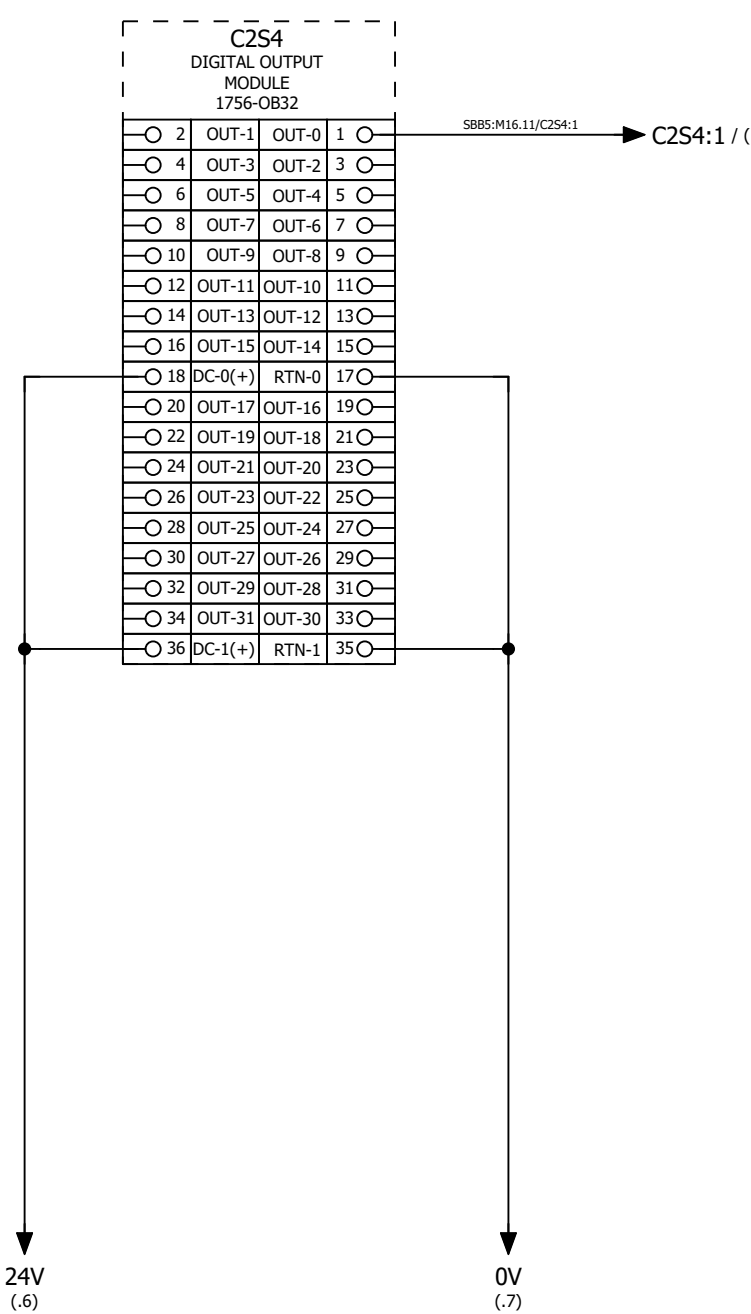
IS-JB-608

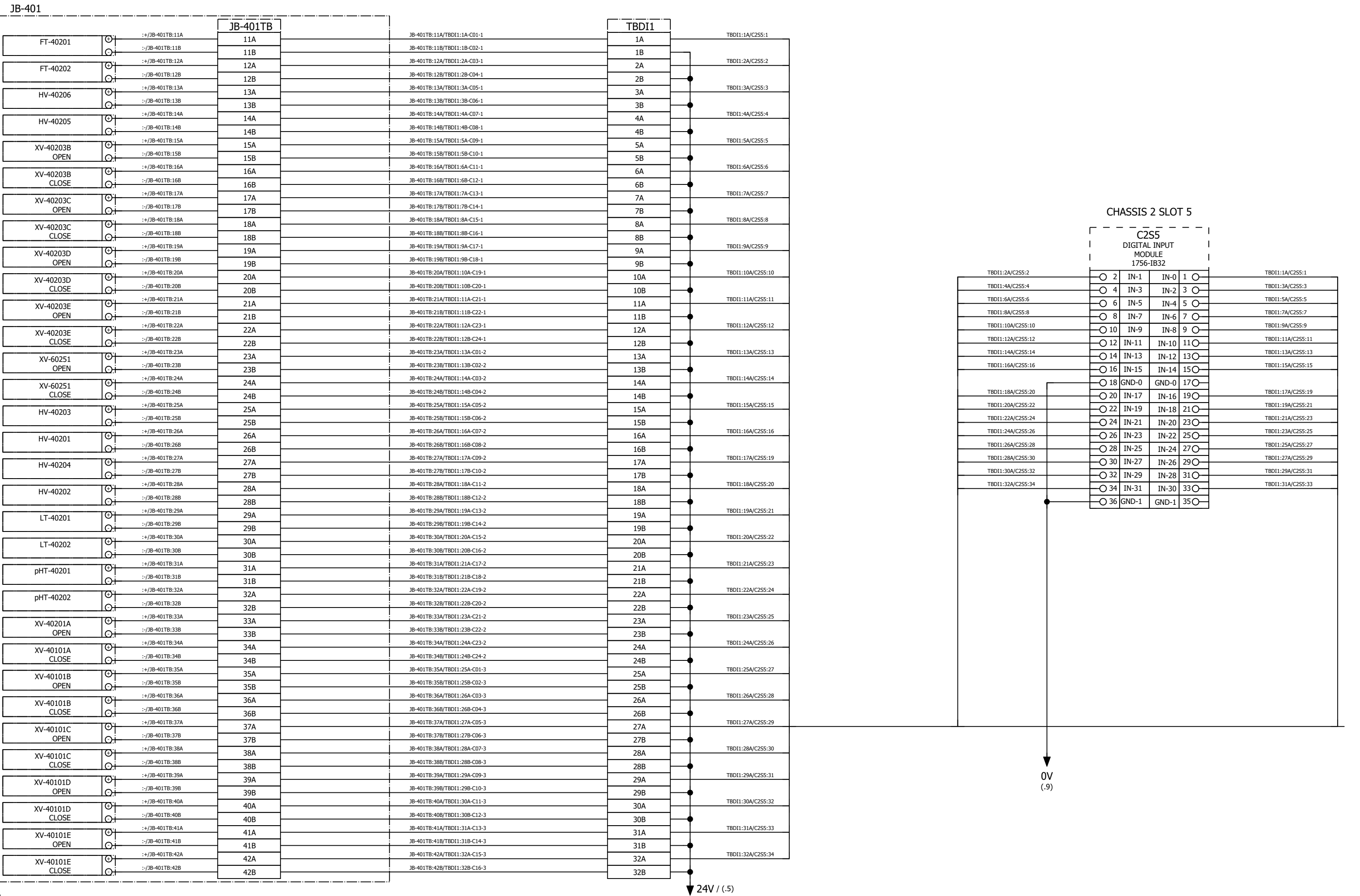


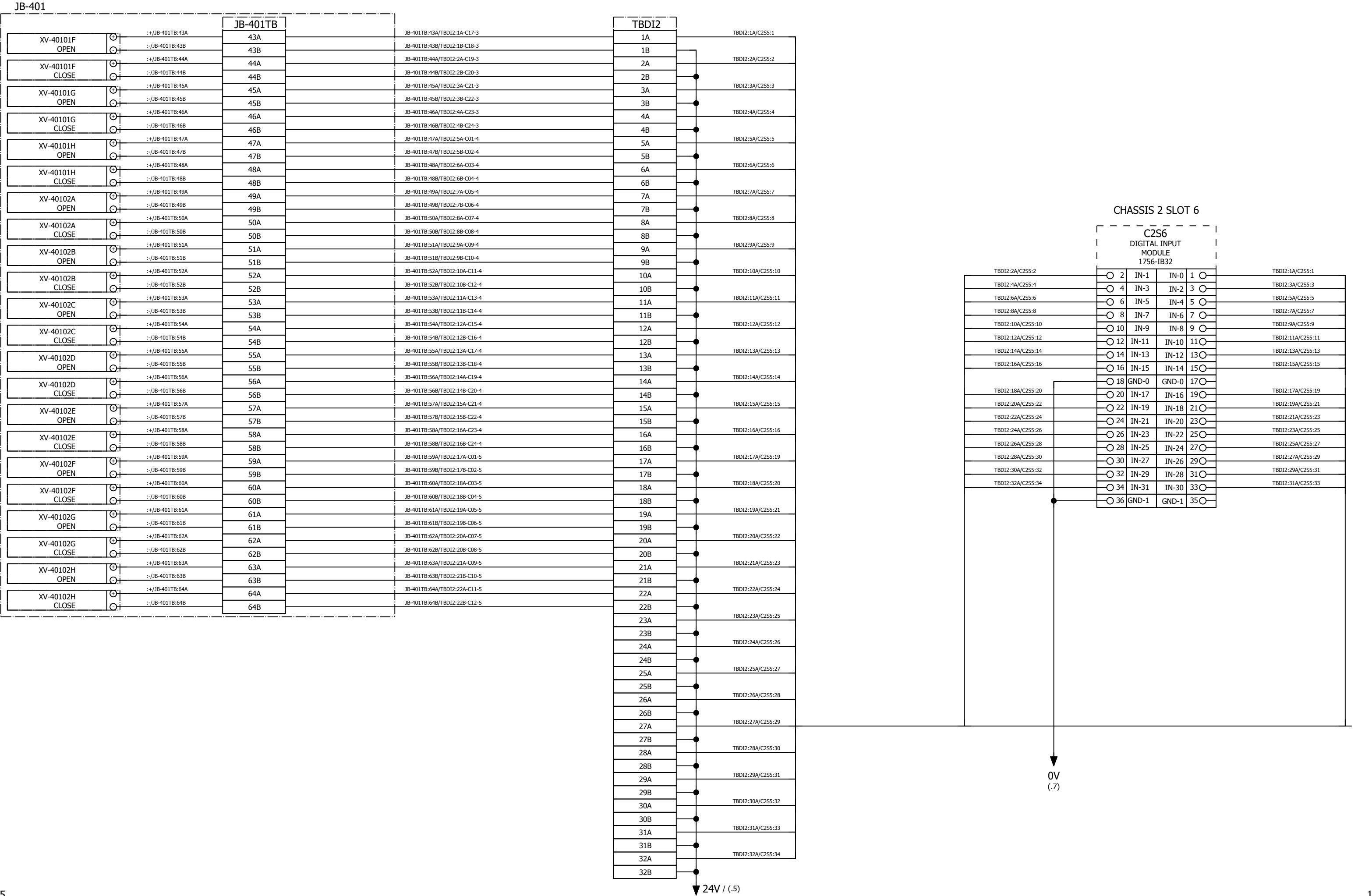
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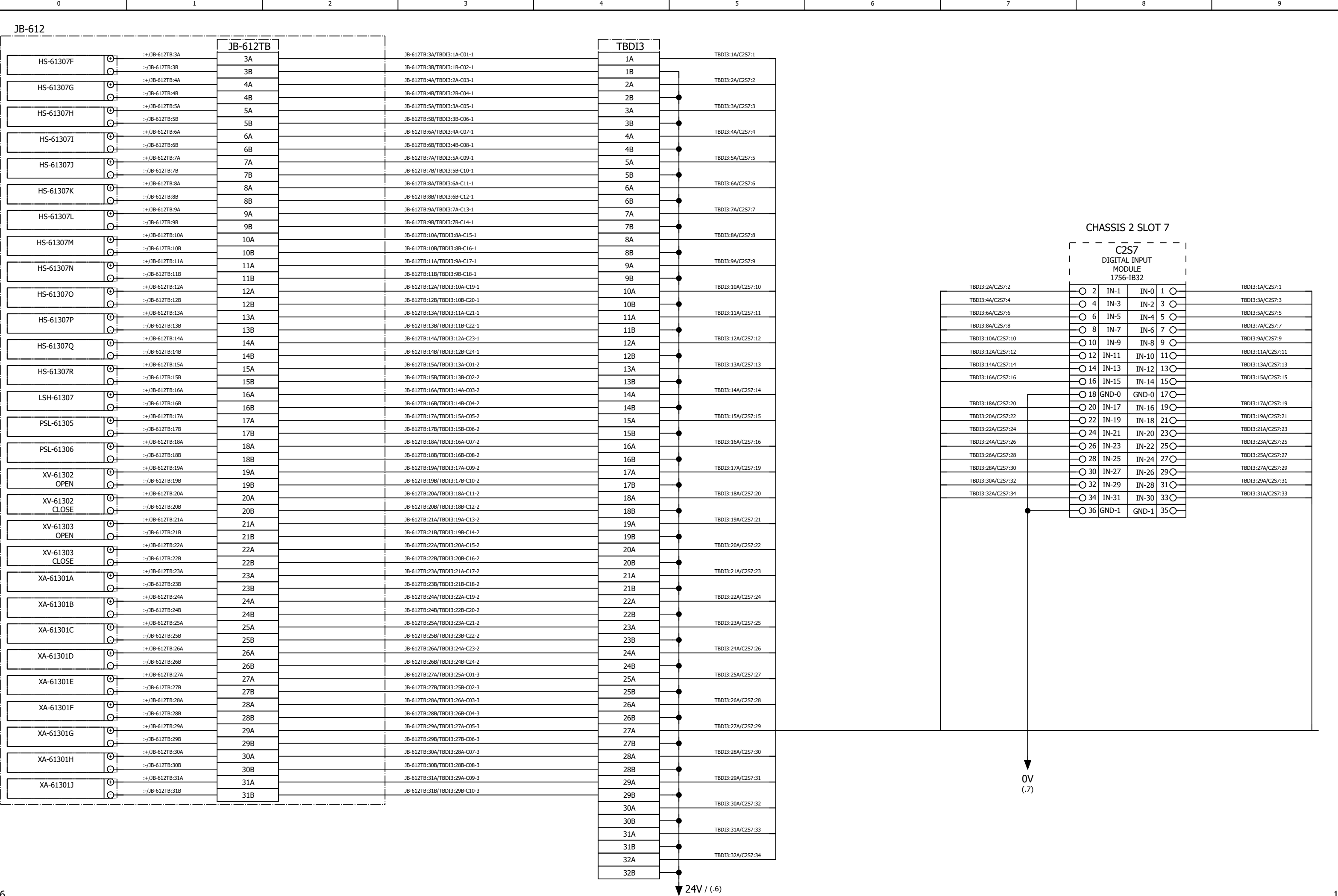
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			Ed.	THUMWH		
			Appr.			
					Basic project according to NFPA standard using a metric grid	
Modification	Date	Name	Original		Replacement of	Replaced by

CHASSIS 2 SLOT 4

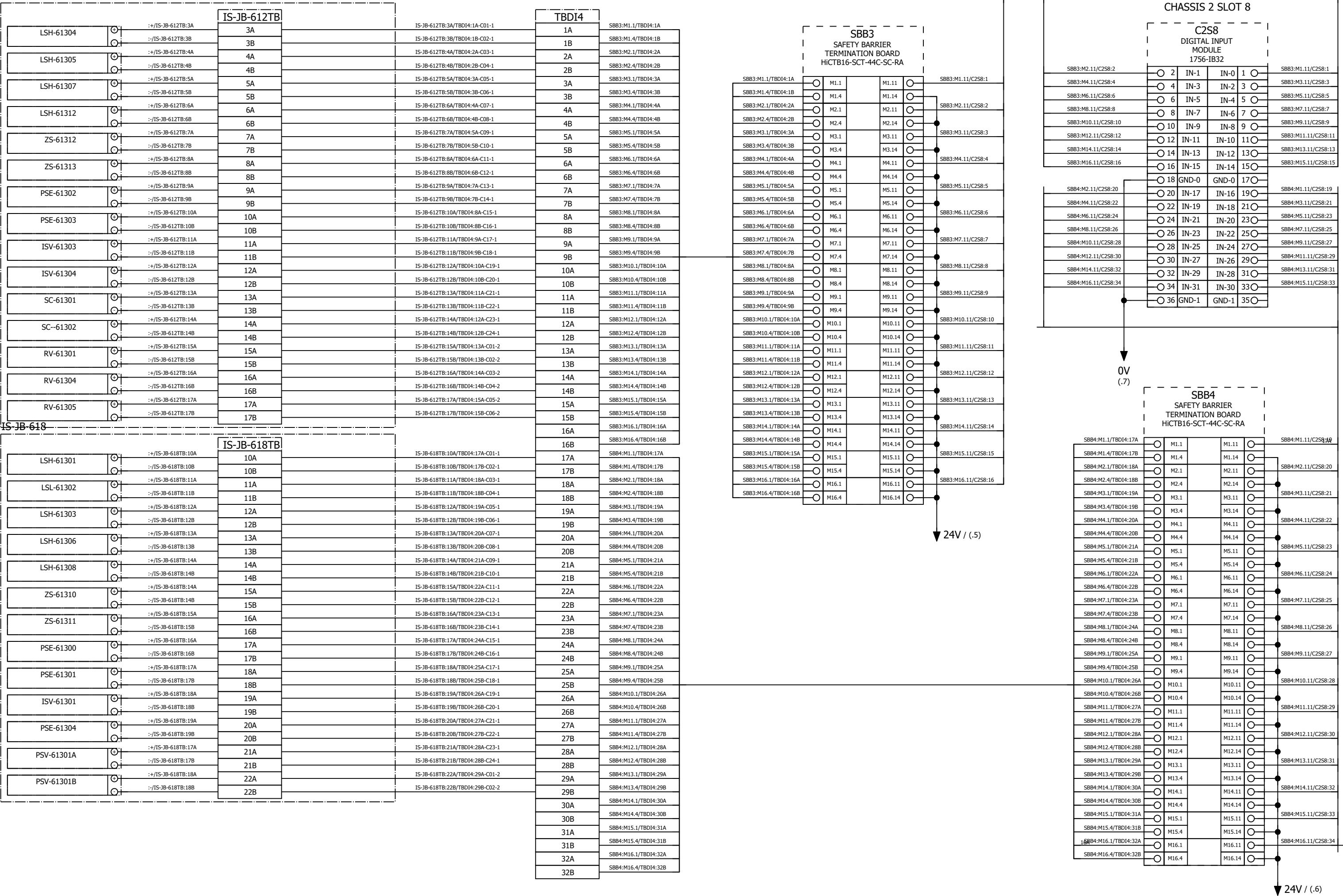


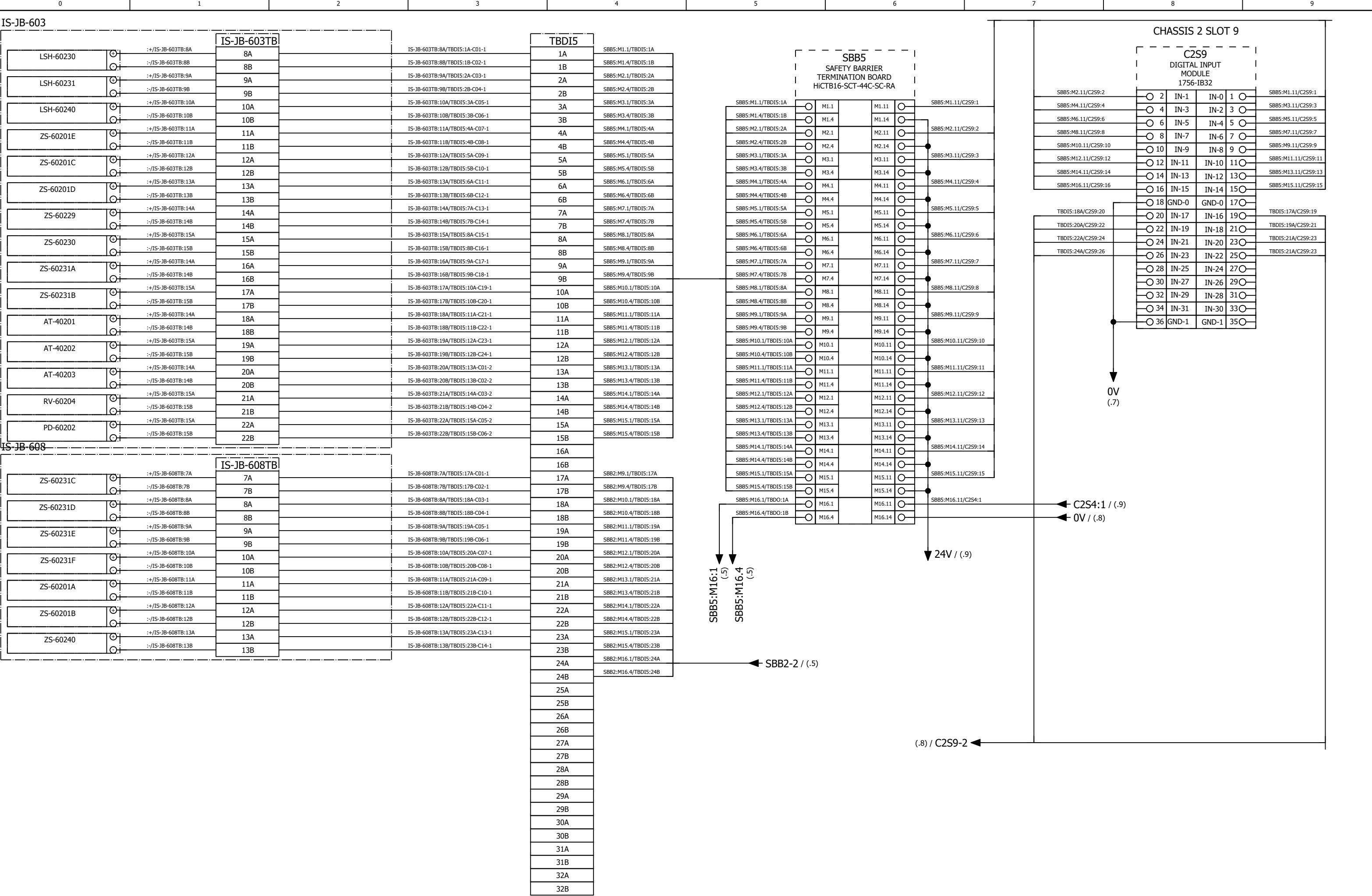


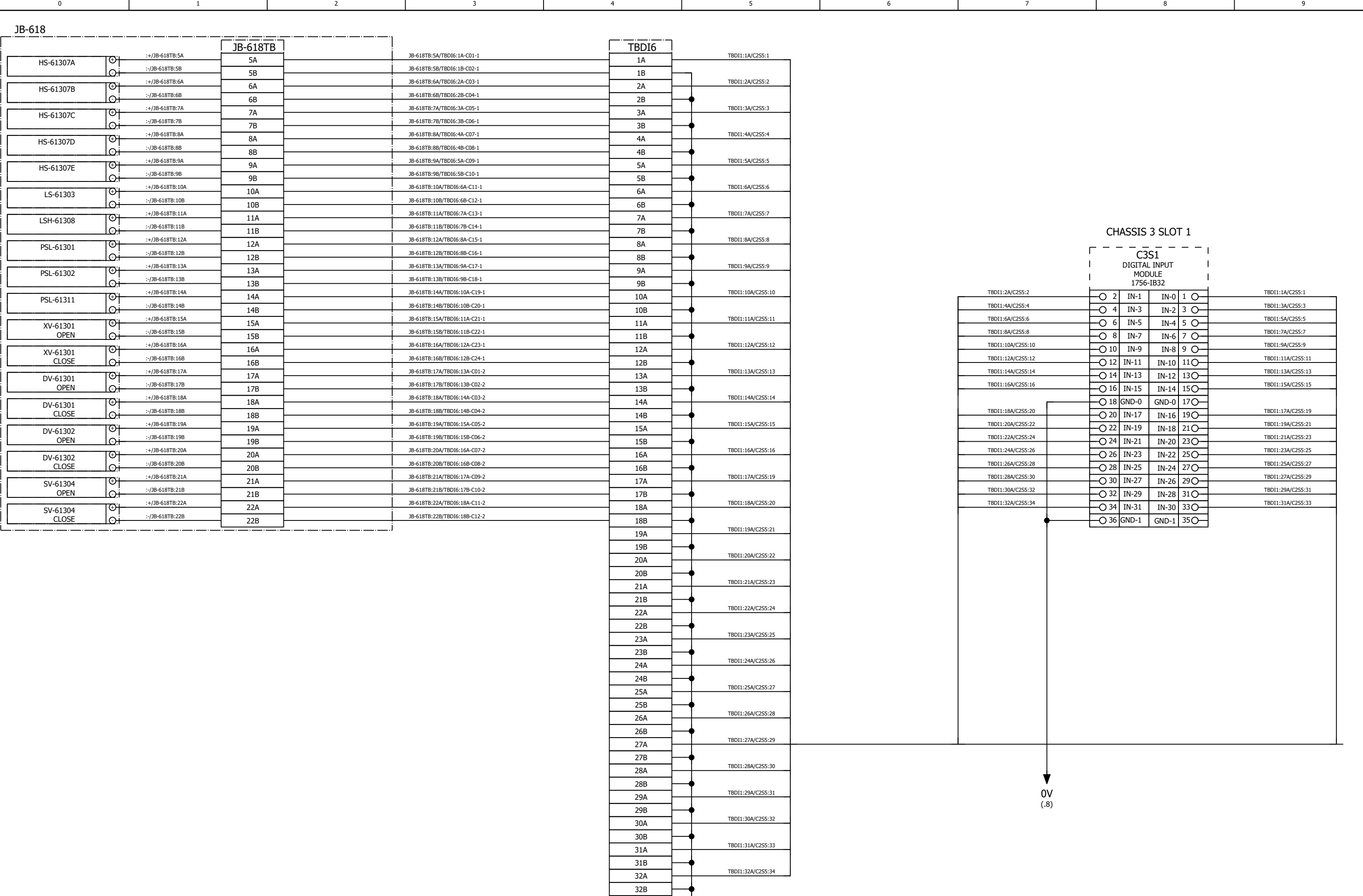


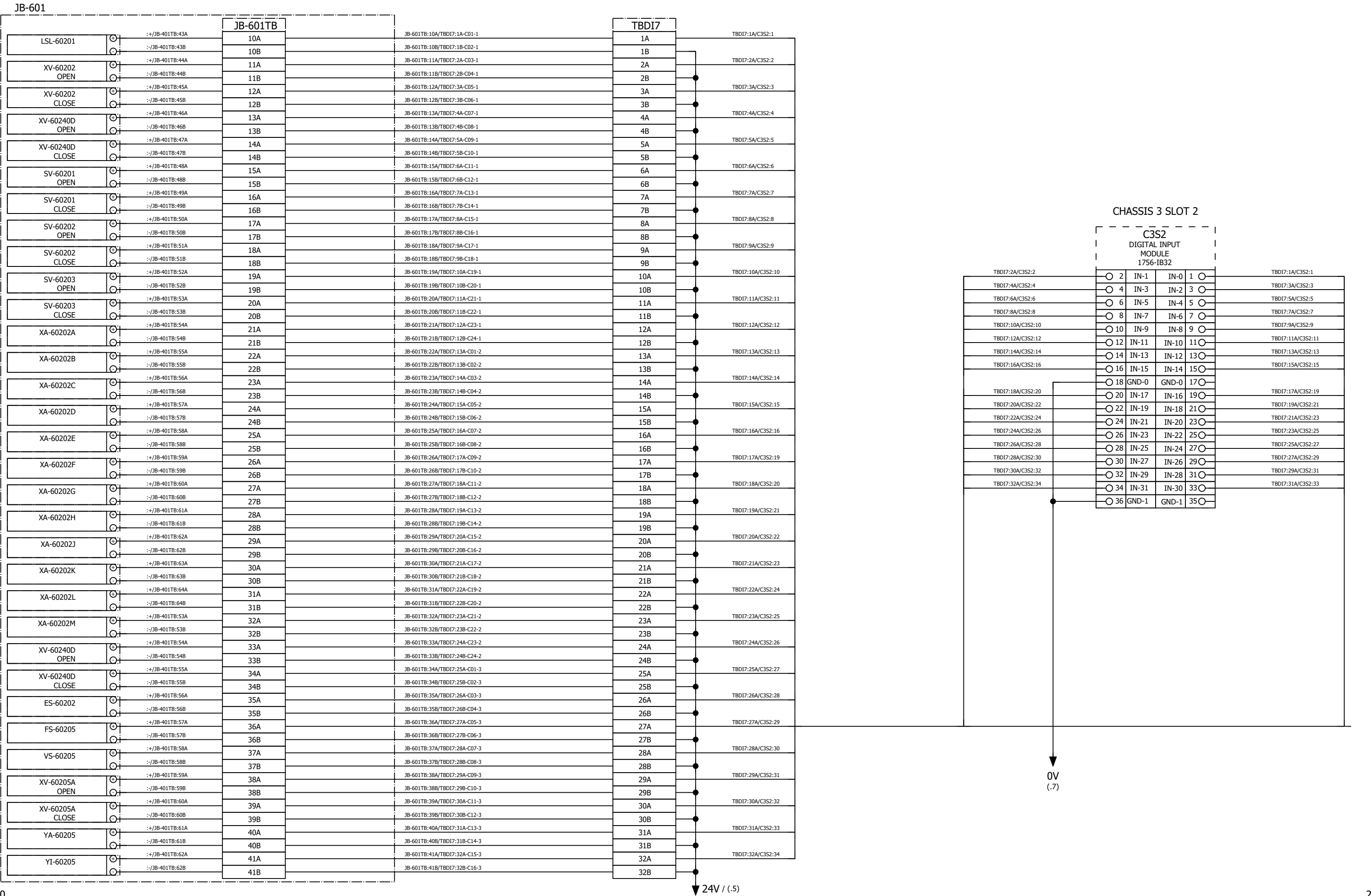


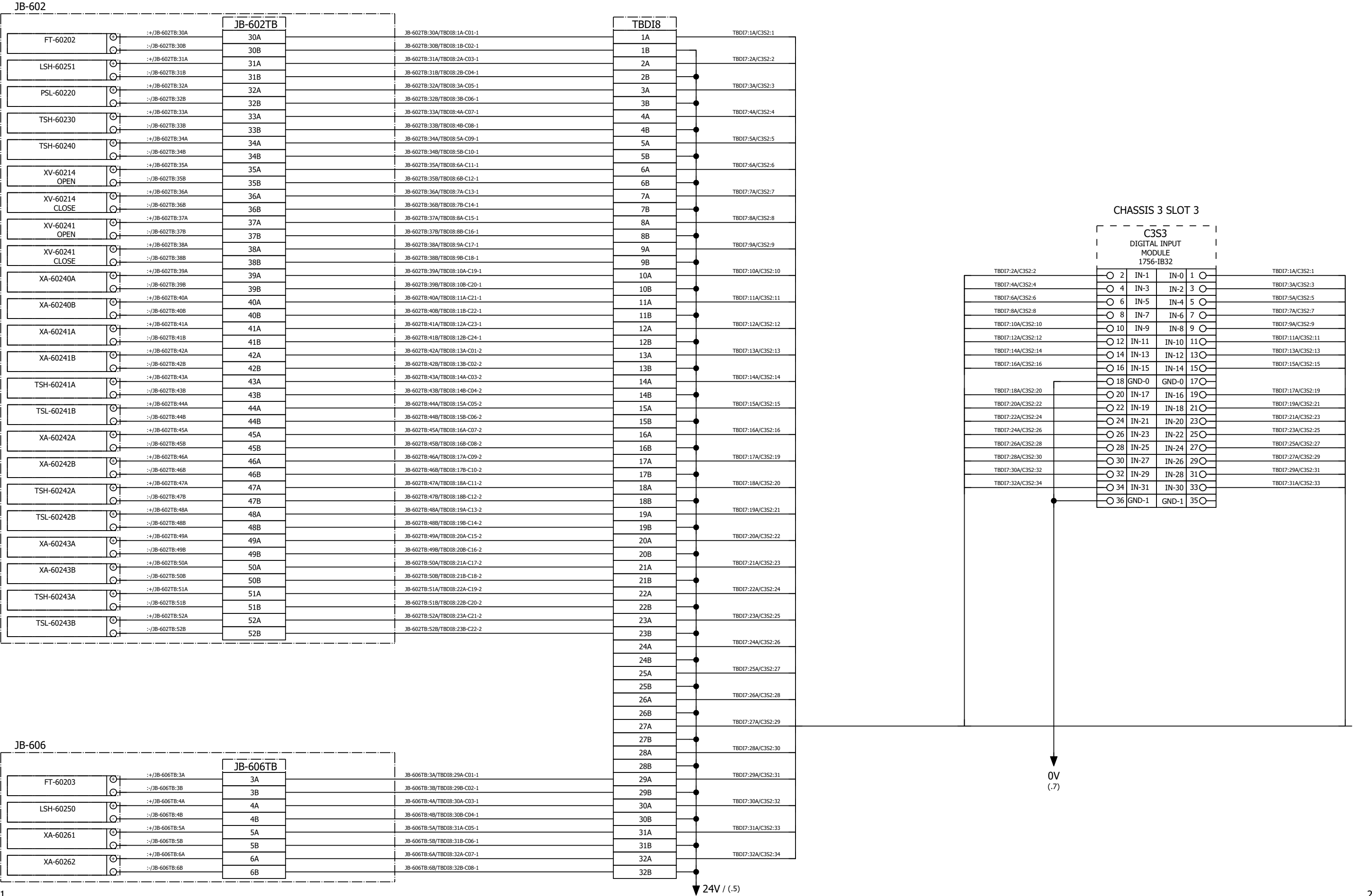
IS-JB-612

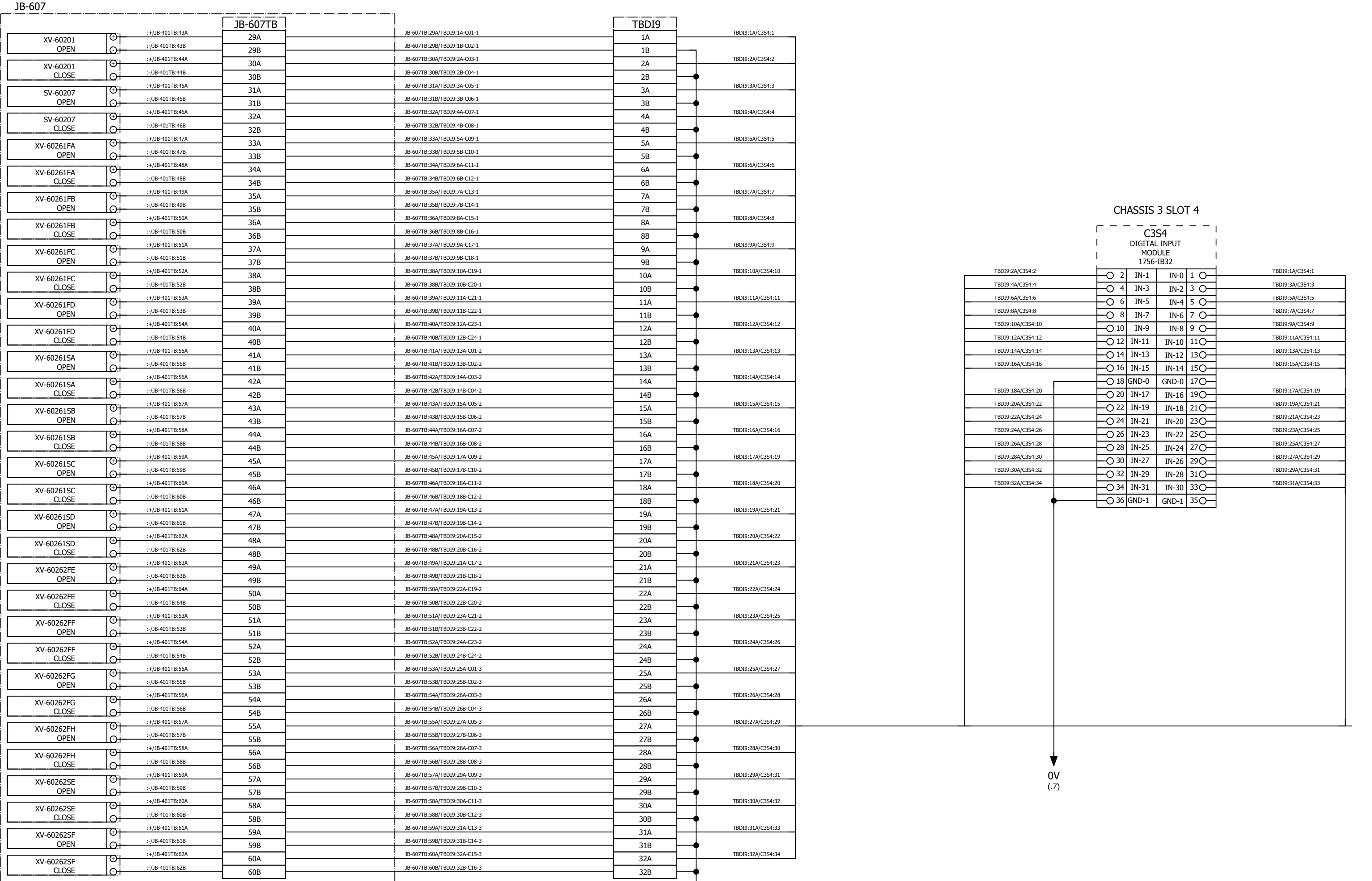




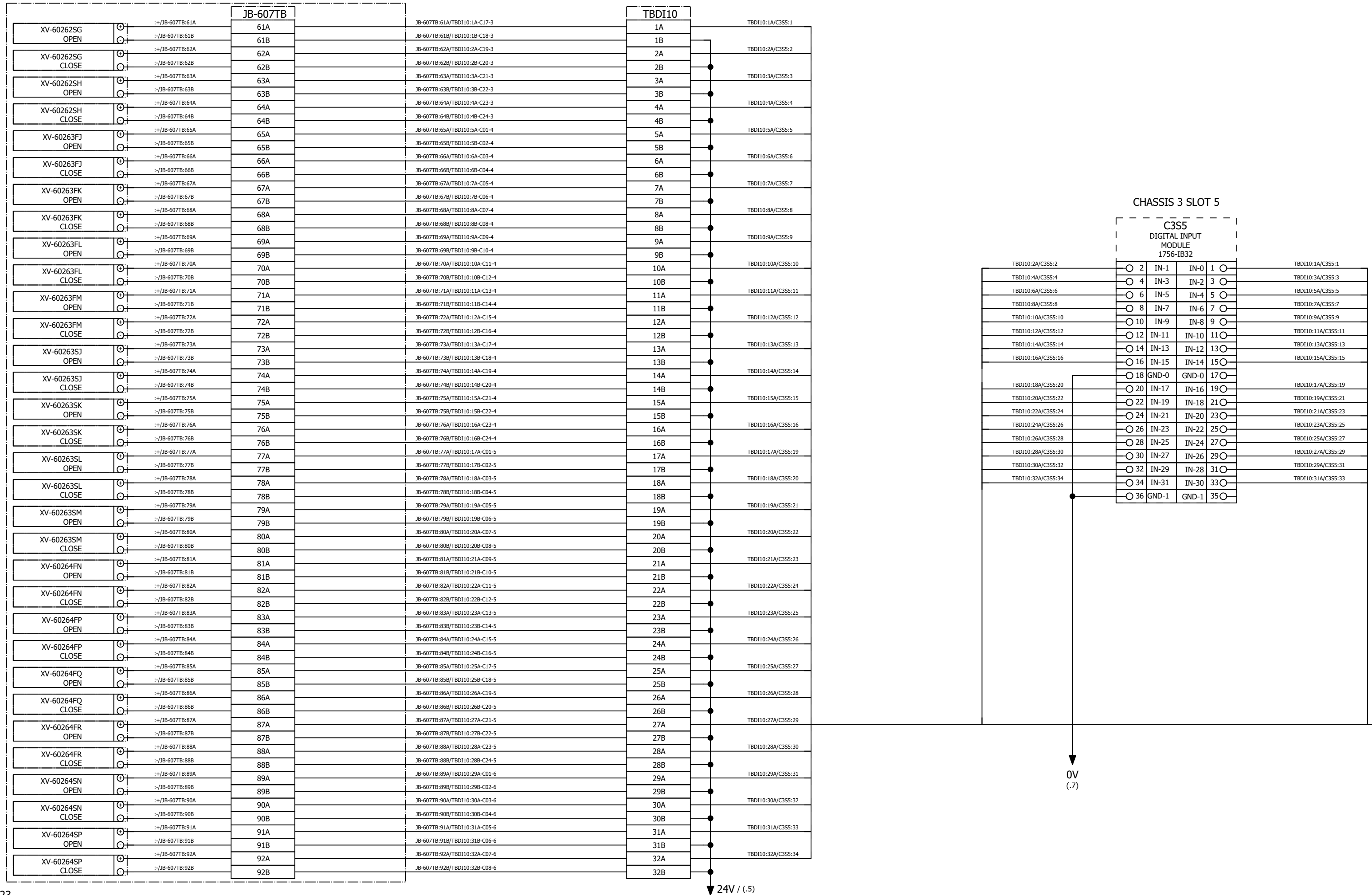


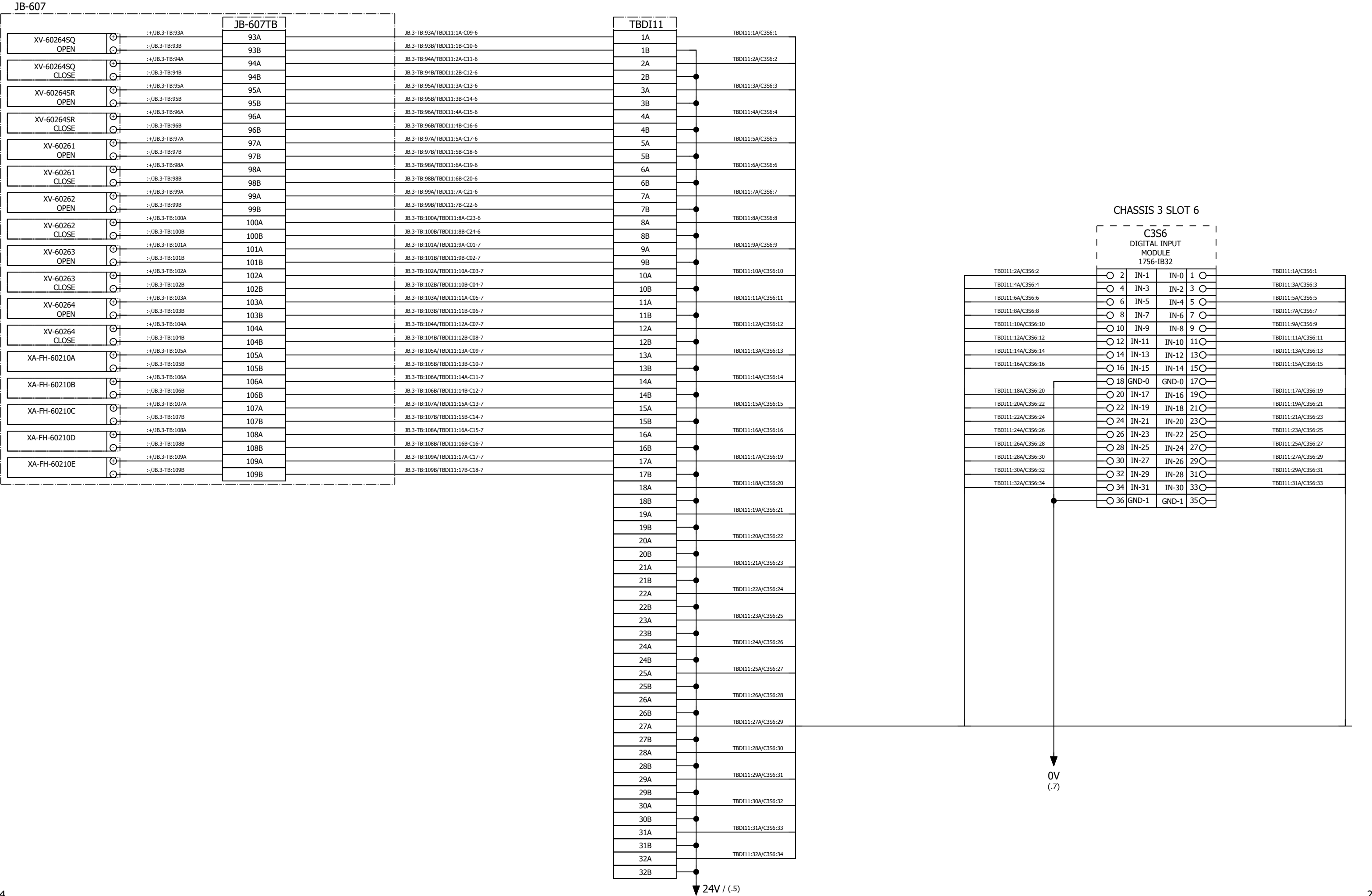


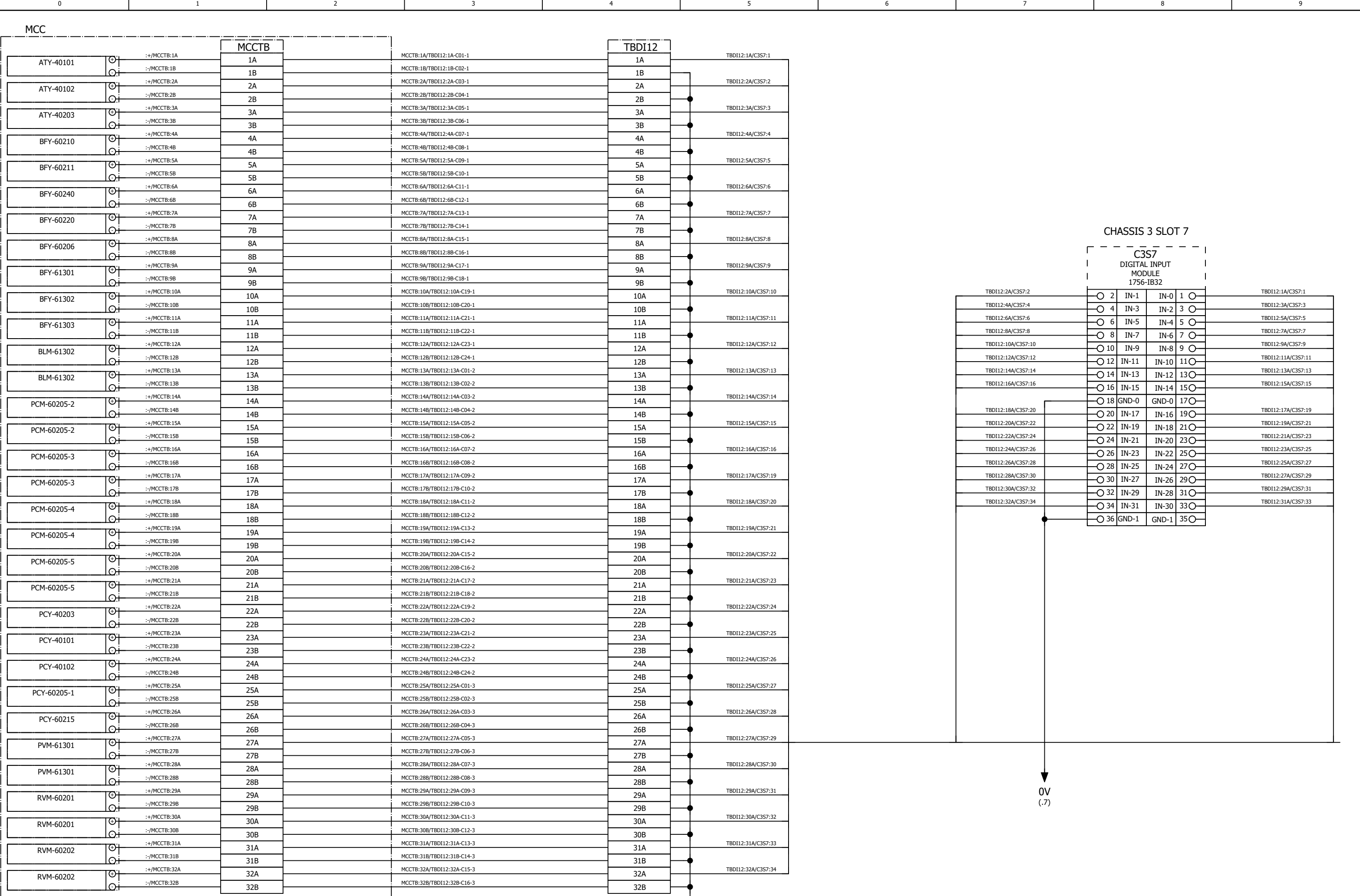


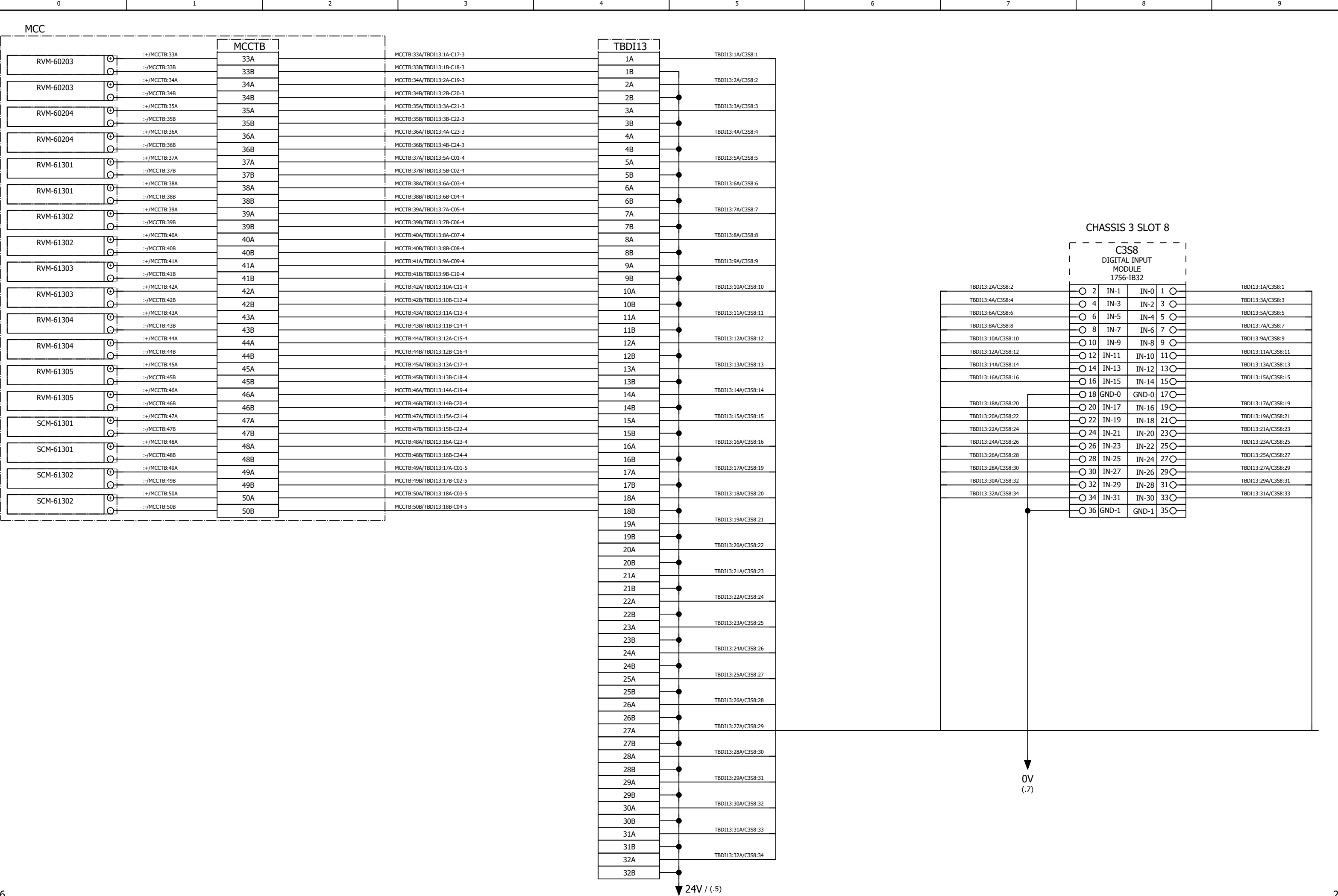


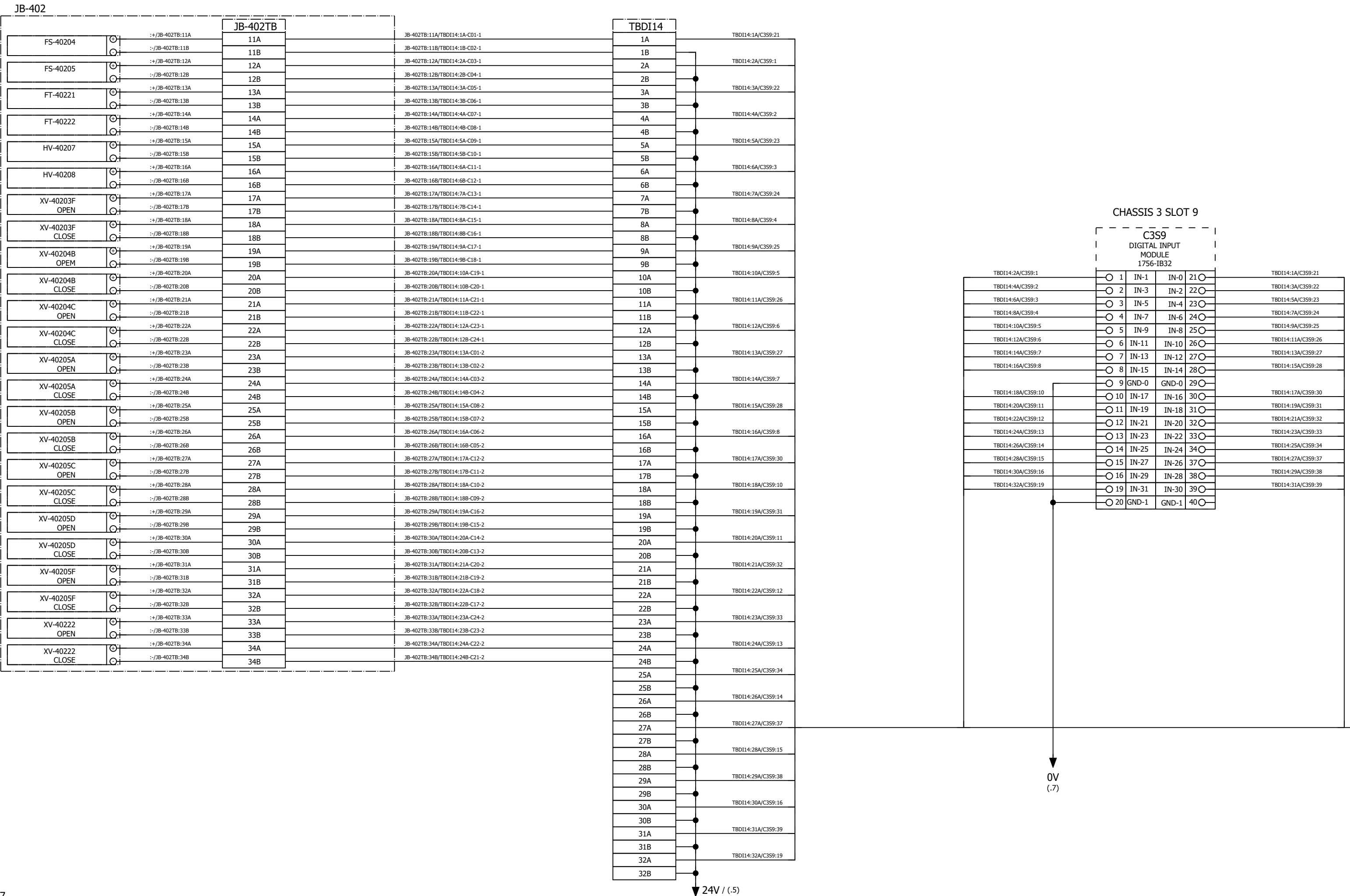
JB-607



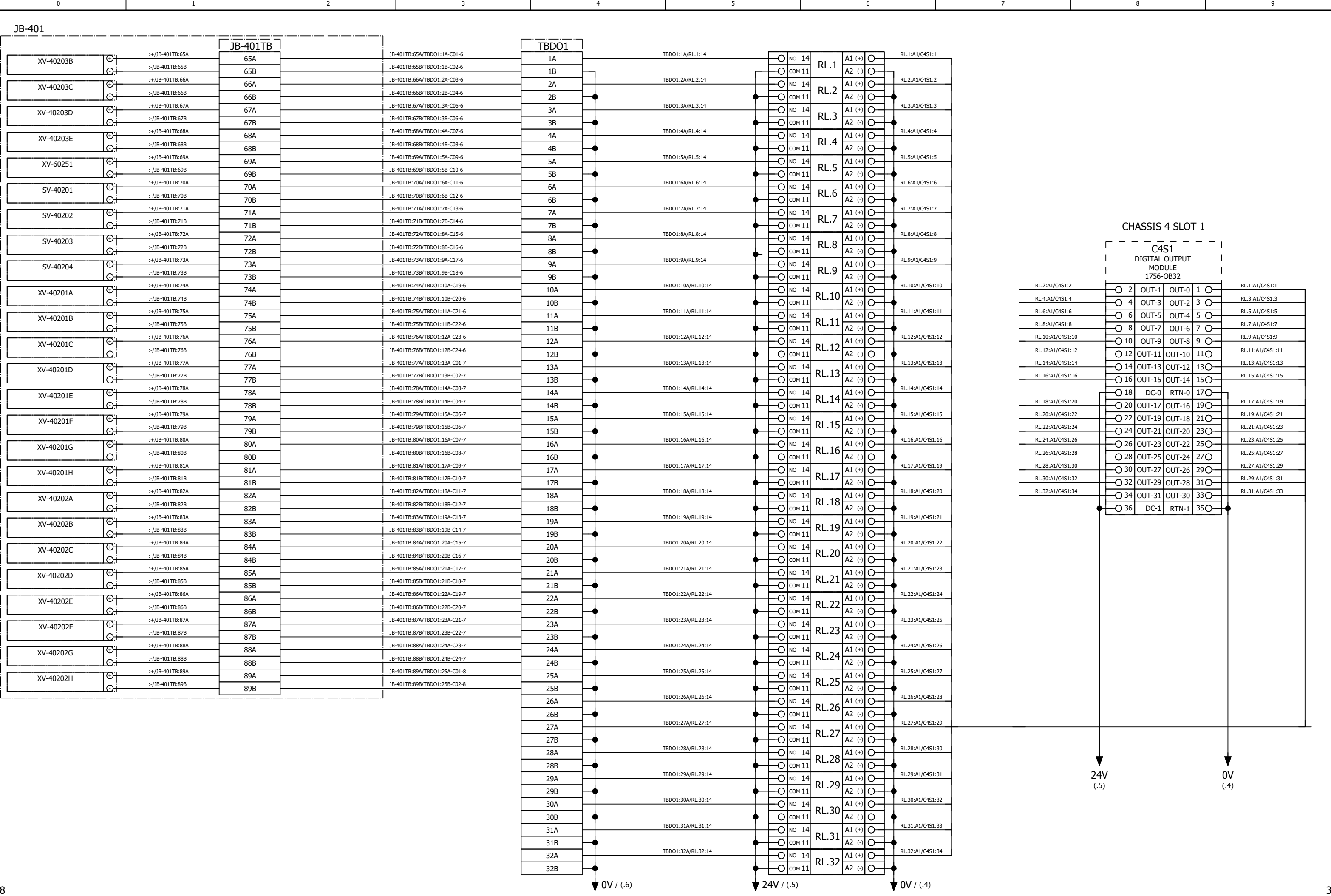




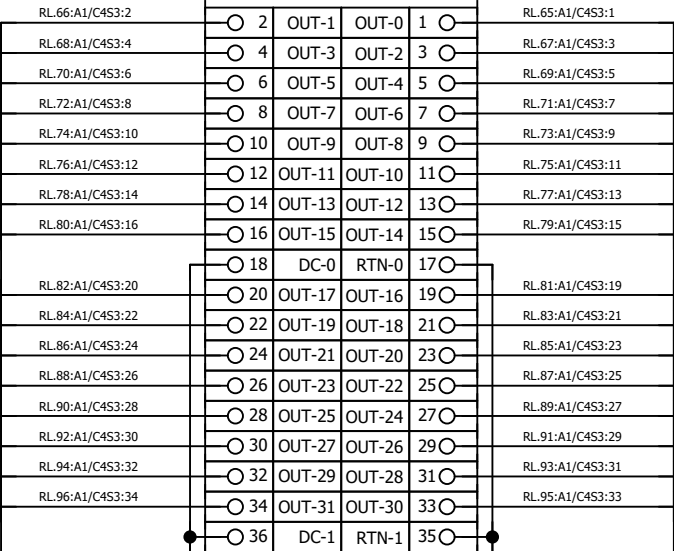
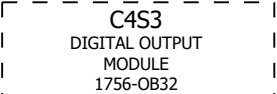
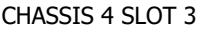
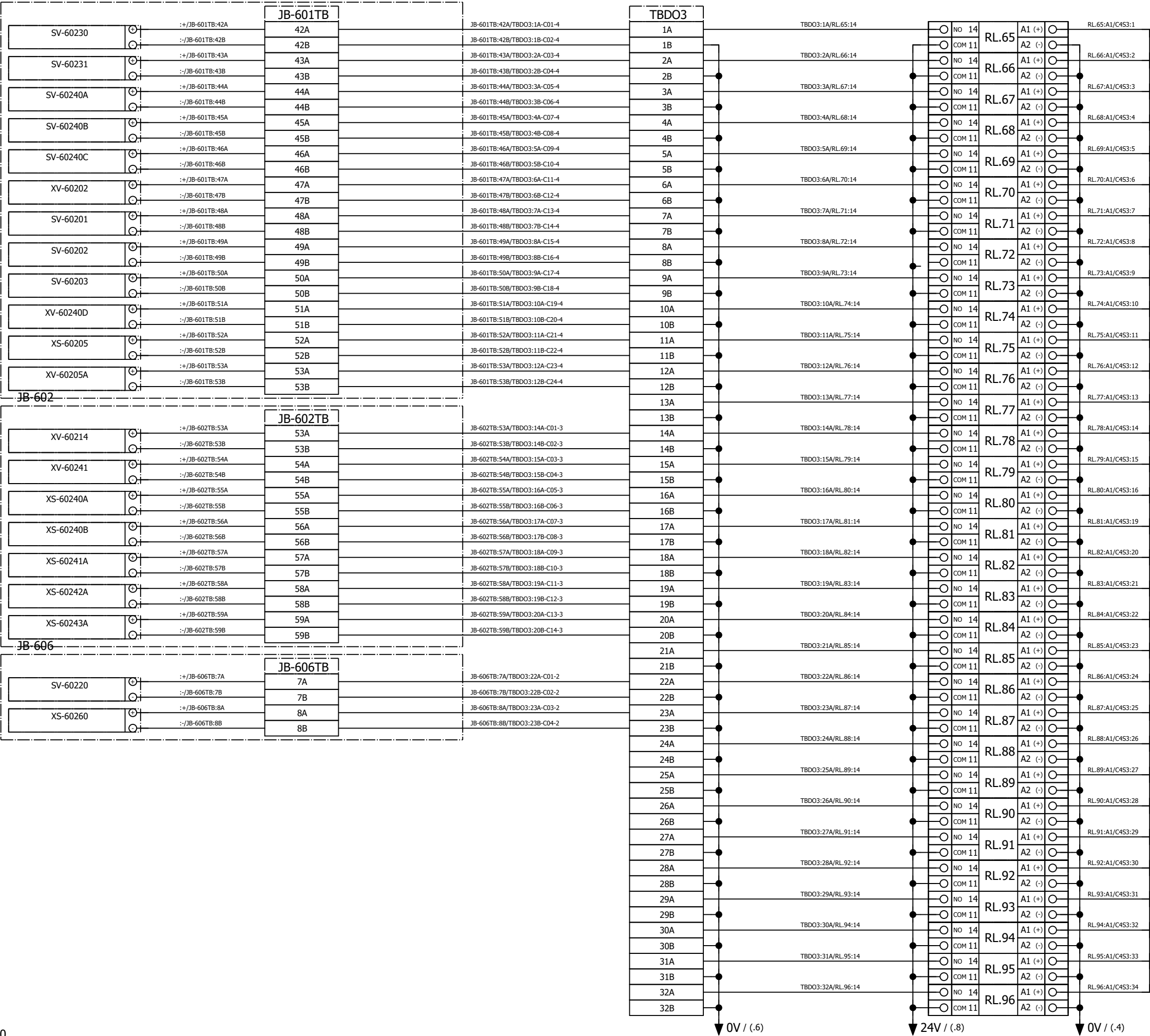




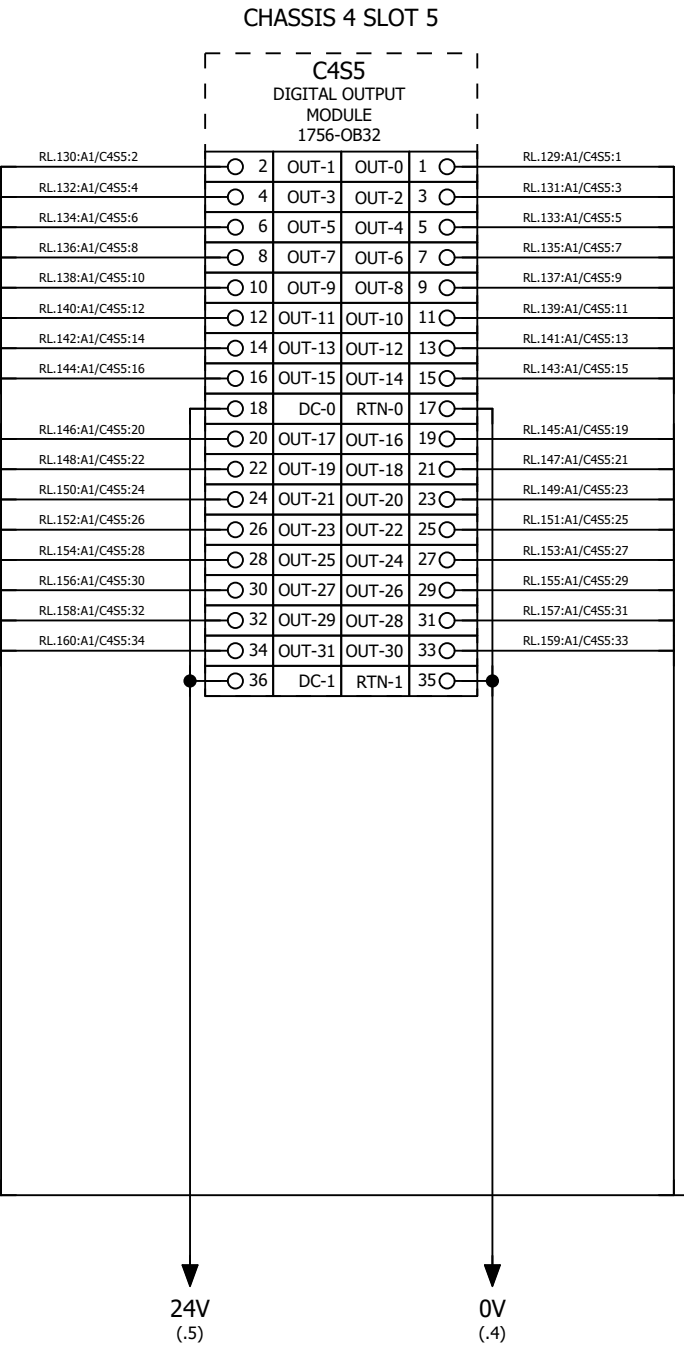
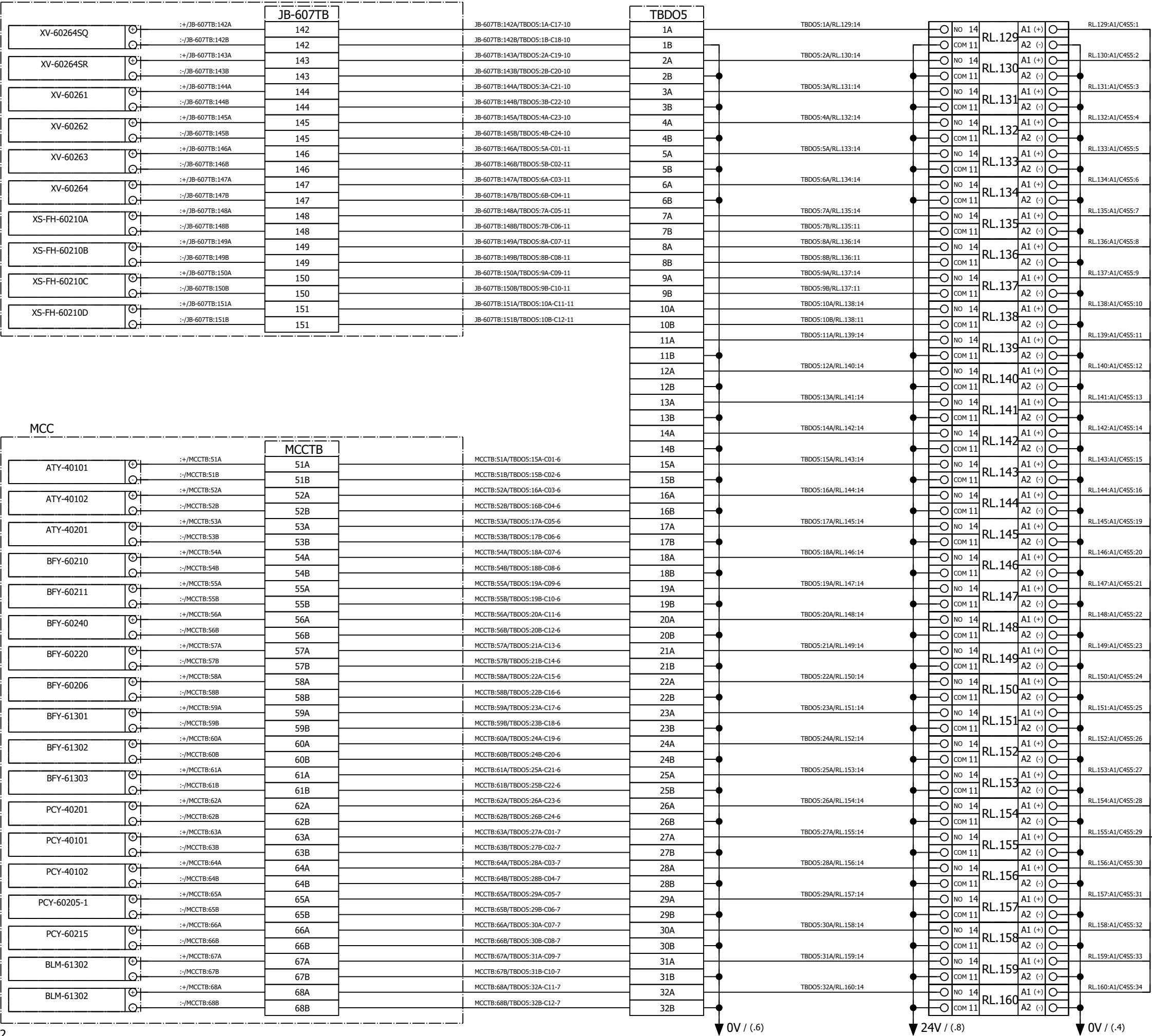
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			Ed.	THUMWH							+	
			Appr.						Basic project according to NFPA standard using a metric grid			
Modification	Date	Name	Original		Replacement of	Replaced by			NFPA_mm_bas001		Page	2
											Page	29 / 39

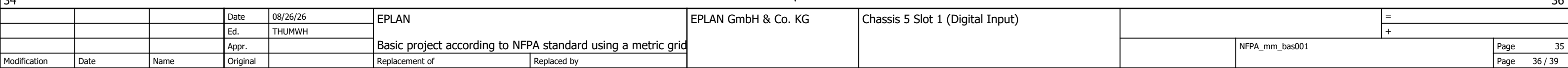


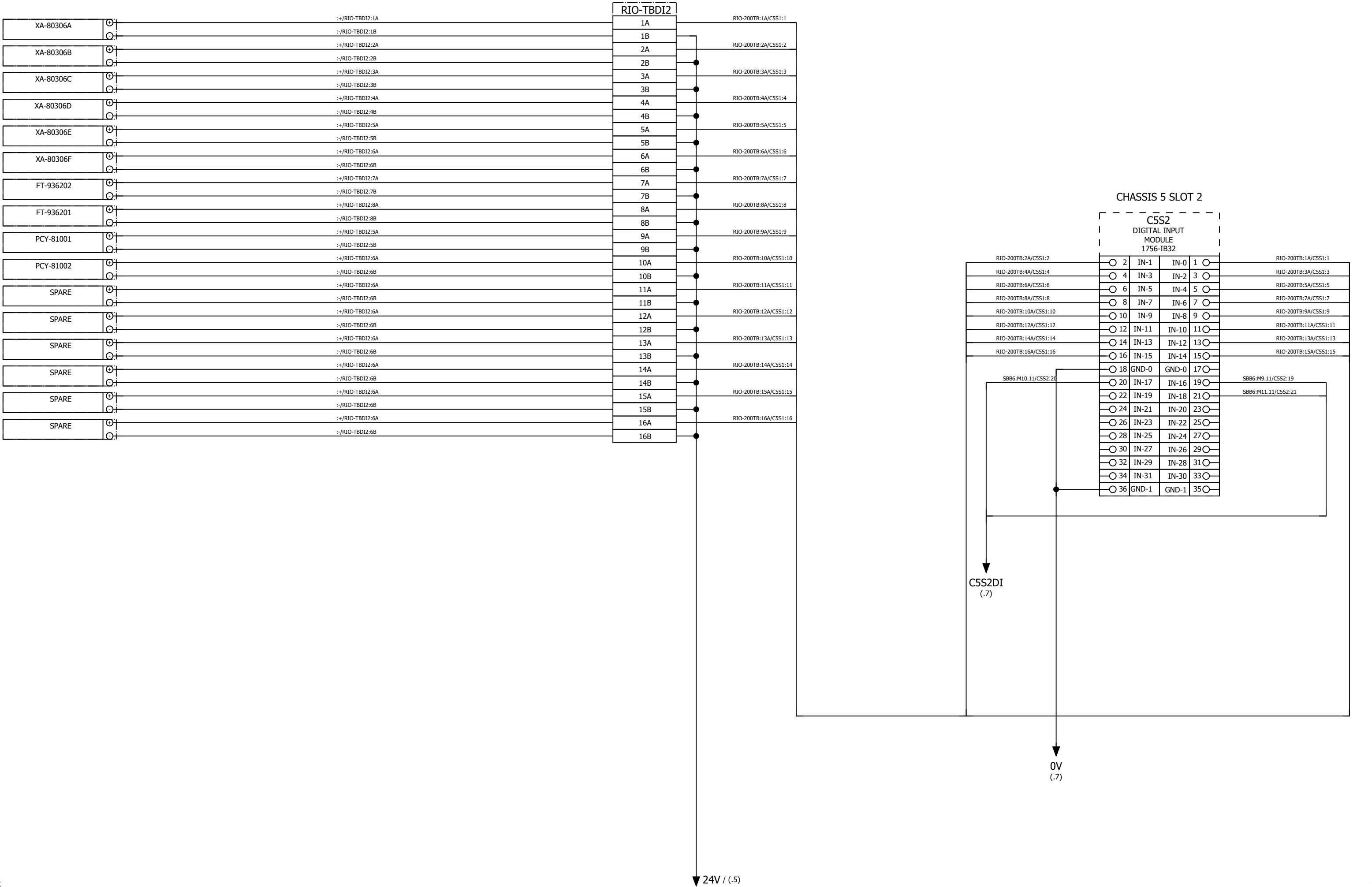


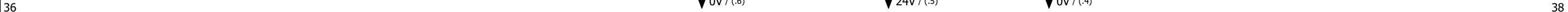


JB-607







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